

Algebra and discrete mathematics

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

Give one student page at a time; allow exploration before hints. The accompanying check programs verify finite instances and identities. General statements need the arguments or supplied facts in the keys. Preparation, timing and classroom engagement remain untested.

AD-01	Build a world that defeats a rule	4
	Every/some language; witness objects and a complete four-type argument	
AD-02	Can your list catch my pattern?	8
	Position matching and mismatch proofs; infinity is a separate advanced page	
AD-03	A collection with no card inside another	12
	Set containment, counting to sixteen, and a chain-cover upper bound	
AD-04	Send a network in a tiny message	17
	Leaves, changing labels, and inverse reasoning; general n uses induction	
AD-05	How many connecting networks survive?	23
	Connectedness and complete case counts; determinants are separately gated	
AD-06	Do two calculation routes agree?	29
	Least/greatest bounds and nested operations; no arithmetic required	
AD-07	Everything your starting clues force	34
	Positive if-then rules and reasoning about every completed run	
AD-08	Which numbers may share a name?	38
	Addition on a four-ring, independent representatives, and pair translation	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Contents continued

Facilitator guide and complete solutions / Remaining atlas entries / September 2026

The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

AD-09	Can both schedules be right?	43
	Remainders and paired clocks; gcd and Bézout are a separate advanced proof	
AD-10	Can the next match hide between our answers?	47
	Algebra and integer descent for completeness; counter arrangements are an earlier entry	
AD-11	Choose a rule that never loses a symbol	51
	Bit coefficients, distribution and inverse proofs; field certification is algebraic	
AD-12	Which cube volumes can these tools reach?	55
	Field-degree theorem and cubic irreducibility; exact geometry is an earlier entry	
AD-13	Move a corner, change the whole staircase	60
	Infinite-grid boundary proof; polynomial spanning and independence for the quotient	
AD-14	Can one calculation remember a rate?	65
	Rational polynomial algebra; derivative meaning uses the explicit polynomial power rule	
AD-15	Encode a rational point with one slope	69
	Exact fractions and quadratic factorization; completeness includes the missing point	
AD-16	Can these finite points make one cycle?	73
	Mod-five inventory; projective coordinates and a supplied group theorem for point addition	

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AD-17	Design the round when closeness begins	78
	Fractional updates and geometric powers; weighted invariants prove exact convergence times	
AD-18	Can you preserve area without preserving shape?	82
	Coordinate area and orientation; projection and matrix algebra support the full proof	
AD-19	Choose bases that reveal what an arrow does	87
	Bases, kernel and image; independent source/target coordinates are essential	
AD-20	Find what commutes, then explain the cancellations	91
	Matrix multiplication and symbolic cancellation; a concrete test leads to general identities	
AD-21	Make a catalog that remembers every choice	96
	Named pairs and matching colors; function uniqueness is the categorical continuation	
AD-22	Which face cards erase the same edge pattern?	101
	Even-degree edge patterns; vector spaces and quotient classes form the formal continuation	
AD-23	How much free space completes this module?	107
	Vector spaces and module actions; idempotents justify the two-inventory model of K0	
AD-24	Count necklaces when every bead count is fixed	112
	Rotation classes and exact counting; the general formula uses gcd and fixed-pair reasoning	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Build a world that defeats a rule

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate with oral facilitation. Quantifier language is the real gate; stop before the last page if every/some remain unstable.

A smallest counterworld plus a four-type explanation of which claims are forced.

Prerequisites and preparation

Core: Distinguish four color/shape types; understand every, some and not; give witnesses and a complete four-type argument. Count at most two objects.

Extensions: The arrow-world extension requires coordinating two quantifiers, without symbolic first-order notation.

Reading and representation: Adult may read all rules. Keep them visible. R/B labels plus different shapes avoid reliance on color.

Materials: Twelve cards: three each of red circle, red square, blue circle, blue square; two world mats; four-type panel. Preparation: about 8 minutes.

0-8 freely sort cards; 8-20 counterworlds; 20-35 inference cards; 35-40 move to four property corners; 40-55 missing witnesses; 55-60 explain one smallest world. Pages are staged.

Launch and facilitation

Build a world with a red object. Every red object must be a circle. Make “Every circle is red” fail.

Student page 1: Make the rule survive

Core: spoken every/some rules

Every card is red or blue, and a circle or square. Duplicates are allowed. An every-rule fails only when a card breaks it.

1 / prompt and solution

Build a world with a red object. Keep “Every red object is a circle” true and make “Every circle is red” false. Use the fewest cards possible. Mark the rule-breaker.

Solution: Use one red circle and one blue circle. The red circle supplies the required red object; the blue circle defeats the conclusion. One card cannot be both red and blue, so two is minimal.

Build a world that defeats a rule

1 / hints to offer only after exploration

1. What circle could break the second rule?
2. Can that card also supply the red object?

2 / prompt and solution

Choose an extra card that keeps the first rule true and one that breaks it. Describe every allowed card type.

Solution: A red square breaks it. Red circles, blue circles and blue squares are all allowed. This complete type check applies to worlds of any finite size and to duplicates.

2 / hints to offer only after exploration

1. Check the added card by itself.

Student page 2: Which claims are forced?

Core: examples versus guarantees

The promises now are: Every red object is a circle. There is at least one square. Claim cards: A Every square is blue. B Every blue object is a square. C There is a red object. D There is a blue object.

3 / prompt and solution

Choose a claim that can fail and build the smallest counterworld. Sort all four claims into forced or can fail. Give a witness world for each failure.

Solution: A and D are forced. B fails with one blue square and one blue circle; two are necessary because the required square and blue non-square witness are different objects. C fails with one blue square, the smallest possible world because a square is required. A holds because red squares are forbidden. D follows from the required square, which must be blue.

3 / hints to offer only after exploration

1. Can a red square satisfy the promises?
2. A there-is claim fails when that entire kind is missing.

4 / prompt and solution

Mark forbidden and required types in the four-type panel. Use it to explain why your forced claims survive a world with 100 objects.

Solution: Red square is forbidden and at least one blue square is required. Red circles and blue circles are optional. Every square must therefore be blue, and a blue object exists. More cards add no new type, so the proof covers every finite world satisfying the promises.

Build a world that defeats a rule

4 / hints to offer only after exploration

1. Large worlds still use only four card types.

Student page 3: A missing witness

Core continuation: negation and empty classes

No red object is promised on this page. An empty tray is allowed as a game world.

5 / prompt and solution

Test “Every red object is a circle” and “There is a red circle” on one blue square and on the empty tray. Explain the difference.

Solution: In both worlds every red object is a circle is true: no red noncircle exists. There is a red circle is false in both worlds. The blue-square world has a nonempty domain with an empty red class, so no empty-domain convention is needed for the main distinction.

5 / hints to offer only after exploration

1. What witness would make each claim fail?

6 / prompt and solution

Compare “Not every red object is a circle” with “Every red object is a square.” Build a nonempty world where just the first is true. Can you make just the second true?

Solution: A red circle plus a red square makes only the first true. One blue square makes only the second true: no red objects exist. If existence of a red object were required, the second statement would imply the first, and that second separation would be impossible. Not every red object is a circle means some red square exists.

6 / hints to offer only after exploration

1. For one direction keep both red shapes.
2. For the other, try having no red objects.

Optional facilitator extension 1

Gate: Coordinate two quantifier orders.

Two people may draw arrows to people, including themselves. Can everyone point to someone while no one receives arrows from everyone? Find the smallest nonempty world.

Solution: Two people with only their self-arrows work. One person cannot: its required self-arrow also makes it a recipient from everyone.

Build a world that defeats a rule

Mathematics and prior use

Finite unary structures, semantic consequence and countermodels. Four types certify this restricted setting; no general first-order decision procedure is claimed.

Prior use: Related to year-1 truth puzzles and Week 5 consistency, but no exact current-week problem is reused. The accepted ten have no finite-countermodel investigation. Planned, not taught.

Source: Open Logic Project, Sets, Logic, Computation

§7.4, Definition 7.11, printed pp. 119-120; §7.7 semantic notions.. Finite card-world interpretation; actual examples and minimality arguments developed here. Evidence: Relevant satisfaction definition revisited on 2026-09-25. Source structures have nonempty domains; an empty tray is explicitly a game convention on page 3.

<https://slc.openlogicproject.org/slc-screen.pdf>

Can your list catch my pattern?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for finite escape. The infinity page is advanced/specialist; changing four counters is not itself an uncountability proof.

A missing word with four mismatch certificates and a method valid against every four-row list.

Prerequisites and preparation

Core: Track row/position labels and give one guaranteed disagreement per row. Count to sixteen; symbolic algebra is unnecessary.

Extensions: Final page: infinite sequences indexed by positive integers and a proof about an arbitrary enumeration. Finite play does not establish this readiness.

Reading and representation: 0/1 can be two counter colors. An experienced facilitator should read the infinite-stage statements.

Materials: Two-color counters; supplied 4-by-4 grid, answer strips and row-to-position witness arrows; sixteen blank word cards. Preparation: about 5 minutes.

0-8 invent patterns; 8-20 escape a list; 20-35 trade lists; 35-40 four people act as positions; 40-55 prove the finite guarantee and boundary. Infinity is a separately gated continuation.

Launch and facilitation

Make a four-place pattern absent from this collection, then write your own list for a partner.

Student page 1: Escape a collection

Core: matching positions

Every pattern has four places. One mismatch is enough to make two patterns different. The supplied collection is 0011, 0101, 1010, 1111.

1 / prompt and solution

Choose a missing pattern. Mark a place where it differs from each row. Could another answer also win?

Solution: Any of the twelve unlisted words wins. Example 1000 differs from the four rows at positions 1,1,3,2 respectively. There are sixteen possible words and four distinct listed words, so twelve are missing.

1 / hints to offer only after exploration

1. One mismatch per row suffices.

Can your list catch my pattern?

2 / prompt and solution

Write four patterns for a partner; repeats are allowed. Trade lists and invent a method that always escapes without searching through every possible word.

Solution: Assign a distinct output position to each row and choose the opposite bit there. Each row then has a guaranteed mismatch. Any permutation of the four positions works, and repeated rows do no harm.

2 / hints to offer only after exploration

1. Reserve a different answer place for each row.

Student page 2: Make a guarantee

Core: one witness for each row

After trying your own idea, use this method: match row 1 to position 1, row 2 to position 2, and so on. Choose the opposite bit in each matched place.

3 / prompt and solution

Apply the method to the supplied collection. Why can changes outside the matched input places not spoil it? What if you rearrange the rows and run the method again?

Solution: Diagonal bits 0,1,1,1 give 1000. Its position i differs from row i . Off-diagonal input changes preserve those witnesses. After a row rearrangement, rerunning the method may change the answer but again gives a witness for every row.

3 / hints to offer only after exploration

1. Point to row 3's protected mismatch.

4 / prompt and solution

Must a list of five four-place patterns still miss something? What is the fewest rows that could catch every pattern? Build a systematic complete collection.

Solution: Five rows miss at least eleven of sixteen words, although five different positions cannot be assigned in a four-place word. Sixteen rows are necessary and sufficient. Use prefixes 00,01,10,11, each with suffixes 00,01,10,11. Fewer than sixteen rows cannot contain all sixteen distinct words. The limit of one recipe differs from the limit of escape itself.

4 / hints to offer only after exploration

1. Count the two choices at each of four places.
2. Repeats do not catch more distinct patterns.

Can your list catch my pattern?

Student page 3: A list with no final row

Advanced: infinite sequences and arbitrary indices

Each infinite binary pattern gives a bit at every positive-integer position. A proposed complete list has rows $s_1, s_2, \dots$, each infinite. A rule at every position defines a sequence without completing an endless drawing.

5 / prompt and solution

Define a new sequence b from the list. If someone says b is row 137, where can you find a mismatch? Explain why no row number works.

Solution: Set $b(i) = 1 - s_i(i)$ for every positive integer i . At position 137 it differs from row 137; at position k it differs from row k . Thus b is no listed row. The construction applies to any proposed list, proving no positive-integer-indexed list contains all infinite binary sequences.

5 / hints to offer only after exploration

1. Give row i its own witness at position i .

6 / prompt and solution

The list maker puts b at the top and moves all old rows down. Is the collection now complete? What does your earlier proof actually promise?

Solution: The old b is now listed; there is no fixed b defeating every list. For the new list t construct $c(i) = 1 - t_i(i)$. It is absent from every new row and differs from b at position 1. The omitted sequence depends on the proposed list.

6 / hints to offer only after exploration

1. Use the new row numbers.

Optional facilitator extension 1

Gate: Three symbols; powers only for the optional count.

Does the finite method work with three symbols?

Solution: Yes: change $0 \rightarrow 1$, $1 \rightarrow 2$, $2 \rightarrow 0$ at the assigned coordinates. Only disagreement matters. There are 3^n length- n words; an n -row list misses the constructed word for positive n .

Can your list catch my pattern?

Mathematics and prior use

Diagonalization and certificates of inequality. Binary sequences avoid decimal-expansion ambiguities.

Prior use: Week 6 identifies hidden codes, whereas this constructs omissions. GA-29 in the accepted ten already has infinite diagonalization for Cantor addresses; skip or explicitly compare that continuation if recently used.

Source: Open Logic Project, Sets, Logic, Computation

§4.6 Theorem 4.17, printed pp. 57-59 (PDF pp. 78-80).. Finite adversarial-list game with a separate infinite argument. Evidence: Binary-sequence proof revisited on 2026-09-25; Theorem 4.18 concerns the power set instead.

<https://slc.openlogicproject.org/slc-screen.pdf>

A collection with no card inside another

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate after containment is secure. Chain construction may need a delayed hint; the proof is the mathematical destination.

Six chosen cards with a complete six-chain upper-bound certificate.

Prerequisites and preparation

Core: Distinguish containment from overlap; count sixteen cards; use a chain partition as an upper bound.

Extensions: Five-symbol extension: permutations, factorial counting and fractions, or an independently built ten-chain certificate.

Reading and representation: Picture symbols can replace letters. Adult can read the pairwise rule.

Materials: Full sixteen-card subset deck of A,B,C,D; counters; six blank chain strips. Preparation: about 12 minutes.

0-10 sort; 10-15 compare overlap with containment; 15-30 maximize; 30-35 movement: grow a group one member at a time; 35-50 chain certificates; 50-60 a forced-card variant.

Launch and facilitation

Choose as many cards as possible. Never choose two when every symbol on one also appears on the other. Sharing some symbols is allowed.

Student page 1: Make a large collection

Core: set containment

The supplied deck has every subset of A,B,C,D once, including empty. A card is inside another when all its symbols occur on the other. A legal collection never contains two distinct selected cards when one is inside the other. Sharing only some symbols is allowed.

1 / prompt and solution

Build a large legal collection. Try to improve it by trading cards, and record your best result. Explain why every pair is legal.

Solution: The six pairs AB,AC,AD,BC,BD,CD are legal: distinct equal-size sets cannot contain one another. A successful improvement is {AB,AC,BC} to all six pairs. Many initial collections are legitimate exploration; page 2 proves six optimal.

A collection with no card inside another

1 / hints to offer only after exploration

1. Group cards by how many symbols they show.

2 / prompt and solution

What happens if you keep the empty card or ABCD? Build a legal collection with different card sizes.

Solution: Either empty or ABCD excludes every other card. A mixed-size example is {A,BC,BD,CD}, size four. Four is the largest mixed-size antichain: apart from empty/full, mixed size forces a singleton or triple; page 3 bounds either case by four. That upper bound is facilitator support, not required before exploration.

2 / hints to offer only after exploration

1. Keep A and look for cards with no A.

Student page 2: Convince a skeptic

Core: complete upper-bound certificate

A chain is a row where each card is inside the next. Divide the whole deck into chains, with no cards omitted or repeated. Blank chain strips are supplied; single-card chains are allowed.

3 / prompt and solution

Why can a legal collection take at most one card from a chain? Build a chain partition whose bound matches your best collection.

Solution: Any two cards in a chain are comparable. A certificate is $\emptyset < A < AB < ABC < ABCD$; $B < BC < BCD$; $C < AC < ACD$; $D < AD < ABD$; BD ; CD . All sixteen cards occur once, and there are six chains. Thus at most six cards can be selected, matching the six pairs.

3 / hints to offer only after exploration

1. Start with a chain from empty to all four symbols.

4 / prompt and solution

Audit a partner's certificate for nesting and full coverage. Explain why it rules out every seven-card collection, not just the ones you tried.

Solution: The supplied certificate has lengths 5,3,3,3,1,1 and exact full coverage. Any seven selected cards in six chains put two in one chain, where they are nested. This applies to every possible collection.

4 / hints to offer only after exploration

1. Assign one color to each chain.

A collection with no card inside another

Student page 3: Keep this card

Core continuation: transfer the bound

The card A must now be chosen. All other rules stay the same.

5 / prompt and solution

Find the largest legal collection that keeps A. Which cards are unavailable, and which remain candidates?

Solution: All other sets containing A and also empty are forbidden. Nonempty subsets of B,C,D remain. $\{A, BC, BD, CD\}$ gives four, as does $\{A, B, C, D\}$.

5 / hints to offer only after exploration

1. Compare every candidate directly with A.

6 / prompt and solution

Prove your new answer is best using chains. Why does prescribing a three-symbol card instead of a singleton give the same maximum?

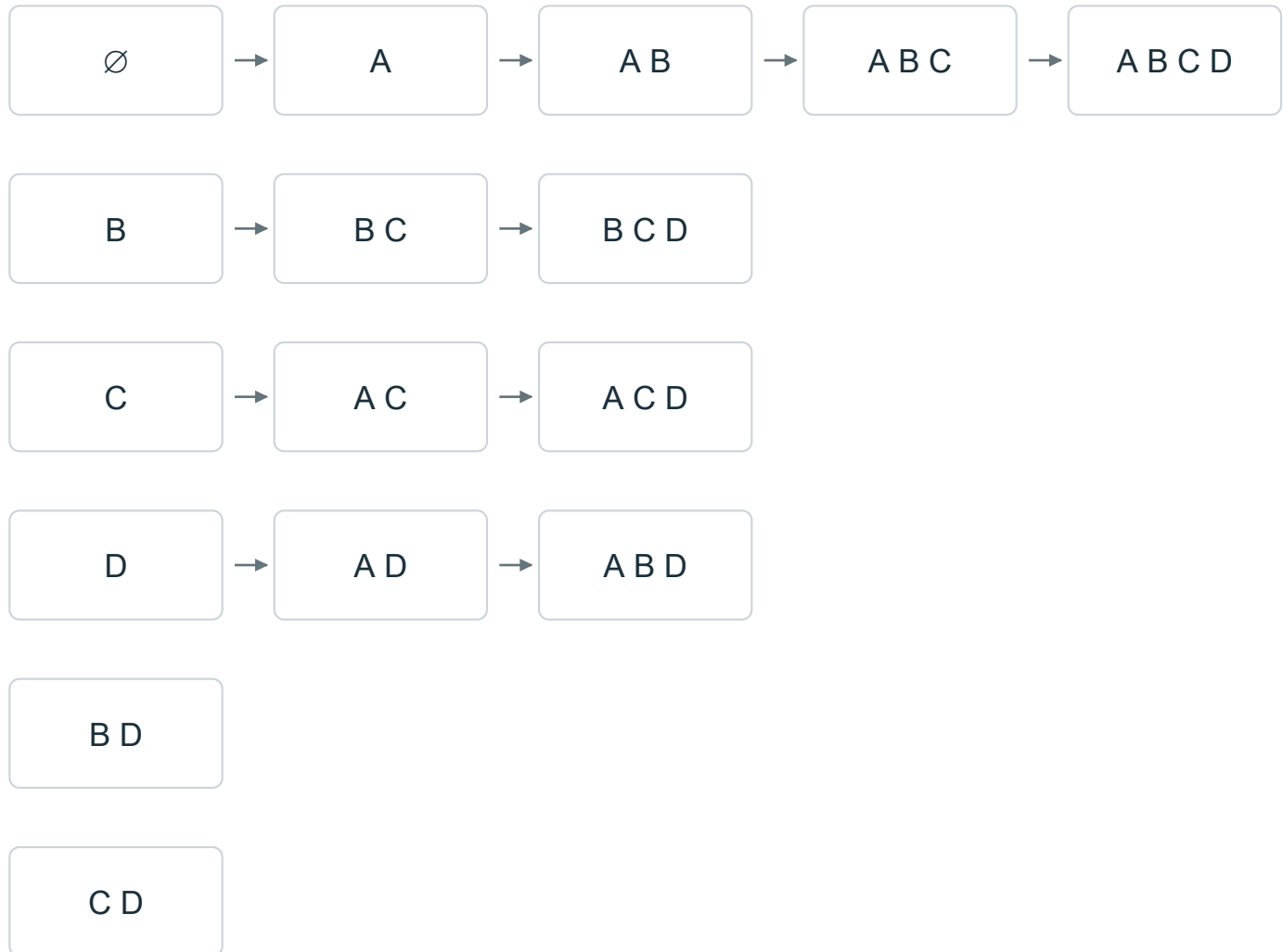
Solution: The remaining seven candidates partition as $B < BC < BCD$; $C < CD$; $D < BD$. At most three join A, giving four total. Complementation reverses inclusion and exchanges singleton with triple, preserving the no-containment rule. Thus a prescribed triple also permits at most four; all four triples attain four.

6 / hints to offer only after exploration

1. The useful remaining cards are a smaller subset deck.
2. Replace each set by its missing symbols.

A complete chain certificate

Worked representation / every card appears exactly once



At most one selected card can come from each row. The six two-symbol cards attain this six-row bound.

A complete chain certificate

Optional facilitator extension 1

Gate: Permutations, factorials and fractions; this is a separate proof.

Find the maximum for five symbols.

Solution: Ten pairs attain ten. For the bound, examine all 120 orders of the symbols and their initial-segment chains. A fixed k -set occurs in $k!(5-k)!$ chains; an antichain occurs at most once per chain. Hence its sum of $1/\text{binomial}(5,k)$ is at most 1. Each denominator is at most 10, so there are at most ten sets.

Mathematics and prior use

Antichains in a Boolean lattice and the $n=4$ case of Sperner's theorem. The finite certificate does not establish every n .

Prior use: No matching Week 1-10 antichain problem. Week 1 shares construction-plus-bound reasoning. AD-07 reuses subset cards for a different mechanism. No exact accepted-ten instance is reused.

Source: Richard P. Stanley, Topics in Algebraic Combinatorics

Chapter 4, Corollary 4.8 and following proof, printed pp. 53-54 (PDF pp. 42-43).. A checked sixteen-card chain partition and new forced-singleton problem. Evidence: Sperner statement and nearby proof inspected on 2026-09-25; both finite bounds independently enumerated.

<https://math.mit.edu/~rstan/algcomb/algcomb.pdf>

Send a network in a tiny message

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. Changing graph/message state is demanding. The degree invariant, rather than a catalog of pictures, is the destination.

A successful exchange plus the reason absent message labels are exactly the current leaves.

Prerequisites and preparation

Core: Recognize trees and leaves; follow changing smallest-label choices; explain why decoding is forced. Ordering labels through five and counting occurrences.

Extensions: General n proof uses induction and the degree/occurrence invariant; powers enter the final count.

Reading and representation: Short algorithm cards; partner may read. A surviving-label ledger reduces memory load.

Materials: Five numbered cards and strings; supplied four-dot and five-dot frames; message strips and survivor ledgers. Preparation: about 8 minutes.

0-10 build networks and remove cycles; 10-25 encode four-label trees; 25-35 reconstruct; 35-40 act out leaf removal; 40-55 prove the inverse and count. Five labels can continue another meeting.

Launch and facilitation

Build a tree on 1,2,3,4. Remove the smallest leaf and record its neighbor, twice. Can a partner recover the edges?

Student page 1: Make a message

Core: leaves and changing labels

A tree connects all named vertices without a cycle. A leaf has one remaining edge. Remove the smallest leaf and its edge; record its neighbor. Repeat until two vertices remain. Record the removed labels. Only numbered circles are vertices; a crossing without a circle is not a vertex.

1 / prompt and solution

Invent and encode a tree on 1,2,3,4. Then encode X with edges 12,13,34 and Y with edges 13,23,24. Show the deletion order.

Solution: Any connected acyclic invented graph is accepted and deterministically checked. X removes 2 then 1, recording (1,3). Y removes 1 then 3, recording (3,2). An edge 12 is undirected.

Send a network in a tiny message

1 / hints to offer only after exploration

1. Recheck the leaves after every removal.

2 / prompt and solution

Give only your message to a partner. Try to reconstruct one another's trees. What would identify the first removed leaf? Test your idea on (3,1).

Solution: Choose the smallest surviving label absent from the remaining message. For (3,1), remove 2 and add 23; then remove 3 and add 13; join 1 and 4. Edges {23,13,14} encode back to (3,1). The decoder is supplied on the next page after the attempt.

2 / hints to offer only after exploration

1. Could a label still appearing as a neighbor disappear now?

Student page 2: Why the receiver knows

Core proof: occurrences and degree

Decoder: choose the smallest surviving label absent from the remaining message. Join it to the first entry. Remove that label and entry. Repeat, then join the final two labels. Sender and receiver both use smallest-leaf order.

3 / prompt and solution

Choose any two-entry message, perhaps a repeated one. Decode and re-encode it. Explain why every message works and why two different trees cannot send the same message.

Solution: With m surviving labels, the suffix has $m-2$ entries, so an absent label exists. The next entry is a different survivor: a previously removed label was absent from the entire suffix at its removal and cannot occur later. Read construction backward, attaching each removed vertex by one edge to the already built tree, so every word decodes to a tree. In the residual tree built from any suffix, vertex v receives one incident edge for each occurrence of v in that suffix, and one further edge when v itself is removed or is joined in the final pair. These edges are distinct. Its residual degree is therefore its suffix-occurrence count plus one. The residual leaves are exactly the absent labels, so encoding this tree removes the same smallest label and records the same next entry at every step: $\text{encode}(\text{decode}(\text{word})) = \text{word}$. Conversely, in an original tree v is recorded once for every incident edge lost before its last edge, hence $\text{degree}-1$ times. Its current leaves are exactly the absent labels, so decoding its code recovers each deleted leaf and neighbor by induction: $\text{decode}(\text{encode}(\text{tree})) = \text{tree}$. Both inverse directions establish existence and uniqueness.

Send a network in a tiny message

3 / hints to offer only after exploration

1. Read the construction backward.
2. How many incident edges disappear before a vertex has only one left?

4 / prompt and solution

Count the trees described by two-entry messages on 1,2,3,4. What do repeated entries tell you? Why is counting drawings or rotations unreliable?

Solution: Four choices at each place give 16 messages and thus 16 labeled trees. The four messages (r,r) give stars centered at r; the other twelve give paths. Rotations do not change labeled edges. Ignoring labels gives only two shapes, star and path.

4 / hints to offer only after exploration

1. Count choices in the two message places.

Student page 3: Predict before drawing

Continuation: five labels; general proof is advanced

For five labels use three-entry messages. Degree is the number of edges touching a vertex. The same rules apply.

5 / prompt and solution

Decode (4,4,2). Before drawing, predict its leaves and all vertex degrees from the message. Explain the rule.

Solution: Steps add 14,34,24, then 25. Occurrences for labels 1-5 are 0,1,0,2,0, so degrees are 1,2,1,3,1 and leaves are 1,3,5. Each vertex is recorded for all incident edges lost before its last edge, hence degree equals occurrences plus one.

5 / hints to offer only after exploration

1. One final edge does not record this vertex as a neighbor.

Send a network in a tiny message

6 / prompt and solution

Design a five-label message with exactly two leaves. Then, if you can justify the inverse for every $n \geq 2$, give the general count.

Solution: Use three distinct entries, for example (1,2,3), decoding to {14,12,23,35}. Exactly two labels are absent, so there are two leaves. Repeated entries leave at least three labels absent. The induction in prompt 3 works for every $n \geq 2$, giving a bijection with length $n-2$ words and $n^{(n-2)}$ trees. At $n=2$ the empty word gives one edge. Generality requires that proof, not extrapolation.

6 / hints to offer only after exploration

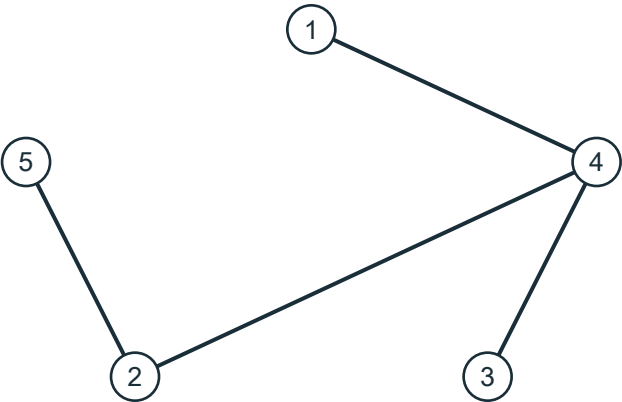
1. Leaves are the absent labels.

Read the message backward

Worked five-label decoding / message (4, 4, 2)

survivors	suffix	smallest absent	edge
1 2 3 4 5	4 4 2	1	1-4
2 3 4 5	4 2	3	3-4
2 4 5	2	4	4-2
2 5	empty	final pair	2-5

Reconstructed labeled tree



Degrees 1, 2, 1, 3, 1 equal message occurrences plus one. Reading construction backward attaches one new leaf each time.

Read the message backward

Optional facilitator extension 1

Gate: Same finite operations.

What if the sender removes largest leaves but the receiver uses smallest?

Solution: Tree $\{12,13,34\}$ sends $(3,1)$ with largest-leaf order, but smallest-leaf decoding returns $\{23,13,14\}$. Either consistent convention works; mixing them does not.

Mathematics and prior use

Prüfer correspondence and Cayley's count. Rotating a drawing does not change its labeled edge set.

Prior use: Week 10 studies routes; accepted-ten GA-25 studies grid trees and edge exchange. This is an invertible code for all labeled trees. AD-05 restricts available edges.

Source: Richard P. Stanley, Topics in Algebraic Combinatorics

Chapter 9 appendix, first proof (Prüfer), printed p. 180 (PDF p. 146).. Use smallest rather than the source's largest leaf, consistently in both directions; new messages supplied. Evidence: Coding and inverse passage read on 2026-09-25. Reversing label order converts the conventions.

<https://math.mit.edu/~rstan/algcomb/algcomb.pdf>

How many connecting networks survive?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the exact count and closure surprise. The determinant page is advanced/specialist and not needed to make the core substantial.

Eight classified trees, then a proof that closing a perimeter road destroys more of these choices than closing the diagonal.

Prerequisites and preparation

Core: Distinguish a connected minimal network from one with a cycle; count small cases and prove a complete classification by a choice of edges.

Extensions: Page 4 requires matrices and 3-by-3 determinants. The general Matrix-Tree theorem is supplied there; proving it requires incidence matrices and Cauchy-Binet.

Reading and representation: Core can be read aloud from labeled roads. Advanced page uses matrix notation and signed products.

Materials: Supplied square A,B,C,D with diagonal AC; five removable road strips; blank network frames; pencils in two colors. Preparation: about 8 minutes.

0-10 connect and break roads; 10-30 classify minimal networks; 30-35 human cycle movement; 35-50 investigate closures; 50-60 explain the most disruptive closure. Added roads and determinants are staged continuations.

Launch and facilitation

Keep all four places connected using only the five pictured roads. No kept road may be unnecessary: removing it must disconnect your network.

Student page 1: Choose the roads

Core: connectedness and unnecessary edges

Available undirected roads are AB,BC,CD,DA and AC. The four vertices are the only places. Two drawings are the same network when they keep the same named roads. A minimal connected network joins all four vertices, and removing any one kept road disconnects it.

How many connecting networks survive?

1 / prompt and solution

Choose a minimal connected network. Try to make a different one. How many roads does each use, and why?

Solution: Each uses three roads. A connected graph on four vertices needs at least three: starting with four components, one edge can reduce the component count by at most one. A minimal connected graph has no cycle, since a cycle edge could be removed. An acyclic connected graph on four vertices has three edges: grow a connected explored set along its unique new-vertex attachments, or remove leaves inductively. Concrete trials may precede this proof.

1 / hints to offer only after exploration

1. Start with four separate places and add roads one at a time.

2 / prompt and solution

Find every minimal connected network. Choose an organizing rule that helps you avoid both duplicates and omissions. Record roads, not just pictures.

Solution: Without AC, choose any three of the four square edges: four trees. With AC, choose exactly one of AB/BC to attach B and one of CD/DA to attach D: four trees. These are all possibilities because a tree has three edges, and with AC present both B and D need an incident selected road. Total eight.

2 / hints to offer only after exploration

1. Separate networks by whether they contain AC.

Student page 2: Close one road

Core: compare classified cases

A road closure removes that road from the available graph. A previously recorded network survives exactly when it did not use that road.

3 / prompt and solution

Choose a road to close so that as many of your networks as possible survive. Predict, then count survivors. Do all five closures have the same effect?

Solution: Closing AC leaves the four trees obtained from the square. Closing any perimeter edge leaves three trees, so closing AC preserves the most choices. Each perimeter edge occurs in five of the eight trees, while AC occurs in four. Geometric symmetry makes the four perimeter edges equivalent.

3 / hints to offer only after exploration

1. Mark appearances of one edge in your completed list.

How many connecting networks survive?

4 / prompt and solution

Explain the difference without recounting all eight pictures. For closure of AB, organize the surviving trees by whether they use AC.

Solution: Without AC, only the path BC,CD,DA survives: one. With AC, B must attach via BC; D can attach via CD or DA: two. Thus three survive an AB closure. For AC closure, any one of four square edges can be omitted, giving four. By square symmetries preserving AC, every perimeter closure has the same three-survivor count.

4 / hints to offer only after exploration

1. After AB closes, what is B's only way into a diagonal-containing tree?

Student page 3: Add a road

Core continuation: a larger case

Now add BD. Its crossing with AC is not a new vertex: roads pass over one another. All six pairs of the four vertices are available.

5 / prompt and solution

Count the new minimal networks by separating those with no diagonal, exactly one diagonal, and both diagonals. Check that none is counted twice.

Solution: No diagonal: four trees. AC only: four; BD only: four, giving eight with exactly one. Both diagonals: they initially make two components $\{A,C\}$ and $\{B,D\}$; any one perimeter road joins them, giving four. Total $4+8+4=16$. This matches AD-04 if that code investigation was done.

5 / hints to offer only after exploration

1. With both diagonals selected, what remains to connect?

6 / prompt and solution

In the six-road graph, does it matter which single road closes? Give a reason and the number of surviving trees.

Solution: All six edges are equivalent under relabeling of K_4 . The sixteen trees use three roads each, for 48 edge appearances; dividing equally among six edges gives eight appearances per edge and eight surviving trees after any one closure. Alternatively deleting any edge gives a relabeled copy of the original five-road graph and hence eight. No probability model is needed.

6 / hints to offer only after exploration

1. Can any pair of vertex names be relabeled as any other pair?

How many connecting networks survive?

Student page 4: A second counting language

Advanced: matrices and determinants

Supplied theorem: for a finite connected simple graph (no loops or parallel edges), form L with vertex degrees on the diagonal and -1 for adjacent distinct vertices (0 otherwise). Delete the same row and column. The determinant equals the number of spanning trees. This page applies the theorem; it does not prove it.

7 / prompt and solution

For the original five-road graph, build L in order A,B,C,D. Delete D's row and column and compute its determinant. Compare with your classification.

Solution: $L = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}$. The D-deleted matrix is $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{bmatrix}$. Expanding along its first row gives $15 - 4 - 3 = 8$. Agreement checks the same graph in two representations, not a proof of the general theorem.

7 / hints to offer only after exploration

1. A diagonal entry counts incident roads.

8 / prompt and solution

Delete A's row and column instead. Why should the new determinant agree? What happens if you take $\det(L)$ without deleting anything?

Solution: The A-deleted matrix is $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, with determinant $2(6-1)-2=8$. The supplied theorem says either principal cofactor counts the same trees. The full L has row sums zero, so it sends $(1,1,1,1)$ to zero and is singular: $\det(L)=0$. Thus the row/column deletion is essential.

8 / hints to offer only after exploration

1. Add the entries in a row of the full matrix.

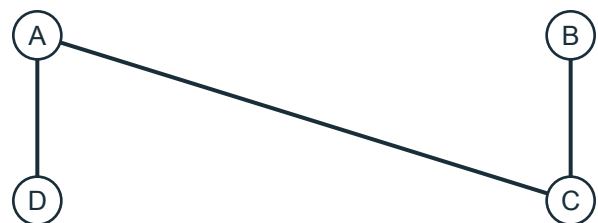
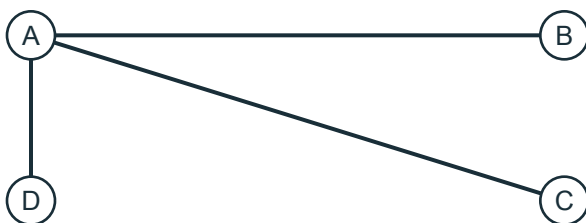
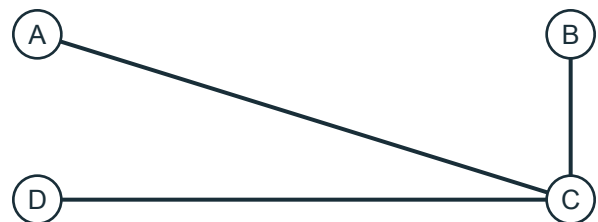
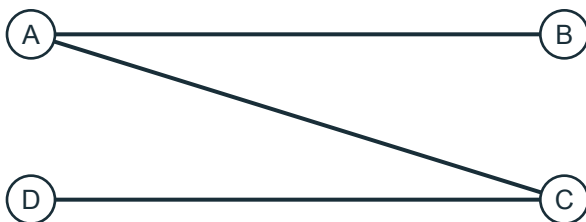
The complete road catalog

Worked eight-tree classification / original five-road graph

Without AC



With AC



The complete road catalog

Optional facilitator extension 1

Gate: Incidence matrices, determinant expansion, rank and Cauchy-Binet; advanced/specialist.

Give the reason behind the supplied Matrix-Tree theorem.

Solution: Orient edges arbitrarily and delete one row from the signed incidence matrix B . The corresponding Laplacian cofactor is $B B^T$. Cauchy-Binet expresses $\det(B B^T)$ as the sum of squared determinants of square submatrices with $n-1$ edge columns. A cycle gives a signed column dependence, hence determinant zero. If $n-1$ chosen edges are acyclic they form a spanning tree; successive leaf expansion gives determinant ± 1 . Thus exactly the spanning trees contribute one. Cauchy-Binet is the supplied linear-algebra theorem in this proof sketch.

Mathematics and prior use

Spanning trees, edge participation and the Matrix-Tree theorem. Road-closure counts compare choices, not traffic volume or real reliability without a probability model.

Prior use: Week 10 minimizes repeated travel, not connecting-subgraph choices. Accepted-ten GA-25 uses grid trees and duality; acknowledge that overlap and emphasize the edge-closure comparison. AD-04 codes all labeled trees.

Source: Richard P. Stanley, Topics in Algebraic Combinatorics

Chapter 9, Lemmas 9.6-9.7 and Theorem 9.8, printed pp. 175-177 (PDF pp. 141-143).. The theorem supplies an optional second counting representation; road-closure comparisons and all small instances are checked directly. Evidence: Laplacian definition and incidence/Cauchy-Binet proof revisited on 2026-09-25. No weighted or probabilistic theorem is assumed.

<https://math.mit.edu/~rstan/algebraic/algebraic.pdf>

Do two calculation routes agree?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. The five-tile diagram is compact, but least/greatest and nested operations are abstract; tangible result markers are essential. The final repair is optional depth.

Two well-defined calculation routes end at different elements in the five-element diamond, with a justified intermediate step at every operation.

Prerequisites and preparation

Core: Follow upward comparisons, distinguish a least upper bound from any upper bound, and evaluate nested operations. No arithmetic; 0 and 1 are names.

Extensions: General set identity uses membership reasoning. Subspace continuation needs linear algebra. The core counterexample requires neither.

Reading and representation: Use spoken floor/ceiling names or meet/join symbols; all intermediate results remain visible.

Materials: Eight subset cards of A,B,C; supplied five-tile and six-tile order diagrams; two movable markers. Preparation: about 10 minutes.

0-10 explore subset cards; 10-25 compare operation routes; 25-35 discover a failure on the five-tile diagram; 35-40 people act as bounds; 40-55 distinguish a broken identity from a missing operation; 55-60 show one counterexample.

Launch and facilitation

For two sets, collect everything in either set, or keep only what they share. Compare two recipes before trying the same operations on a new diagram.

Student page 1: Two recipes with sets

Core: union and intersection

For sets, JOIN collects every symbol in either set; MEET keeps only symbols in both. Supplied cards show all subsets of A,B,C. Recipe L: $\text{meet}(X, \text{join}(Y,Z))$. Recipe R: $\text{join}(\text{meet}(X,Y), \text{meet}(X,Z))$.

1 / prompt and solution

Choose three set cards and carry out both recipes, keeping each intermediate result visible. Try to make the answers different. Then test $X=AB$, $Y=AC$, $Z=BC$.

Solution: All choices agree for sets. For the specified triple, $\text{join}(Y,Z)=ABC$ and the left result is AB ; $\text{meet}(X,Y)=A$ and $\text{meet}(X,Z)=B$, whose join is AB . If learners report a difference, check copied operations or intermediate results rather than announcing a theorem before the search.

Do two calculation routes agree?

1 / hints to offer only after exploration

1. Draw or place one marker for each intermediate set.

2 / prompt and solution

Explain why any symbol belongs to the left answer exactly when it belongs to the right answer. Your explanation should cover every choice of three cards.

Solution: A symbol is on the left iff it is in X and in at least one of Y, Z . This is equivalent to being either in both X, Y or in both X, Z , which is right-side membership. Since membership agrees for each possible symbol, the sets agree. This proves the identity for all sets, not only this eight-card deck.

2 / hints to offer only after exploration

1. Follow one symbol through both recipes.

Student page 2: Try a new diagram

Core: greatest floor and least ceiling

The diagram has 0 below a, b, c , and 1 above a, b, c ; no comparisons connect different middle tiles. Travel only upward to compare. Every tile is also above and below itself. JOIN is the least tile above both inputs; MEET is the greatest tile below both.

3 / prompt and solution

Choose several pairs and find their JOIN and MEET. Include equal inputs, two different middle tiles, and pairs involving 0 or 1. Does every pair have both operations?

Solution: Yes. Equal inputs return that input. For different middle tiles, join is 1 and meet is 0. With 0, $\text{join}(0, x) = x$ and $\text{meet}(0, x) = 0$; with 1, $\text{join}(1, x) = 1$ and $\text{meet}(1, x) = x$. These pair types exhaust the diagram, so it is a lattice.

3 / hints to offer only after exploration

1. List all common upper bounds before choosing the least one.

4 / prompt and solution

Choose X, Y, Z from the five tiles and try to make recipes L and R disagree. Record every intermediate result and explain whether the failure means an operation is missing.

Solution: Take $X=a, Y=b, Z=c$. $\text{join}(b, c) = 1$, so $L = \text{meet}(a, 1) = a$. Both $\text{meet}(a, b)$ and $\text{meet}(a, c)$ are 0, so $R = \text{join}(0, 0) = 0$. All operations exist; only the proposed distributive identity fails. Any ordering of the three distinct middle elements gives the analogous failure.

Do two calculation routes agree?

4 / hints to offer only after exploration

1. Try three different middle tiles after your own choices.

Student page 3: Does the operation exist?

Core continuation: least versus minimal bounds

New diagram: 0 lies below p, q ; each of p, q lies below each of r, s ; r, s lie below 1. The pairs p, q and r, s are incomparable. Only upward paths count.

5 / prompt and solution

Find all common upper bounds of p and q . Is one of them the least? Find the matching difficulty for $\text{MEET}(r, s)$.

Solution: Common upper bounds of p, q are $r, s, 1$. Neither r nor s is below the other, and 1 is not below them. Thus there is no least upper bound. Common lower bounds of r, s are $0, p, q$, with neither p nor q greatest. This poset is not a lattice. Having upper bounds is insufficient.

5 / hints to offer only after exploration

1. Least must be below every other upper bound.

6 / prompt and solution

Add one comparison between r and s , in either direction. Redraw or move tiles so all comparison arrows point upward. Does this repair both missing operations? Check every kind of pair to decide whether the new diagram is a lattice (every pair has both JOIN and MEET).

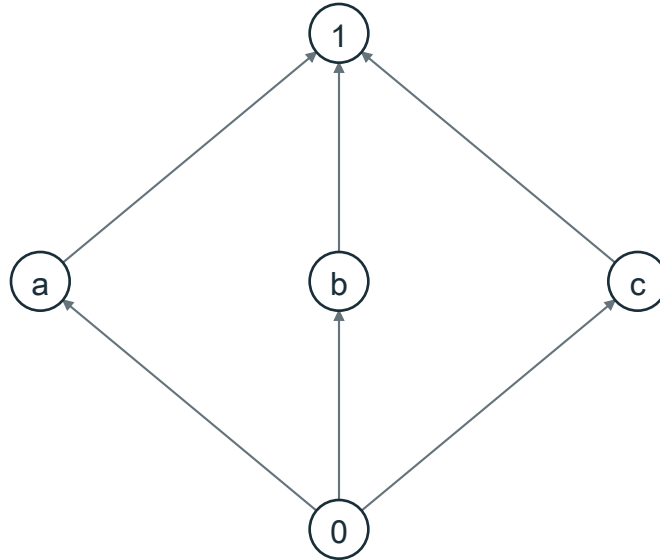
Solution: Choose $r < s$ (the reversed choice is symmetric). Then $\text{join}(p, q) = r$, $\text{meet}(p, q) = 0$; r and s are now comparable. Every other unequal pair is comparable except p, q , so it has the smaller input as meet and larger as join. Thus all pairs have unique operations and the repair gives a lattice. If the drawing retains redundant arrows, interpret upward paths rather than assuming every drawn arrow is a cover.

6 / hints to offer only after exploration

1. After adding $r < s$, which pair is still incomparable?

Existing operations, different results

Worked five-element diamond counterexample



Recipe L	Recipe R
$\text{join}(b,c) = 1$	$\text{meet}(a,b) = 0; \text{meet}(a,c) = 0$
$\text{meet}(a,1) = a$	$\text{join}(0,0) = 0$

Both routes are legal. Their different answers disprove the identity; they do not make JOIN or MEET undefined.

Existing operations, different results

Optional facilitator extension 1

Gate: Linear subspaces, intersections and spans; advanced/specialist.

How does the diamond arise from vector subspaces?

Solution: In a two-dimensional vector space choose three distinct lines a, b, c through the origin. Add zero subspace 0 and the whole plane 1 . Distinct lines intersect in 0 and span the plane, giving precisely the same five-element sublattice and the same distributivity failure. Join is vector-space sum, not set union.

Mathematics and prior use

Lattices, distributivity, and the distinction between failure of an identity and nonexistence of an operation. The diagram commonly called M3 is called M5 in this source.

Prior use: No exact Week 1-10 match. AD-03 also uses containment, and AD-07 develops another lattice from closure. Do not assume that Week 1 flip diagrams automatically carry lattice structure.

Source: Burris and Sankappanavar, A Course in Universal Algebra

Chapter I §§1 and 3, Definition 3.1 and Figure 5, printed pp. 5-7, 12-13 (PDF pp. 21-23, 28-29).. Use concrete set operations, the five-element diamond, and a separately built six-element nonlattice to distinguish two failures. Evidence: Definitions and distributivity example revisited on 2026-09-25. Source names its five-element diamond M5; M3 is our conventional alternative name.

<https://math.hawaii.edu/~ralph/Classes/619/univ-algebra.pdf>

Everything your starting clues force

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for combining finished collections; promising with specific gates for the order-independence proof. Distinguish “no new token can be added” from merely choosing not to use an available rule.

Two completed collections combine to force a new token, and a least-closed-set argument explains why every completed run has the same outcome.

Prerequisites and preparation

Core: Follow visible positive if-then rules without deleting tokens; reason about forced additions, all possible orders and intersections. Count up to four tokens.

Extensions: General finite proof uses induction on additions. A lattice-of-closed-sets explanation requires containment and least/greatest bounds, developed in AD-06.

Reading and representation: Three short rule cards can be read aloud; use pictures for a,b,c,d if desired.

Materials: Tokens a,b,c,d in two colors; supplied rule cards, start/finish mats, sixteen subset cards for an optional complete catalog. Preparation: about 8 minutes.

0-10 informal recipe play; 10-25 compare legal orders; 25-35 combine completed collections; 35-40 act as prerequisite tokens; 40-55 minimal starts and proof; 55-60 challenge a changed rule.

Launch and facilitation

Choose starting tokens. Add a token when a rule permits it, never remove tokens, and continue until no new token can be added. Can different choices of rule change the finished collection?

Student page 1: Choose a starting collection

Core: positive rules

Rules: a gives b; b together with c gives d; d gives c. Adding does not use up prerequisites. Never remove a token or add a duplicate. Stop only when no new token can be added. Your finished set is the closure of your start.

1 / prompt and solution

Choose a starting set and finish it. Then both partners start from {a,d}, choosing different first additions. Record both runs. Do they finish differently?

Solution: From {a,d}, either add b first then c, or c first then b; both finish at {a,b,c,d}. The b-and-c rule adds nothing new once d is already present. Any chosen start is checked by successive legal additions and by testing all three rules at the end. All possible completed outcomes are unique, justified on page 3.

Everything your starting clues force

1 / hints to offer only after exploration

1. Which two rules can add something immediately?

2 / prompt and solution

Find the smallest starting collections that eventually produce all four tokens. Find every smallest one, and explain why no smaller start works.

Solution: The two smallest starts are $\{a,c\}$ and $\{a,d\}$. Since no rule produces a , every generator contains a . Starting with a alone gives only $\{a,b\}$; a second token must be c or d . Both work, so the minimum size is two and these are all size-two generators. Every other generator is a superset of at least one of them.

2 / hints to offer only after exploration

1. Which token can never be produced by a rule?

Student page 2: Combine finished collections

Core: closed sets and containment

A collection is closed when no new token is forced. Union keeps every token in either collection; intersection keeps only tokens present in both.

3 / prompt and solution

Find two closed collections whose union is not closed. Show the new forced token. Then find all closed collections, using the subset cards if useful.

Solution: For example $\{a,b\}$ and $\{c\}$ are closed, but their union $\{a,b,c\}$ forces d . The full closed list is $\emptyset, \{b\}, \{c\}, \{c,d\}, \{a,b\}, \{b,c,d\}, \{a,b,c,d\}$. To prove completeness, if a is present then b is; if d is present then c is; if b and c are present then d is. Splitting by a present/absent yields exactly the listed seven.

3 / hints to offer only after exploration

1. A two-token prerequisite can be split between two finished collections.

4 / prompt and solution

Try to make the intersection of two closed collections unfinished. If it cannot happen, explain why. What is the smallest closed collection containing both $\{a,b\}$ and $\{c\}$?

Solution: It cannot happen: if all prerequisites of a rule occur in the intersection, they occur in each closed set, so the required output occurs in each and hence in the intersection. The smallest closed set containing $\{a,b\}$ and $\{c\}$ is $\{a,b,c,d\}$, the closure of their union. In general intersection is meet and closure of union is join among closed sets.

Everything your starting clues force

4 / hints to offer only after exploration

1. Follow one rule's prerequisites through both sets.

Student page 3: Why the order does not matter

Core proof: all completed runs

A run may choose any rule that adds a new token. It is complete only when no rule can add anything. The rules remain the same throughout a run.

5 / prompt and solution

Prove that any completed run ends in the same set. Use two facts to organize your explanation: a run cannot keep adding forever, and every forced addition must belong to every closed set containing the start.

Solution: At most four minus the starting size new tokens can be added, so every run that continues whenever possible terminates closed. Let C be any closed set containing the start. Induct on additions: prerequisites already lie in C , so their rule's output lies in C . Thus each terminal set is contained in every closed superset of the start. For two terminal sets T, U , apply this with $C=U$ and $C=T$ to obtain $T \subseteq U$ and $U \subseteq T$; hence equality. This proof works for any finite set of positive addition-only implications.

5 / hints to offer only after exploration

1. Compare a run with an arbitrary finished collection that contains its starting tokens.

6 / prompt and solution

Change the rules: from $\{a\}$, you may replace a by b or replace a by c , removing a . Follow each first choice until no rule is possible. Explain exactly which part of the earlier guarantee fails.

Solution: The two terminal sets are $\{b\}$ and $\{c\}$. Removing a makes the other rule unavailable, so the order/choice changes the result. The previous proof assumes additions preserve all earlier tokens and premises. It does not apply to replacement rules, even though this tiny replacement game still terminates.

6 / hints to offer only after exploration

1. Can the second replacement still happen after the first?

Everything your starting clues force

Optional facilitator extension 1

Gate: The least-closed-superset argument from page 3.

Show that closure is extensive, monotone and idempotent for any finite positive addition system.

Solution: Extensive: no starting tokens are removed. Monotone: if $S \subseteq T$, the closure of T is closed and contains S , so leastness gives $\text{cl}(S) \subseteq \text{cl}(T)$. Idempotent: a closure is already closed, so starting there adds nothing. These statements rely on completing all possible additions.

Mathematics and prior use

Finite implication closure, least fixed points, and a lattice whose join is closure of union. Positive, addition-only rules are essential to the proof.

Prior use: Week 5 asks whether clues force a unique solution; here explicit addition rules generate a closure. AD-03 supplies a reusable subset representation; AD-06 introduces the meet/join language. No exact accepted-ten instance is reused.

Source: Burris and Sankappanavar, A Course in Universal Algebra

Chapter I §5, Definition 5.1 and Theorem 5.2, printed p. 21 (PDF p. 37).. Four-token implication system and minimal generators are solved directly; closure vocabulary connects the experiments. Evidence: Closure axioms and meet/join statement revisited on 2026-09-25. The order-independence argument is given in full for finite positive rules.

<https://math.hawaii.edu/~ralph/Classes/619/univ-algebra.pdf>

Which numbers may share a name?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific gates. Independent representative choices and the pair-translation proof are substantial; parity familiarity alone is not the payoff.

All three valid renamings of addition modulo four, proved by translating an identified pair, with a concrete failure witness for an invalid renaming.

Prerequisites and preparation

Core: Use addition on a four-position ring or supplied table; choose representatives independently; prove a classification by translating identified pairs. Small sums and parity.

Extensions: Six-ring classification requires cyclic translation and subgroup/coset reasoning or a separately checked finite case argument.

Reading and representation: Bag labels and short rules can be read aloud; the supplied table avoids an arithmetic-memory barrier.

Materials: Duplicate number cards 0,1,2,3; bag outlines; four-position ring and addition table; blank quotient table. Preparation: about 8 minutes.

0-10 play ring addition; 10-25 invent and challenge bag assignments; 25-35 repair by merging; 35-40 human representatives; 40-55 classify all possibilities; 55-60 test an added machine.

Launch and facilitation

Put numbers into bags, exactly one bag per number. To add two bags, choose one number from each and add around the ring. The answer bag must not depend on those choices.

Student page 1: Rename the numbers

Core: independent choices of representatives

Addition wraps on the ring: for example $3+2=1$. Each number belongs to one nonempty bag. To add two bags, choose a number from each, allowing the same number twice. A renaming works only if all choices give the same answer bag.

Which numbers may share a name?

1 / prompt and solution

Invent a split into two bags and test it. Try to produce a decisive failure witness: the same input bags but two different answer bags. Test $\{0,1\}$ and $\{2,3\}$.

Solution: The supplied split fails: using its first bag twice, $0+0=0$ stays in that bag, while $1+1=2$ lands in the other. The valid two-bag split is $\{0,2\}\{1,3\}$; other two-bag choices fail by the classification on page 3. For an invented partition, one pair of inconsistent outcomes suffices to disprove it, while successful sampled sums alone are not a proof.

1 / hints to offer only after exploration

1. Hold the input bags fixed while changing their representatives.

2 / prompt and solution

Find a two-bag renaming that survives every choice. Make its bag-addition table and explain why you did not need to inspect infinitely many sums.

Solution: Let $E=\{0,2\}$, $O=\{1,3\}$. $E+E=E$, $E+O=O$, $O+E=O$, $O+O=E$, since ring addition preserves parity. Alternatively check all sixteen table entries, grouping the four choices in each bag pair. There are only four representatives, so the finite table is exhaustive.

2 / hints to offer only after exploration

1. What information is unchanged when you replace 0 with 2?

Student page 2: Repair by merging

Core: translation of an identified pair

You may repair a failed renaming by merging bags, but may not split bags. Use the fewest merges needed to make addition well-defined.

3 / prompt and solution

Start with $\{0,2\}\{1\}\{3\}$. What additional merge is forced? Give a same-input witness before repair and prove the repair works.

Solution: Since 0 and 2 share a bag, adding 1 gives 1 and 3, which must share a bag. Before repair, the first bag plus $\{1\}$ can output $\{1\}$ via $0+1$ or $\{3\}$ via $2+1$. Merge $\{1\}$ and $\{3\}$, yielding the parity partition, which prompt 2 proves valid. One merge is necessary and sufficient.

3 / hints to offer only after exploration

1. Add the same number to both identified numbers.

Which numbers may share a name?

4 / prompt and solution

Start with $\{0,1\}\{2\}\{3\}$. Determine the smallest possible repair. Explain why your argument also applies whenever neighboring ring positions share a bag.

Solution: From $0\sim 1$, adding 1 gives $1\sim 2$, then $2\sim 3$, then $3\sim 0$. Transitivity forces all four into one bag, which always works. Two bag merges are required from this three-bag partition. Any adjacent identified pair similarly propagates around the four-cycle under repeated $+1$, forcing one bag.

4 / hints to offer only after exploration

1. Keep adding 1 to both sides of the same-bag statement.

Student page 3: All possible renamings

Core proof: complete finite classification

Every bag assignment is a partition: bags do not overlap, and all numbers are present. A valid renaming must survive addition of any fixed number to equivalent representatives.

5 / prompt and solution

Classify every valid renaming, allowing one, two, three or four bags. Give a reason that no missing case could work.

Solution: Exactly three partitions work: four singleton bags; parity $\{0,2\}\{1,3\}$; one bag $\{0,1,2,3\}$. If no distinct numbers share a bag, it is the singleton partition. Otherwise an identified pair is adjacent or opposite. Adjacent forces one bag by prompt 4. Opposite forces the other opposite pair by adding 1, so at least parity is merged; any additional merge joins an adjacent pair and forces one bag. Each of the three listed partitions is directly valid.

5 / hints to offer only after exploration

1. Every distinct pair on a four-ring is neighboring or opposite.

6 / prompt and solution

Add a second machine f with outputs $f(0)=0$, $f(1)=0$, $f(2)=1$, $f(3)=1$. A renaming must now make both addition and f independent of representatives. Which of your valid renamings survive?

Solution: Singletons and the single bag survive every operation. Parity fails: 0 and 2 share E , but $f(0)=0$ lies in E while $f(2)=1$ lies in O . Therefore exactly the singleton and indiscrete partitions survive. Validity is always relative to the specified operations.

6 / hints to offer only after exploration

1. Send two representatives of E through the machine.

Every valid four-ring renaming

Worked classification / addition modulo four

0 1 2 3

0 2

1 3

0

1

2

3

+	E	O
E	E	O
O	O	E

$E=\{0,2\}$; $O=\{1,3\}$. An adjacent identification forces all four together. An opposite identification forces parity unless further merged.

Every valid four-ring renaming

Optional facilitator extension 1

Gate: Modular addition, translating equivalences, subgroup/coset reasoning; advanced continuation.

Classify all valid renamings for addition modulo six.

Solution: Exactly four: singletons; $\{0,3\}\{1,4\}\{2,5\}$; $\{0,2,4\}\{1,3,5\}$; and one bag. For a proof let H be the class of 0. Translation compatibility and symmetry make H a subgroup: sums and additive inverses remain equivalent to 0. Every class is a coset of H . Subgroups of $\mathbb{Z}/6$ are $\{0\}$, $\{0,3\}$, $\{0,2,4\}$, and all residues: if H contains 1 or 5 it is all; if it contains 2 or 4 it contains $\{0,2,4\}$; if it also contains 3 it is all; otherwise only $\{0\}$ or $\{0,3\}$ remains. Each listed coset partition is compatible with addition.

Mathematics and prior use

Congruences and quotient operations for the cyclic addition algebra. An extra operation can invalidate a previously valid partition.

Prior use: Week 4 uses cyclic arithmetic. This central question is whether identification makes an operation well-defined, not orbit length. Accepted-ten GA-11 also studies algebraic rules, with very different prerequisites.

Source: Burris and Sankappanavar, A Course in Universal Algebra

Chapter II §5, Definition 5.1 and compatibility discussion, printed pp. 38-39 (PDF pp. 54-55).. Bag representatives realize the congruence condition; four- and six-ring classifications are derived and checked independently. Evidence: Definition and explanation of well-defined quotient operations revisited on 2026-09-25.

<https://math.hawaii.edu/~ralph/Classes/619/univ-algebra.pdf>

Can both schedules be right?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with specific remainder gates. Test whether learners can hold two clock roles fixed; the general theorem is advanced number theory.

Classify every compatible four/six report pair and explain both the parity obstruction and why each compatible pair has one answer per twelve-minute period.

Prerequisites and preparation

Core: Remainders, counting to twenty-four, ordered pairs and a completeness argument. Moving counters is the entry; compatibility and uniqueness within a period are the core.

Extensions: General theorem: gcd, lcm, integer divisibility and a supplied Bézout identity. Three-clock continuation uses repeated congruence solving.

Reading and representation: An adult may read reports. Learners must distinguish the four-clock from the six-clock and a unique residue from a unique integer.

Materials: Printed four- and six-position clock faces, two counters per pair, blank timeline and report grid; optional eight/twelve faces. No custom physical clocks needed. Preparation: about 6 minutes.

5-10 minutes to move both counters; 15 to invent and repair reports; 15 for a complete compatibility certificate. General gcd proof may be a separate 25-minute continuation.

Launch and facilitation

Both counters start at zero and move once each minute. Choose one report from each clock and challenge another pair to decide whether the reports can occur together.

Student page 1: Invent a report

Core: remainders and synchronized movement

Clocks have positions 0,1,2,3 and 0,1,2,3,4,5. Both start at 0 and advance one step at every integer time $t \geq 0$. A report (a,b) names the four-clock first and six-clock second.

1 / prompt and solution

Choose reports that can occur together and reports that cannot. Explain each decision. Then solve the supplied report (3,5) and describe every time when it occurs.

Solution: For example (0,0) occurs at $t=0,12,24,\dots$, while (0,1) cannot occur because the first residue forces even t and the second odd t . The report (3,5) first occurs at 11; all times are $11+12k$ for integers $k \geq 0$. To check an arbitrary choice, use the same-parity rule proved on page 2; do not accept a failed short search as impossibility.

Can both schedules be right?

1 / hints to offer only after exploration

1. List several times matching just one report.
2. Keep each clock in its named role.

2 / prompt and solution

A notice says (2,5). You may change exactly one reported number by one clock step, forward or backward, wrapping at zero. Find every repair, and the first time for each. Is the repaired notice uniquely determined?

Solution: The four possible changes give (1,5),(3,5),(2,4),(2,0), first occurring at 5,11,10,6 respectively. They are all compatible. The original notice is impossible by parity; the repairs are not unique. Each is one allowed edit, so each has minimum positive edit cost.

2 / hints to offer only after exploration

1. Try both directions on each of the two clocks.

Student page 2: Explain all reports

Core proof: a complete period

A first joint return means the first positive time when both counters are back at zero. Organize the possible reports in any way that makes omissions and repetitions visible.

3 / prompt and solution

Find the first joint return. Classify every report that appears before it, without checking each pair by a separate search. Why can no report repeat earlier?

Solution: The first joint return is $\text{lcm}(4,6)=12$. Times 0-11 give (0,0),(1,1),(2,2),(3,3),(0,4),(1,5),(2,0),(3,1),(0,2),(1,3),(2,4),(3,5). Every same-parity pair appears. Any time difference preserving both reports is divisible by 4 and 6, hence by 12, so the twelve reports are distinct. There are 2·3 even/even plus 2·3 odd/odd pairs, twelve in total; this proves completeness. Opposite-parity pairs are impossible.

3 / hints to offer only after exploration

1. Compare the parity of each report.
2. A repeated report would give a smaller joint return by taking the time difference.

Can both schedules be right?

4 / prompt and solution

Replace the clocks by eight and twelve positions. Predict the exact compatibility test, then test reports (5,9) and (2,9). Classify the compatible reports and their repetition period.

Solution: The residues must agree modulo 4. Report (5,9) occurs at $21+24k$; (2,9) fails because 2 and 9 differ modulo 4. There are 4 choices of common residue mod 4, two eight-clock representatives and three twelve-clock representatives, giving 24 compatible reports. Times 0-23 produce distinct reports because the first common multiple is 24; every produced report is compatible, so all compatible ones occur exactly once.

4 / hints to offer only after exploration

1. Which smaller clock can be read consistently from both larger clocks?

Student page 3: Build a general solver

Advanced: gcd, divisibility and Bézout

Let $m, n > 0$ and $d = \gcd(m, n)$. Supplied fact: if $\gcd(M, N) = 1$, there are integers u, v with $uM + vN = 1$. Write the reports as $t \equiv a \pmod{m}$, $t \equiv b \pmod{n}$.

5 / prompt and solution

Prove that a solution exists exactly when d divides $b - a$. When it exists, build one using the supplied fact. Prove uniqueness modulo mn/d , and explain how to choose the first nonnegative time.

Solution: Necessity: both m and n divide appropriate differences, so d divides $t - a$ and $t - b$, hence $b - a$. For sufficiency put $m = dM, n = dN, b - a = dc$. Then $\gcd(M, N) = 1$. Choose $uM + vN = 1$ and set $t_0 = a + mu$. This is a modulo m ; modulo n , $t_0 - b = d(mu - c) = -dNv$, so it is b . If t, t' both work, their difference is divisible by dM and dN . Write $t - t' = dMk$; divisibility by dN gives $N|Mk$, and Bézout gives $N|k$. Thus mn/d divides the difference. Conversely adding mn/d preserves both residues. Reducing t_0 to $0 \leq r < mn/d$ gives the first nonnegative solution r ; all nonnegative solutions are $r + k mn/d$, $k \geq 0$. This also handles $m = 1$ or $n = 1$.

5 / hints to offer only after exploration

1. After dividing by d , solve $Mk \equiv (b - a)/d \pmod{N}$.

6 / prompt and solution

Use the construction for reports 7 modulo 10 and 1 modulo 6. Then create your own compatible pair with a shared factor and show your construction agrees with a direct check.

Solution: Here $d = 2, M = 5, N = 3, c = -3$ and $u = 2$ works because $2 \cdot 5 - 3 \cdot 3 = 1$. Thus $t_0 = 7 + 10 \cdot 2 \cdot (-3) = -53$, which reduces to 7 modulo 30. Check $7 \equiv 7 \pmod{10}$ and $7 \equiv 1 \pmod{6}$. Every nonnegative answer is $7 + 30k$. One created example is $t \equiv 4 \pmod{6}, t \equiv 7 \pmod{9}$, with $d = 3$ and first answer 16 modulo 18; direct substitution confirms it. Arbitrary choices are checked by the proved divisibility condition and substitution.

Can both schedules be right?

6 / hints to offer only after exploration

1. A negative intermediate solution is allowed; reduce it to one period.

Optional facilitator extension 1

Gate: Remainders, coprimality and the two-clock theorem.

Solve the three reports $t \equiv 3 \pmod{4}$, $t \equiv 5 \pmod{6}$, $t \equiv 2 \pmod{5}$. Prove that your full list of nonnegative answers has no omissions.

Solution: The first two combine to $t = 11 + 12k$. Modulo 5 this is $1 + 2k \equiv 2$, hence $k \equiv 3 \pmod{5}$. Thus $t = 47 + 60j$ for $j \geq 0$. Conversely every such time works. Uniqueness modulo 60 follows from divisibility by 4,6,5; repeated two-clock solving gives existence rather than assuming simultaneous consistency.

Mathematics and prior use

Generalized Chinese remainder theorem: compatibility modulo the gcd, existence by reduction to coprime moduli, uniqueness modulo the lcm.

Prior use: Week 4 already investigates modular orbits and Week 9 common multiples. This is a revisit of movement with a new destination: consistency and repairing simultaneous reports. AD-08 has congruences as renamings, not combined schedules.

Source: Keith Conrad, The Chinese Remainder Theorem

Theorem 1.1 and §2, printed pp. 1-2; §3, pp. 2-3. The coprime proof supplies the modular-inverse/Bézout mechanism. The noncoprime schedule and reduction are derived here. Evidence: Read the existence and uniqueness proofs and the pairwise-coprime warning on 2026-09-25; the general gcd statement is proved explicitly below.

<https://kconrad.math.uconn.edu/blurbs/ugradnumthy/crt.pdf>

Can the next match hide between our answers?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist for the satisfying proof; the counter entry is a promising earlier activity but must be labeled as such.

Produce the first triangular-square total above one million, and justify that the recurrence skipped no positive solution using a decreasing inverse with a base case.

Prerequisites and preparation

Core: Figurate counting is the entry. The satisfying new core uses polynomial expansion, a two-variable recurrence, integer parity and positive-integer descent. Keep row count separate from square side.

Extensions: General Pell theory is not assumed. A recurrence for the matching totals uses algebra and second differences; optional exact calculator arithmetic is permitted.

Reading and representation: Equation reading and inequalities are required for the completeness proof. A learner stopping with counters has an earlier concrete result, not a proof of the infinite classification.

Materials: At most 36 counters, triangular-row and square mats, graph paper and a calculator for exact integer products. Do not prepare thousands of counters. Preparation: about 6 minutes.

10 minutes of arrangements; 15 for re-encoding and checking a proposed recipe; 20 for descent. The beyond-one-million challenge and proof can be a second meeting.

Launch and facilitation

Choose counts to test as both complete triangular rows and a square. Then challenge a recipe that claims to find the next match without checking every intervening number.

Student page 1: Choose counts to compare

Entry: rows and square arrays; algebra follows

A triangle has rows of lengths $1, 2, \dots, m$. A square has n rows of n counters. Both m and n are positive integers; matching totals satisfy $m(m+1)/2 = n^2$.

1 / prompt and solution

Find a match larger than one using at most 36 counters. Explain the triangle count with a picture or pairing argument. Why is adding one to both side labels not a reliable way to find the next match?

Solution: The larger match is $m=8, n=6$, with 36 counters. Two copies of rows 1 through m pair to m rows of $m+1$, giving $m(m+1)/2$. Completeness within 36 follows from triangular totals 1,3,6,10,15,21,28,36 and squares 1,4,9,16,25,36. Increasing both labels from 8,6 gives 45 versus 49, so the same update does not preserve equality.

Can the next match hide between our answers?

1 / hints to offer only after exploration

1. Choose which of the two lists is shorter to search.

2 / prompt and solution

Set $x=2m+1$ and $y=2n$. Find an equation relating x and y . Prove both directions: a triangle-square match gives such a pair, and every positive integer pair satisfying your equation gives a match.

Solution: Expanding yields $x^2-2y^2=4m(m+1)+1-8n^2=1$. Conversely $x^2-2y^2=1$ forces x odd since x^2 is odd, then modulo 4 forces y even. Thus $m=(x-1)/2, n=y/2$ are integers. Since $y>0$ it is at least 2 and $x\geq 3$, hence $m, n\geq 1$. Reversing the expansion gives the matching-total equation.

2 / hints to offer only after exploration

1. Multiply the matching-total equation by 8, then add 1.

Student page 2: Test a machine that makes matches

Algebra core: identities and growth

Proposed forward machine: $U(x,y)=(3x+4y, 2x+3y)$. Work only with positive integer solutions of $x^2-2y^2=1$. A successful forward rule alone does not show that no solutions were skipped.

3 / prompt and solution

Prove that U preserves the equation and produces a strictly larger positive pair. Start with $(3,2)$, use the machine to reach a triangular-square total above one million, and check that total exactly.

Solution: $(3x+4y)^2-2(2x+3y)^2=x^2-2y^2$ by expansion. Both new coordinates are positive integers and exceed the old coordinates. Starting from $(3,2)$ gives $(17,12), (99,70), (577,408), (3363,2378)$. These correspond to $(m,n)=(1,1), (8,6), (49,35), (288,204), (1681,1189)$, and totals $1, 36, 1225, 41616, 1413721$. Exactly $1681\cdot 1682/2=1413721=1189^2$. At this stage the construction proves a match above the threshold, not yet that it is the first.

3 / hints to offer only after exploration

1. Convert back only after generating each new x,y pair.

Can the next match hide between our answers?

4 / prompt and solution

Solve $X=3x+4y$ and $Y=2x+3y$ for x,y . Use that reverse machine on two generated examples. What must still be proved before it can certify that every possible match comes from the starting pair?

Solution: The inverse is $D(X,Y)=(3X-4Y,3Y-2X)$, since both matrix products are the identity. It sends $(99,70)$ to $(17,12)$ and $(17,12)$ to $(3,2)$. It also preserves x^2-2y^2 . Completeness still requires proving that every positive solution except the base has a positive integer predecessor of smaller size, and that descent cannot terminate elsewhere.

4 / hints to offer only after exploration

1. Eliminate one variable; the determinant is 1.

Student page 3: Make “next” a theorem

Advanced proof: inequalities and integer descent

Use $D(x,y)=(3x-4y,3y-2x)$. Positive solutions have x odd and y even. Examine $(3,2)$ separately before treating $y>2$.

5 / prompt and solution

Prove that every positive solution other than $(3,2)$ has a positive predecessor with smaller y . Explain why repeated descent reaches $(3,2)$, and why reversing the steps proves that U lists every positive solution exactly once.

Solution: If $y=2$, the equation gives $x=3$. For $y>2$, $x^2=2y^2+1$ gives $4y/3<x<3y/2$: the lower inequality follows from $2>16/9$, and the upper from $1<y^2/4$. Hence $3x-4y>0$ and $3y-2x>0$. Also $x>y$, so $3y-2x<y$. D has integer coefficients and preserves the equation, so it gives a smaller positive solution. Positive integer y cannot decrease forever; the only point where the $y>2$ step ceases is $y=2, x=3$. The base maps to $(1,0)$, outside the positive- y domain. Since D and U are inverse, every original pair is a finite iterate of U on $(3,2)$. U strictly increases y , so no iterate repeats and their order is increasing in square side and total.

5 / hints to offer only after exploration

1. To prove $3y-2x>0$, compare x^2 with $(3y/2)^2$.
2. A decreasing positive integer cannot continue indefinitely.

Can the next match hide between our answers?

6 / prompt and solution

Now decide whether your displayed total above one million is the first. Give the triangle row count and square side of the first match after it, and separate the roles of computation and proof.

Solution: The complete increasing list through the threshold is 1,36,1225,41616,1413721, so 1413721 is the first above one million. Applying U to (3363,2378) gives (19601,13860), hence $m=9800, n=6930$, total 48024900; check $9800 \cdot 9801/2 = 6930^2$. Arithmetic checks particular matches; the descent proof rules out unseen matches between them.

6 / hints to offer only after exploration

1. Use the completeness result before using the word “first.”

Optional facilitator extension 1

Gate: Quadratic irrational arithmetic, conjugates and linear recurrences.

Show that the matching totals N_k , starting $N_0=0, N_1=1$, satisfy $N_{k+2}=34N_{k+1}-N_k+2$. Does this recurrence alone prove completeness?

Solution: Let $x_k + y_k\sqrt{2} = (3+2\sqrt{2})^k$, so $N_k = y_k^2/4$ and $x_k^2 - 2y_k^2 = 1$. Put $r = (3+2\sqrt{2})^2 = 17+12\sqrt{2}$, $r^{-1} = 17-12\sqrt{2}$; $r+r^{-1}=34$. The expression $N_k = (r^k + r^{-k} - 2)/32$ follows by squaring $y_k = ((3+2\sqrt{2})^k - (3-2\sqrt{2})^k)/(2\sqrt{2})$. Each r -power sequence obeys the homogeneous recurrence with coefficient 34; the constant $-2/32$ yields the extra 2. This formula generates the same sequence but does not independently exclude other matches; completeness still uses descent.

Mathematics and prior use

Pell equation $x^2 - 2y^2 = 1$, a norm-preserving invertible integer transformation and a positive-integer descent proving completeness.

Prior use: Year-1 figurate-number material and the original AD-10 card already use 36 and the forward recurrence. Those are openly revisited as entry. The new destination is descent/completeness and a certified large threshold, not another unproved next-term calculation.

Source: Keith Conrad, Pell's Equation, I

§3, Theorem 3.1, printed pp. 3-4; §4, multiplication of solutions. Use the triangular-square re-encoding and quadratic multiplication. The specific decreasing inverse and threshold proof are written independently here. Evidence: Read the correspondence, forced parity and product mechanism on 2026-09-25. The source is not cited as a substitute for the descent proof.

<https://kconrad.math.uconn.edu/blurbs/ugradnumthy/pelleqn1.pdf>

Choose a rule that never loses a symbol

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with explicit algebra gates; pair addition is accessible, while distributivity and solving a general two-report problem need preparation.

Derive every nonzero inverse for the successful rule and use it to prove that two distinct input-output reports determine a unique affine machine.

Prerequisites and preparation

Core: Two-bit addition, distributive multiplication and inverse reasoning. The field claim uses the supplied polynomial-quotient ring fact plus a complete inverse classification.

Extensions: Linear decoding over a finite field; Frobenius extension uses polynomial substitution. No claim of the classification of all finite fields is needed.

Reading and representation: Four symbols $0, 1, t, u$ with $u = 1 + t$. Learners must keep coefficient bits separate from ordinary integers modulo 4.

Materials: Four labeled two-bit cards, coefficient mat and blank multiplication tables for two rule choices; a separate modulo-four card. Paper and pencil suffice. Preparation: about 8 minutes.

10 minutes of bit addition; 15 choosing and testing a t -square rule; 15 explaining invertibility; 15-20 for a two-report decoder.

Launch and facilitation

Keep the four coefficient cards fixed. Choose whether t^2 becomes t or $t+1$. Can every nonzero multiplier be undone under your choice?

Student page 1: Keep the cards, change multiplication

Core: bits and distribution

Cards are $0=(0,0), 1=(1,0), t=(0,1), u=(1,1)=1+t$. Add coefficient bits modulo 2, so equal terms cancel. Multiply by distributing, then use ONE chosen rule: A has $t^2=t$; B has $t^2=t+1$. These are formal symbols, not residues $0, 1, 2, 3$ modulo 4.

1 / prompt and solution

Choose A or B and derive enough products to decide whether every nonzero card has a partner whose product is 1. Trade your evidence with a pair using the other rule.

Solution: For A: $t^2=t, tu=t+t^2=0, u^2=1+t=t+1=u$. Thus t and u are nonzero zero divisors and have no inverses; only 1 is invertible. For B: $t^2=u, tu=t+(t+1)=1, u^2=1+t^2=t$. Thus $1^{-1}=1, t^{-1}=u, u^{-1}=t$. The identity and zero give the remaining products in either table. This covers every card, not just a successful example.

Choose a rule that never loses a symbol

1 / hints to offer only after exploration

1. Expand $t(1+t)$ before reducing.
2. A nonzero card multiplying another nonzero card to 0 cannot have an inverse.

2 / prompt and solution

For each rule, try to find two different inputs that multiplication by t sends to the same output. Explain either your collision or why no collision can exist.

Solution: For A, $t \cdot 0 = 0 = t \cdot u$, and $t \cdot 1 = t = t \cdot t$. For B no collision exists: if $ta = tb$, multiply both sides by u , using $ut = 1$, to obtain $a = b$. Explicitly multiplication by t sends $0, 1, t, u$ to $0, t, u, 1$. An inverse is a universal no-collision certificate.

2 / hints to offer only after exploration

1. The output list is useful evidence; an inverse explains all inputs at once.

Student page 2: Certify the arithmetic

Algebra proof: quotient rings and inverses

Supplied fact: polynomials over the two-element field, reduced modulo a monic quadratic, form a commutative ring with identity; each class has a unique remainder $a+bt$. Associativity and distributivity come from polynomial arithmetic. A field is such a ring in which every nonzero element is invertible.

3 / prompt and solution

Identify the quadratic modulus for each rule. Explain why the facts supplied here together with your inverse work certify B as a field and reject A. Test the two possible roots of each modulus as another explanation.

Solution: A is the quotient by $T^2+T=T(T+1)$; B is by T^2+T+1 . In B the complete inverse list from prompt 1, together with the supplied ring fact, satisfies the field definition. In A, t, u are distinct nonzero classes by unique remainders but $tu=0$, so it is not a field. The A polynomial has roots $0, 1$; the B polynomial evaluates to 1 at both $0, 1$. A quadratic over a field factors nontrivially exactly when it has a root, so B is irreducible, consistent with the general quotient theorem. We do not infer associativity from a few table entries.

3 / hints to offer only after exploration

1. In characteristic 2, subtracting a term is the same as adding it.

Choose a rule that never loses a symbol

4 / prompt and solution

Could a clever renaming make B into ordinary arithmetic modulo 4 while preserving both addition and multiplication? Give a property that any such renaming would have to preserve and show the mismatch.

Solution: No. In B every nonzero element has an inverse and the product of nonzero elements cannot be 0. Modulo 4 the nonzero residue 2 satisfies $2 \cdot 2 = 0$ and has no inverse. A bijection preserving both operations must preserve 0 and inverses (the unique additive and multiplicative identities are determined by the operations), so this mismatch is impossible. Also every B element added to itself is 0, while $1+1=2 \neq 0$ modulo 4.

4 / hints to offer only after exploration

1. A renaming must preserve equations, not merely the number of labels.

Student page 3: Decode a hidden machine

Continuation: affine equations over B

Use rule B from now on. A hidden machine has form $f(z)=az+b$, with a,b among $0,1,t,u$. Choose two distinct input cards and ask for their output cards. Arithmetic is the four-symbol arithmetic throughout.

5 / prompt and solution

Make an affine machine and give a partner two distinct input-output reports. Recover your partner's a,b . Prove that every pair of output reports at distinct inputs determines exactly one affine machine, including equal outputs.

Solution: For inputs $r \neq s$ and outputs h,k , subtracting is addition, so $a(r+s)=h+k$. The nonzero card $r+s$ has an inverse; hence $a=(h+k)(r+s)^{-1}$ and $b=h+ar$. These satisfy both equations and are forced, proving existence and uniqueness. If $h=k$ then $a=0$ and $b=h$, a valid constant machine. For example $a=t, b=u$ gives $f(0)=u, f(1)=1$; those reports recover $a=t, b=u$.

5 / hints to offer only after exploration

1. Add the two equations so the b terms cancel.

6 / prompt and solution

Decode the reports $f(1)=t$ and $f(t)=0$, then predict $f(u)$. Repeat these same reports under rule A: can they determine a unique machine? Use an explicit calculation.

Solution: In B, $(1+t)a=u$ $a=t$, so $u^{-1}=t$ gives $a=t^2=u$; then $b=t+a=t+u=1$. Therefore $f(u)=u^2+1=t+1=u$. In A, the equations give $b=ta$ and $a+ta=t$, i.e. $u a=t$. But multiplication by u has outputs only $0,u$, never t : $u \cdot 0=0, u \cdot 1=u, u \cdot t=0, u \cdot u=u$. Thus no A machine fits these reports. As a separate collision example, the A machines 0 and tz agree at inputs $0,u$, showing lack of uniqueness can also occur for other reports.

Choose a rule that never loses a symbol

6 / hints to offer only after exploration

1. Check the image of multiplication by $1+t$ under each rule.

Optional facilitator extension 1

Gate: Distributivity, characteristic 2 and function composition.

For rule B, classify the map $F(z)=z^2$ and prove that it preserves addition and multiplication. Does it equal the identity?

Solution: It fixes 0,1 and exchanges t,u because $t^2=u, u^2=t$. Thus F^2 is the identity but F is not. Commutativity and characteristic 2 give $(a+b)^2=a^2+b^2$, and $(ab)^2=a^2b^2$. It preserves 1 and is bijective, so it is an automorphism. This proof checks the ring rules as well as the finite permutation.

Mathematics and prior use

$GF(4)$, irreducible versus reducible quadratic quotients, invertible multiplication and two-point interpolation.

Prior use: Week 2 binary representations and Week 4 modular arithmetic are preparations. AD-08 studied quotient compatibility; this family uses a polynomial quotient and contrasts two different multiplication rules on the same four cards. The original AD-11 arithmetic is revisited, then applied to reversible decoding.

Source: Keith Conrad, Finite Fields

§1, Theorem 1.1 and warning, printed p.1; Example 2.6, p.4. Use polynomial remainders as four-element arithmetic. Both quadratic choices and every decoding instance are checked directly. Evidence: Read the quotient construction and warning that $\mathbb{Z}/(p^n)$ is not the field on 2026-09-25; no general finite-field classification is inferred from four labels.

<https://kconrad.math.uconn.edu/blurbs/galoistheory/finitefields.pdf>

Which cube volumes can these tools reach?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist for the intended payoff. Exact square-root geometry is an accessible earlier stop and must not be advertised as the impossibility proof.

Prove exactly which positive integer cube volumes have classically constructible edges, using a stated degree theorem and a complete cubic irreducibility argument.

Prerequisites and preparation

Core: Exact compass/straightedge work is an entry. The actual obstruction requires rational polynomials, irreducibility of cubics, degree of an algebraic number and the supplied constructible-degree theorem.

Extensions: Classify positive integer volumes; a separate angle-trisection extension uses a cubic cosine identity and irreducibility. Calculus is not required.

Reading and representation: Read polynomial equations and distinguish exact construction from decimal approximation. The unit length and unmarked-tool restrictions must be kept throughout.

Materials: Compass, unmarked straightedge, printed unit segment, blank construction area and supplied line/circle equation cards; calculator for cubing rational bounds. Preparation: about 8 minutes.

10-15 minutes of exact square-root construction; 15 of coordinate-intersection reasoning; 20 for the cubic obstruction and volume classification. Earlier geometry and full field-degree proof are different stopping points.

Launch and facilitation

An exact segment can certify a square area, yet the same drawing may not certify a cube volume. Choose a construction and identify the equation its length really solves.

Student page 1: Certify the equation of a length

Geometric entry: exact unit constructions

Start from a unit segment. Legal tools are an unmarked straightedge and compass, used for finitely many line/circle intersections. A physical measurement may suggest a length, but its construction needs an exact argument.

Which cube volumes can these tools reach?

1 / prompt and solution

Choose and carry out an exact construction of $\sqrt{2}$ and then $\sqrt{3}$. Explain each length. If your $\sqrt{3}$ segment were the edge of a cube, what would its volume be? Does it solve the cube-volume 3 problem?

Solution: Construct a unit square; its diagonal has square 2, hence length $\sqrt{2}$. Construct a right triangle with legs $\sqrt{2}$ and 1; its hypotenuse has square 3 and positive length $\sqrt{3}$. Perpendiculars and copying lengths are legal classical constructions. A cube with edge $\sqrt{3}$ has volume $(\sqrt{3})^3=3\sqrt{3}$, not 3. The edge for volume 3 must solve $x^3=3$.

1 / hints to offer only after exploration

1. Use Pythagoras to certify the construction, not a ruler reading.

2 / prompt and solution

A partner proposes the rational edge $1442/1000$ for volume 3. Check its cube exactly. Find a nearby rational edge whose cube lies on the other side of 3. Explain what the two checks establish and what they do not.

Solution: $1442^3/1000^3=2998442888/1000000000<3$. The nearby $1443/1000$ gives $3004685307/1000000000>3$. Thus the positive cube root lies strictly between 1.442 and 1.443 , since cubing is strictly increasing on positive numbers. Both endpoints are rational and constructible, but neither is the exact root. Arbitrarily fine rational brackets do not by themselves provide a finite exact construction.

2 / hints to offer only after exploration

1. Compare integer numerators with $3 \cdot 1000^3$.

Student page 2: What can one legal step add?

Advanced: algebraic degree and supplied theorem

Supplied theorem: coordinates made from a unit segment by finitely many legal intersections lie in a tower of extensions of degree 1 or 2 over $\mathbb{Q}$. Hence any constructible real algebraic number has degree 2^r over $\mathbb{Q}$ for some integer $r \geq 0$. Degree means the degree of its irreducible minimal polynomial. This is a necessary condition, not a general sufficiency test.

Which cube volumes can these tools reach?

3 / prompt and solution

Explain the quadratic step behind the theorem. If known coordinates lie in a field K , why do line-line, line-circle and two distinct circle intersections introduce at most a square root? Include tangency and intersections that do not exist.

Solution: Lines have linear equations over K , so a unique line-line intersection uses only field arithmetic; parallel distinct lines give none. A circle has $x^2+y^2+dx+ey+f=0$ over K because its center and squared radius are in K . Substituting a nonvertical line gives a quadratic; a vertical line fixes x first. Two distinct circle equations subtract to a nonzero linear equation, unless concentric with different radii, when subtraction is a nonzero constant and there are no intersections. Thus remaining intersections reduce to the line-circle case. Quadratic roots use at most one square root; a repeated tangent root lies in K , and a negative real discriminant gives no real new point. Coincident lines/circles do not specify an isolated new intersection. The full tower-law conclusion remains the supplied theorem.

3 / hints to offer only after exploration

1. Subtract two circle equations before solving them.

4 / prompt and solution

Use the theorem to decide whether the exact positive edge α with $\alpha^3=3$ is constructible. Prove the needed irreducibility rather than merely calling α irrational.

Solution: A rational root of the monic integer polynomial X^3-3 must be an integer dividing 3, so candidates are $\pm 1, \pm 3$; none is a root. A reducible cubic over $\mathbb{Q}$ has a linear factor and hence a rational root, so X^3-3 is irreducible. Thus $[\mathbb{Q}(\alpha):\mathbb{Q}]=3$, which is not 2^r for any $r \geq 0$. The necessary theorem excludes a finite exact construction with these tools. Irrationality alone would not exclude $\sqrt{2}$, so it is not the obstruction.

4 / hints to offer only after exploration

1. For a cubic, any nontrivial factorization has a degree-one factor.

Student page 3: Classify all integer requests

Proof continuation: positive integer cube volumes

A request is a positive integer volume N , measured relative to the unit cube. Its positive edge is the real solution of $x^3=N$. Do not include negative volumes; if you discuss $N=0$, label it separately.

Which cube volumes can these tools reach?

5 / prompt and solution

Classify all positive integers N for which the exact edge is constructible. Prove both directions. Test your classification on $N=8,9,27$ and 64 .

Solution: Exactly the perfect cubes $N=k^3$ with positive integer k work; edge k is constructible by concatenating unit segments. If N is not a perfect cube, a rational root of X^3-N would be an integer k (rational-root theorem), forcing $N=k^3$. Therefore no rational root exists; the cubic is irreducible and its positive root has degree 3, excluding construction. Thus $8,27,64$ work with edges $2,3,4$; 9 does not. If $N=0$ is separately allowed, its edge 0 is rational/constructible, a degenerate cube, and it is not part of the positive-volume statement.

5 / hints to offer only after exploration

1. Prove the converse by asking what a rational root of a monic cubic could be.

6 / prompt and solution

A learner says, "Every irrational root is forbidden," and another says, "The supplied theorem alone proves that every algebraic number of power-of-two degree is allowed." Explain the flaw in each claim. Give an explicit constructible root of X^4-2 and certify it by legal square-root steps.

Solution: The first claim fails at $\sqrt{2}$ or $\sqrt{3}$. For X^4-2 the positive root is $\sqrt[4]{2}$; construct $\sqrt{2}$ first, then its square root using the supplied semicircle/geometric-mean construction: take diameter segments 1 and $\sqrt{2}$, erect a perpendicular at their junction, and its height to the semicircle is $\sqrt[4]{2}$. The right-triangle altitude theorem gives $\text{height}^2=1\cdot\sqrt{2}$. The second claim reverses a merely necessary condition: a degree that divides a power of 2 does not alone exhibit a tower of quadratic extensions containing the number. The explicit tower certifies this particular fourth root; the worksheet does not claim every degree-four number is constructible.

6 / hints to offer only after exploration

1. The theorem rules some things out; a construction or an actual quadratic tower is still needed to rule one in.

Which cube volumes can these tools reach?

Optional facilitator extension 1

Gate: Triple-angle identity, rational-root theorem and the constructible-degree theorem.

Assume the identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$. Show that an exact 20° angle cannot be constructed from the unit segment, so there cannot be a classical method trisecting every constructible angle.

Solution: If 20° were constructible, its cosine c would be a constructible coordinate on the unit circle. Since $\cos 60^\circ = 1/2$, $4c^3 - 3c = 1/2$. Put $u = 2c$; then $u^3 - 3u - 1 = 0$. The only rational-root candidates ± 1 give -3 and 1 , so this cubic is irreducible and u has degree 3. Constructibility of c would give constructibility of $2c = u$, contradiction. A 60° angle is constructible via an equilateral triangle. This excludes a universal trisection procedure, not trisection of every individual angle.

Mathematics and prior use

Quadratic coordinate extensions, a necessary constructible-degree condition and irreducible-cubic obstructions.

Prior use: The original AD-12 card treats cube volume 2. We revisit its classical tool restriction but use volume 3, a full positive-integer-volume classification, and a rational approximation contrast. Week 1 shape manipulation does not establish a field-degree obstruction.

Source: Thomas Judson, Abstract Algebra: Theory and Applications

§21.3.1, Lemmas 21.3.3, 21.3.5, 21.3.6 and Theorem 21.3.7; §21.3.2. Use the square-root construction and quadratic-tower obstruction; derive volume 3 and the positive-integer classification. Evidence: Read the intersection argument and theorem on 2026-09-25. State degree 2^r with $r \geq 0$, including rational degree 1, correcting the page's $k > 0$ wording. The power-of-two condition is used only as necessary.

<https://judsonbooks.org/aata-files/aata-html/fields-section-constructions.html>

Move a corner, change the whole staircase

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for grid-ready learners at the finite-boundary stage; promising with linear-algebra gates for quotient dimension.

Find the unique legal one-step change that adds exactly two survivors, prove there are no other such changes, and, at the algebra gate, prove the surviving monomials form a quotient basis.

Prerequisites and preparation

Core: Nonnegative grid coordinates, right/up reachability and a finite certificate for an infinite region. The quotient-basis stage needs polynomial linear combinations and independence.

Extensions: Multiplication in a polynomial quotient and kernels of linear maps. A two-variable rectangle extension uses exponent counting and a proof of its basis.

Reading and representation: Picture rules permit an oral launch. Upper pages require reading $x^i y^j$ and distinguishing an exponent position from its coefficient.

Materials: Printed coordinate grids from 0 to 6, four movable corner markers and pencil shading; an unbounded-axis arrow is essential. Optional transparent overlay, no exact-count survivor tokens.
Preparation: about 7 minutes.

10 minutes exploring the forbidden rule; 15 finding and certifying the survivors; 15 changing one corner; 20 for the algebraic normal form on a later page.

Launch and facilitation

Each forbidden corner spoils every place northeast of it. Choose an organization that finds all survivors, including any beyond the edge of the paper. Then move one corner to save exactly two more places.

Student page 1: Find the places that survive

Core: grid directions and a complete boundary

Coordinates (i,j) are nonnegative integers. A corner (a,b) forbids itself and every (i,j) with $i \geq a$ and $j \geq b$. Corners are $(3,0),(2,1),(1,3),(0,4)$. The grid continues forever to the right and upward.

1 / prompt and solution

Choose a way to find every surviving position. Show your result on the grid and explain why enlarging the paper cannot reveal more survivors.

Solution: Column $i=0$ has $j=0,1,2,3$; column $i=1$ has $j=0,1,2$; column $i=2$ has only $j=0$. Every $i \geq 3$ is forbidden by $(3,0)$, and every $j \geq 4$ by $(0,4)$. Thus there are eight survivors: $(0,0),(0,1),(0,2),(0,3),(1,0),(1,1),(1,2),(2,0)$. The two outer corners bound the whole infinite search; the two inner corners determine the staircase inside.

Move a corner, change the whole staircase

1 / hints to offer only after exploration

1. Classify a whole column at once.
2. Find corners that exclude every sufficiently large coordinate.

2 / prompt and solution

Is any of the four starting corners redundant, meaning that removing it leaves the forbidden region unchanged? Give a witness for each decision rather than judging the picture.

Solution: None is redundant. Its own position is not northeast of any other corner, so removing that corner frees that position. Specifically $(3,0), (2,1), (1,3), (0,4)$ are pairwise incomparable in coordinatewise order. Each is therefore an independent necessary generator of this forbidden region.

2 / hints to offer only after exploration

1. Can another corner already forbid the corner you plan to remove?

Student page 2: Save exactly two places

Core continuation: classify legal changes

Move exactly one corner one unit right or one unit up. All other corners stay fixed. The goal is to keep only finitely many survivors and increase their number by exactly two. A moved corner may become redundant.

3 / prompt and solution

Find every move that meets the goal. Explain what happens for all eight possible moves, including moves that create infinitely many survivors.

Solution: Moving $(2,1)$ right to $(3,1)$ is the unique success: it frees $(2,1), (2,2)$, giving ten survivors; the moved corner is now redundant because of $(3,0)$. Moving $(3,0)$ right adds only $(3,0)$; moving it up leaves the whole row $j=0$ unbounded and creates infinitely many survivors. Moving $(2,1)$ up adds only $(2,1)$. Moving $(1,3)$ right or up adds only $(1,3)$. Moving $(0,4)$ up adds only $(0,4)$; moving it right leaves the whole column $i=0$ unbounded. These eight cases exhaust the legal choices. Moving corners northeast can only free positions, never forbid a previously surviving one.

3 / hints to offer only after exploration

1. The corner on each coordinate axis is the only one that can stop travel forever along that axis.

Move a corner, change the whole staircase

4 / prompt and solution

Return to the original corners. A right step or an up step sends a survivor to its neighboring position, or to zero if that neighbor is forbidden. Find every surviving position for which BOTH kinds of step go to zero. Explain why your list is complete.

Solution: They are $(0,3), (1,2), (2,0)$. A position killed by an up step is the top survivor of one of the three nonempty columns. Those same three positions have forbidden right neighbors. Every other survivor has an allowed up neighbor, so no other position can qualify. This argument classifies the full infinite grid via its finite staircase.

4 / hints to offer only after exploration

1. First locate the top surviving point in each nonempty column.

Student page 3: What the staircase proves in algebra

Advanced: polynomial quotients and a basis argument

Work over $\mathbb{Q}$. Position (i,j) represents $x^i y^j$. Declare x^3, x^2y, xy^3, y^4 and all their polynomial multiples equal to zero. Thus we work in $\mathbb{Q}[x,y]/I$, where $I = (x^3, x^2y, xy^3, y^4)$. Do not erase terms merely because their total degree is large.

5 / prompt and solution

Prove that every polynomial has exactly one remainder using the surviving monomials. Give both spanning and independence arguments. Reduce $(x+y)^3$, and give an example showing that a simple total-degree cutoff is the wrong rule.

Solution: The survivors are $1, y, y^2, y^3, x, xy, xy^2, x^2$. A monomial is in I exactly when divisible by one displayed generator: the forward direction is clear; conversely every term of a sum of generator multiples is of that divisible type, and coefficient cancellation cannot create a nondivisible monomial. Removing those terms spans the quotient with the eight survivors. No nonzero linear combination of survivors lies in I , since all terms of every element of I have divisible exponents; hence the remainder is unique and the classes are a basis of dimension 8. Expansion gives $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 \equiv 3xy^2 + y^3$. The degree-three monomial x^3 vanishes while y^3 and xy^2 survive, so degree alone cannot determine removal.

5 / hints to offer only after exploration

1. A polynomial is a unique finite sum of monomials before taking the quotient.

Move a corner, change the whole staircase

6 / prompt and solution

Classify every quotient element killed by multiplication by BOTH x and y . Could a linear combination of other survivors create a cancellation that produces an extra answer?

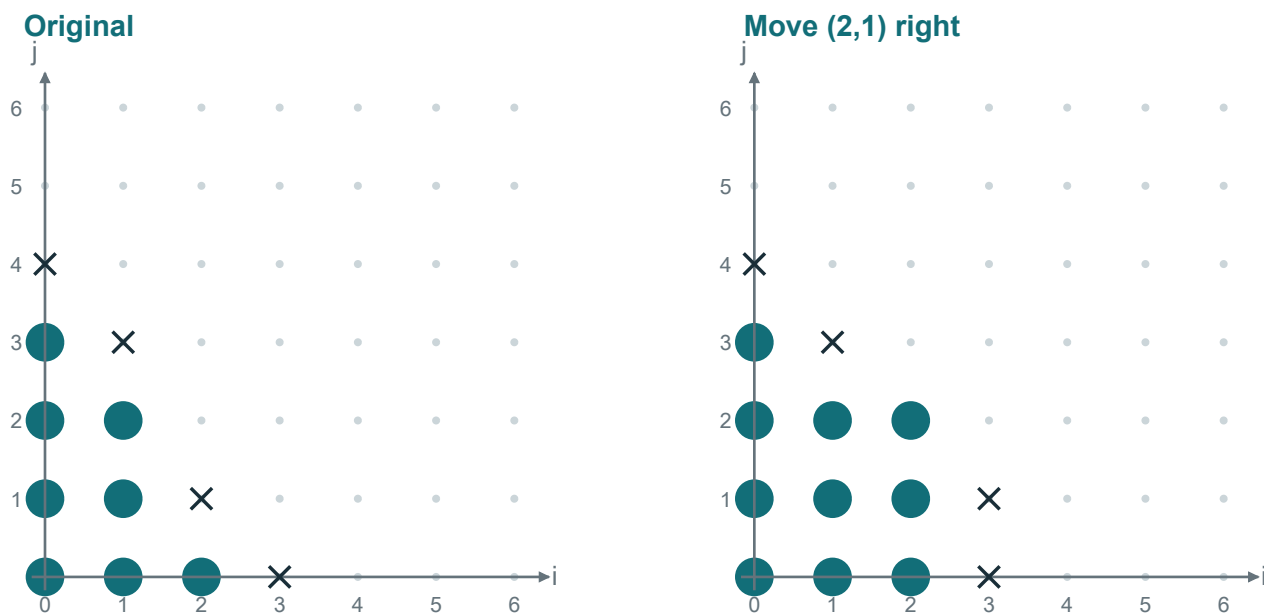
Solution: Exactly a $y^3 + bxy^2 + cx^2$ with $a, b, c \in \mathbb{Q}$ are killed by both. Multiplication by y kills y^3, xy^2, x^2 , while the other surviving basis monomials map to distinct nonzero survivors; therefore its kernel is precisely their span, with no hidden cancellation. Multiplication by x also kills each of those three, so their span is the common kernel. The kernel of x alone is larger, also containing xy . Distinct surviving shifted monomials cannot cancel unless their original exponents were identical.

6 / hints to offer only after exploration

1. Use the unique remainder from prompt 5 to compare coefficients after shifting.

Worked boundary certificates

A filled dot is a survivor. A cross is a forbidden corner. The axes continue beyond these windows.



The move frees (2,1) and (2,2). At $i \geq 3$ the corner (3,0) still forbids everything, and at $j \geq 4$ the corner (0,4) still forbids everything. Thus the new region is finite.

In the original grid, right and up both kill exactly (0,3), (1,2) and (2,0). Their monomials are y^3 , xy^2 and x^2 . Among monomials whose shifted images survive, those images are distinct. Thus cancellation cannot add other joint-kernel elements.

Move a corner, change the whole staircase

Optional facilitator extension 1

Gate: Polynomial basis and coordinatewise exponent shifts.

For positive integers r, s , use only corners $(r, 0)$ and $(0, s)$. Prove the quotient $\mathbb{Q}[x, y]/(x^r, y^s)$ has dimension rs , and classify elements killed by both x and y .

Solution: The survivors are $x^i y^j$ with $0 \leq i < r$ and $0 \leq j < s$. Divisibility by x^r or y^s gives the spanning/independence proof, so dimension is rs . Multiplication by x kills exactly the last-column span and multiplication by y exactly the top-row span. Their common kernel is the one-dimensional span of $x^{r-1}y^{s-1}$. This includes $r=1$ or $s=1$, where one of the two multiplication maps is zero.

Mathematics and prior use

Monomial ideals, standard-monomial normal forms, basis spanning/independence and multiplication kernels.

Prior use: Week 1 boards represent tilings, while these coordinates encode divisibility. The original AD-13 four-survivor example is replaced by a larger staircase and a constrained redesign problem. AD-14 will study one special nilpotent quotient, a related but different question.

Source: J. S. Milne, Algebraic Geometry

Current PDF: Chapter 1, polynomial-ring discussion, printed p.25 (PDF page 25), and ring/ideal framework. Use monomial linear independence and quotient arithmetic. The exact staircase, move classification and basis proof are derived here. Evidence: Read the current polynomial-basis passage and ideal framework on 2026-09-25. The current PDF has different pagination from the original atlas locator; no Gröbner-basis theorem is needed.

<https://www.jmilne.org/math/CourseNotes/AG.pdf>

Can one calculation remember a rate?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with explicit polynomial-algebra gates; advanced/specialist when interpreting derivatives. Do not tell learners ϵ is an ordinary very small nonzero real number.

Use one exact dual-number evaluation to compute a polynomial's value and derivative, then prove the general identity and product rule while identifying information it does not retain.

Prerequisites and preparation

Core: Signed rational algebra, distribution, polynomial evaluation and formal coefficient comparison. The value/derivative payoff needs polynomial differentiation or its formal power rule, stated explicitly.

Extensions: The higher-order extension needs second derivatives and truncated polynomial multiplication. No claim about analytic convergence or actual tiny real numbers is made.

Reading and representation: Read an ordinary coefficient and an ϵ coefficient as separate exact coordinates. Equality requires equality of both coefficients.

Materials: Two-column coefficient mats, optional signed counters for initial integer trials, paper for inverse fractions and a blank calculation graph. Preparation: about 5 minutes.

10 minutes trying products; 10-15 classifying inverses and cancellation; 20 for polynomial evaluation with a rate coefficient. Higher-order information is a separate continuation.

Launch and facilitation

Use pairs $a+b\epsilon$ with the exact rule $\epsilon^2=0$. Choose two pairs and multiply. Can a second coefficient preserve information that would be lost by setting $\epsilon=0$?

Student page 1: Make an unfamiliar arithmetic

Algebra core: exact coefficient rules

A dual number is the formal pair $a+b\epsilon$, with $a,b\in\mathbb{Q}$. Add coefficients separately. Multiply by distribution and replace ϵ^2 by 0. The pair $0+1\epsilon$ is different from $0+0\epsilon$. This is exact arithmetic in $\mathbb{Q}[T]/(T^2)$, not arithmetic with a tiny real ϵ .

Can one calculation remember a rate?

1 / prompt and solution

Choose two nonzero dual numbers whose product is zero, and two whose product is 1. Then find an inverse for $3-2\epsilon$ and explain which choices made your two searches possible.

Solution: For example ϵ and 2ϵ are nonzero pairs but their product is 0. The pair $1+\epsilon$ has inverse $1-\epsilon$. For $3-2\epsilon$, an inverse is $1/3+(2/9)\epsilon$: ordinary product 1 and ϵ coefficient $3\cdot(2/9)-2\cdot(1/3)=0$. Nonzero pure ϵ multiples are zero divisors, while nonzero ordinary coefficients allow inverse construction. Arbitrary proposed pairs must be checked by both coefficient equations.

1 / hints to offer only after exploration

1. Write the ordinary and ϵ coefficients of the product separately.

2 / prompt and solution

Classify all invertible $a+b\epsilon$. Explain exactly why canceling a nonzero factor can fail, even though canceling an invertible factor works.

Solution: $(a+b\epsilon)(c+d\epsilon)=ac+(ad+bc)\epsilon$. Equality to 1 requires $ac=1$, so $a\neq 0, c=1/a$, and $d=-b/a^2$. This also proves sufficiency by substitution. If $a=0$ no product has ordinary coefficient 1. Cancellation fails, for example $\epsilon\cdot 0=\epsilon\cdot\epsilon=0$ although $0\neq\epsilon$. If a factor has an inverse, multiplying both sides by it gives valid cancellation using associativity inherited from the quotient ring.

2 / hints to offer only after exploration

1. The ordinary coefficient gives a necessary condition before the ϵ equation.

Student page 2: Run two numbers through one program

Continuation: polynomial evaluation and derivatives

Use the polynomial program $f(z)=(z^2+1)(z-2)$. Evaluate it by its displayed operations, keeping both coefficients at every intermediate step. Polynomial derivative notation will be used on the next page.

3 / prompt and solution

Choose $z=3+\epsilon$ or $z=3+2\epsilon$ and run the program. Compare the ordinary output with $f(3)$. Predict how the ϵ coefficient changes if the input is $3+b\epsilon$ for any rational b .

Solution: For $3+b\epsilon$: $z^2=9+6b\epsilon$, $z^2+1=10+6b\epsilon$, and $z-2=1+b\epsilon$. Their product is $10+16b\epsilon$. Thus $3+\epsilon$ gives $10+16\epsilon$; $3+2\epsilon$ gives $10+32\epsilon$; ordinary output is always $f(3)=10$. The coefficient scales linearly with b . The calculations are exact because every term with ϵ^2 is zero.

3 / hints to offer only after exploration

1. Do not substitute $\epsilon=0$ before the multiplication is complete.

Can one calculation remember a rate?

4 / prompt and solution

A partner says the two output coefficients determine the whole polynomial. Choose two different polynomials that give the same output at $2+\epsilon$, and show an ordinary input at which they disagree. What information did the program forget?

Solution: For example $f(z)=z^2$ and $g(z)=z^2+(z-2)^2$ both give $4+4\epsilon$ at $2+\epsilon$ because the added term becomes $\epsilon^2=0$. At $z=3$ they give 9 and 10. Their common value and first derivative at 2 agree, but second-order behavior differs. In general every multiple of $(z-2)^2$ is invisible to evaluation at $2+\epsilon$. Thus dual output is local first-order information, not the full polynomial.

4 / hints to offer only after exploration

1. Add a nonzero term that vanishes after substituting $z=2+\epsilon$.

Student page 3: Prove what the second coefficient means

Algebra/calculus proof: polynomial power rule

For a polynomial $f(X)=\sum c_k X^k$ over $\mathbb{Q}$, define $f'(X)=\sum k c_k X^{(k-1)}$, with the constant term contributing 0. This agrees with the usual derivative for real polynomials. Prove statements for every polynomial, not just the chosen program. Supplied algebra fact: a nonzero polynomial over $\mathbb{Q}$ has only finitely many roots.

5 / prompt and solution

Prove $f(a+b\epsilon)=f(a)+b f'(a)\epsilon$ for every rational a,b . Use it to explain why the earlier program output 16 in its ϵ column when the input was $3+\epsilon$.

Solution: For $k \geq 1$, expansion of $(a+b\epsilon)^k$ leaves only the term a^k and the k terms containing exactly one $b\epsilon$, giving $a^k + k a^{(k-1)}b\epsilon$. Terms with at least two ϵ factors vanish. For $k=0$ the output is 1. Summing with coefficients c_k gives the formula, including $a=0$ (handle $k=1$ directly; no negative power is needed). For $f=(X^2+1)(X-2)=X^3-2X^2+X-2$, $f'=3X^2-4X+1$ and $f'(3)=27-12+1=16$. The ring identity is exact; no limiting-size approximation has been used.

5 / hints to offer only after exploration

1. Count how many expansion terms contain exactly one ϵ factor.

Can one calculation remember a rate?

6 / prompt and solution

Use dual multiplication to derive the product rule for two polynomials f, g . Then derive the chain rule by evaluating $f(g(a+b\epsilon))$. State what would need to change to retain second-order information.

Solution: Write $f(a+b\epsilon)=f(a)+b f'(a)\epsilon$ and $g(a+b\epsilon)=g(a)+b g'(a)\epsilon$. Multiplication gives ordinary value $f(a)g(a)$ and ϵ coefficient $b[f'(a)g(a)+f(a)g'(a)]$. Compare with $(fg)(a+b\epsilon)$, taking $b=1$, to obtain $(fg)'=f'g+fg'$ at every rational a , hence as a polynomial identity (a nonzero polynomial has finitely many roots). Composition gives $f(g(a))+b f'(g(a))g'(a)\epsilon$, so $(f\circ g)'=(f'\circ g)g'$ by the same reasoning, or directly as a formal coefficient identity. To retain second-order information, keep an ϵ^2 column and impose $\epsilon^3=0$ instead; simply calling ϵ a smaller real number does not change the algebra.

6 / hints to offer only after exploration

1. Compare coefficients of ϵ after multiplying the two outputs.

Optional facilitator extension 1

Gate: Second polynomial derivatives, binomial expansion and truncated quotient arithmetic.

Now work with $\epsilon^3=0$ while $\epsilon^2\neq 0$. Derive $f(a+b\epsilon+c\epsilon^2)$ through the ϵ^2 term, and test $f(X)=X^3$ at $a=2, b=1, c=3$.

Solution: Expansion gives $f(a)+b f'(a)\epsilon+[c f'(a)+(b^2/2) f''(a)]\epsilon^2$. In each power, an ϵ^2 contribution comes from one $c\epsilon^2$ factor or two $b\epsilon$ factors; counting those choices gives the coefficients. For X^3 at $(2, 1, 3)$, the result is $8+12\epsilon+(36+6)\epsilon^2=8+12\epsilon+42\epsilon^2$. The formula is exact in $\mathbb{Q}[\epsilon]/(\epsilon^3)$, and $1, \epsilon, \epsilon^2$ are independent residue classes.

Mathematics and prior use

Dual numbers, nilpotents, formal first-order Taylor expansion and forward propagation of a derivative coefficient.

Prior use: The original AD-14 inverse/zero-divisor examples are entry. The new core follows a chosen arithmetic program and proves first-order information is exact. AD-13 supplies quotient normal forms, while this family focuses on invertibility and differentiation.

Source: J. S. Milne, Algebraic Geometry

Current PDF, Chapter 4 §f, “Dual numbers,” printed p.90 (PDF page 90), Proposition 4.26; “Derivations,” p.91. Use the exact quotient-ring multiplication and first-order polynomial expansion. Scalar polynomial programs and inverse examples are derived here. Evidence: Read the current dual-number definition, multiplication, Taylor calculation and Leibniz-rule discussion on 2026-09-25. The atlas’s older Chapter 2 locator no longer identifies this passage.

<https://www.jmilne.org/math/CourseNotes/AG.pdf>

Encode a rational point with one slope

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with explicit fractional-algebra gates. The drawing offers a real choice, but completeness must be argued symbolically.

Derive mutually inverse point/slope formulas for every rational point on the ellipse except one, then explicitly include the exceptional basepoint.

Prerequisites and preparation

Core: Fractions, signed coordinates, substitution and factoring a quadratic with one known root. Exact symbolic algebra is needed for completeness.

Extensions: Integer triples and coprimality/parity; an optional projective parameter handles the infinite slope. No general rationality theorem for all curves is assumed.

Reading and representation: Distinguish a schematic drawn point from a pair of exact rational coordinates; read ordered pairs and a slope parameter.

Materials: Printed ellipse $x^2+2y^2=1$ with basepoint $(-1,0)$, ruler, blank exact-coordinate ledger and graph paper. No compass construction of every rational point is assumed. Preparation: about 6 minutes.

10 minutes choosing secants and predicting signs; 15 deriving or checking exact points; 15 decoding and completeness; 15 for integer triples if ready.

Launch and facilitation

Aim a line from the marked point across the ellipse. Choose a rational slope and send only the second point to a partner. Can the partner recover the slope exactly?

Student page 1: Aim, calculate and send

Core: slopes and exact fractions

The curve is $x^2+2y^2=1$. The marked point is $B=(-1,0)$. A nonvertical line of slope t through B has equation $y=t(x+1)$. Use exact fractions; a plotted approximation alone does not certify membership.

1 / prompt and solution

Choose $t=1$ or $t=2/3$, predict where the second intersection lies, and find it exactly by substitution. Give the point to a partner without the slope; let them try to recover your choice.

Solution: For $t=1$, substitution gives $3x^2+4x+1=(x+1)(3x+1)=0$; the second point is $(-1/3, 2/3)$. For $t=2/3$ the equation becomes $17x^2+16x-1=(x+1)(17x-1)=0$, giving $(1/17, 12/17)$. Both satisfy $x^2+2y^2=1$ exactly. Positive slope gives positive y for a second point with $x > -1$. Recovery uses $t=y/(x+1)$, yielding 1 or $2/3$.

Encode a rational point with one slope

1 / hints to offer only after exploration

1. One quadratic root is known before doing any arithmetic.

2 / prompt and solution

Invent a different rational slope and create a point challenge. Then decode the supplied point $(-7/11, 6/11)$, checking both its curve equation and the recovered line.

Solution: Any rational t gives the formula derived next. For example $t=-1/2$ gives $(1/3, -2/3)$, which checks $1/9+8/9=1$. The supplied point checks $49/121+72/121=1$, and its slope is $(6/11)/(4/11)=3/2$. Substituting $y=(3/2)(x+1)$ recovers it. A partner's arbitrary rational point must pass both the equation and the inverse-slope check.

2 / hints to offer only after exploration

1. The horizontal change from B is $x+1$, not x .

Student page 2: Can any rational point escape?

Core proof: known root and inverse

Continue to use finite rational slopes t . A complete parametrization claim needs both directions and every exceptional case.

3 / prompt and solution

Derive the second-intersection formula for general t . Prove that its denominator cannot vanish for rational t and that it always gives a point different from B. Does $t=0$ cause any difficulty?

Solution: Substitution factors as $(x+1)[(1+2t^2)x+(2t^2-1)]=0$. Thus $x=(1-2t^2)/(1+2t^2)$, $y=2t/(1+2t^2)$. The denominator is positive, hence nonzero for every real and rational t . Also $x+1=2/(1+2t^2)>0$, so the second point differs from B. Direct substitution gives $[(1-2t^2)^2+8t^2]/(1+2t^2)^2=1$. At $t=0$ the point is $(1,0)$, with no exceptional denominator.

3 / hints to offer only after exploration

1. Factor out $x+1$ before using the quadratic formula.

Encode a rational point with one slope

4 / prompt and solution

Prove that every rational point on the ellipse other than B is recovered by exactly one rational slope. Describe the missing point geometrically and explain how an infinite-slope card completes the correspondence.

Solution: For a rational point (x,y) with $x \neq -1$, $t=y/(x+1)$ is rational. Its line goes through B and the point, so factoring the substituted equation shows the other point is precisely the formula from prompt 3. This also proves uniqueness of t . If $x=-1$ the ellipse equation forces $y=0$, so B is the sole omitted point. The vertical line $x=-1$ meets the ellipse only at B (a double intersection); it is tangent there. Assigning the formal parameter ∞ to B extends the bijection from $\mathbb{Q}$ to the punctured rational ellipse into a bijection $\mathbb{Q} \cup \{\infty\}$ with all rational points. No finite real t gives B.

4 / hints to offer only after exploration

1. Check the exceptional denominator of the inverse, not just the forward formula.

Student page 3: Trade slopes for integer triples

Continuation: integer algebra and completeness

Write $t=p/q$ with integers p,q and $q \neq 0$. A point with rational coordinates can be written $(A/C, B/C)$, $C \neq 0$; scaling A,B,C together does not change the point.

5 / prompt and solution

Clear denominators in your parametrization to construct integer triples satisfying $A^2+2B^2=C^2$. Choose p,q to make a nontrivial example. Explain in what sense this produces every rational point, including any exception.

Solution: Take $A=q^2-2p^2$, $B=2pq$, $C=q^2+2p^2$. Expansion proves $A^2+2B^2=C^2$. For $p=2,q=3$, this is $(1,12,17)$; for $p=3,q=2$ it is $(-14,12,22)$, reducible to $(-7,6,11)$. Every rational point other than B has rational $t=p/q$, so its coordinate ratios agree with one generated triple. This does not assert every integer triple equals the raw formula without scaling or sign changes. The missing rational point $(-1,0)$ is represented by $(-1,0,1)$; allowing $q=0,p \neq 0$ in the homogeneous formula gives the same point.

5 / hints to offer only after exploration

1. Keep the common denominator instead of reducing the two coordinates separately.

Encode a rational point with one slope

6 / prompt and solution

Does a line through a known point guarantee a rational second point on every polynomial curve? Test $y^2=x^3+x+1$ through $(0,1)$ using the horizontal line $y=1$. Explain exactly which earlier step breaks.

Solution: The point $(0,1)$ lies on this cubic. With $y=1$, the equation becomes $x^3+x=x(x^2+1)=0$. After removing the known root $x=0$, the remaining factor has degree 2 and has no real or rational roots. Thus there is no distinct rational second point for this chosen line. For the conic, removing one root leaves a linear equation with rational coefficients and nonzero leading coefficient; a cubic leaves a quadratic. This one example refutes the claimed general guarantee, but does not by itself classify which cubics admit rational parametrizations.

6 / hints to offer only after exploration

1. Compare the degree left after removing the known factor.

Optional facilitator extension 1

Gate: Prime divisibility, parity and the meaning of a primitive integer triple.

Suppose p, q are coprime integers with $q > 0$. Determine the common divisor of $q^2-2p^2, 2pq, q^2+2p^2$, including the case $p=0$.

Solution: No odd prime can divide all three: it would divide $2q^2$ and $4p^2$, hence both p and q , contradicting coprimality. If q is odd, both $q^2 \pm 2p^2$ are odd, so the common divisor is 1. If q is even then p is odd; $q^2 \pm 2p^2$ are exactly 2 modulo 4, and $2pq$ is divisible by 4, so the gcd is exactly 2. The case $p=0$ forces $q=1$ and gives gcd 1. Thus the raw triple is primitive exactly when q is odd; otherwise dividing by 2 makes it primitive.

Mathematics and prior use

Rational parametrization of a conic with a known rational point, inverse maps and an exceptional parameter.

Prior use: Week 9 rational slopes are preparatory. The original AD-15 circle is replaced by $x^2+2y^2=1$, a point-encoding/decoding task and generalized integer triples. AD-16 changes the ground field instead of parametrizing rational coordinates.

Source: Ravi Vakil, The Rising Sea, January 29, 2015 draft

§6.5.8, Figure 6.1 and circle calculation, printed pp.191-192; §6.5.9. Adapt the line-through-a-known-point method to a new ellipse and derive all formulas and exceptions directly. Evidence: Read the circle substitution, known-root factorization and inverse map on 2026-09-25. The source uses basepoint $(1,0)$ on a circle; this worksheet uses $(-1,0)$ on a different conic.

<https://math.stanford.edu/~vakil/216blog/FOAGjan2915public.pdf>

Can these finite points make one cycle?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with modulo-five gates for enumeration; advanced/specialist for the elliptic group structure. No cryptographic-security conclusion is made from this tiny group.

At the finite entry, certify the complete point set. At the advanced gate, find a point whose repeated addition visits the entire projective set, then solve point equations through its cyclic labels.

Prerequisites and preparation

Core: Modulo-five arithmetic and a complete square-residue search give the finite entry. The new cyclic-structure payoff requires field division, point-addition formulas and a supplied elliptic-curve group theorem.

Extensions: Projective triples, equivalence under nonzero scaling, formal partial derivatives and nonsingularity are explicit advanced prerequisites. Orbit completeness and group-equation solving use modular reasoning.

Reading and representation: Read coordinates as residues 0-4, not ordinary real positions. Never join the modular points into a smooth real curve.

Materials: Blank 5×5 coordinate grid, residue square table, counters without a prescribed count, separate infinity card for the projective stage, and supplied addition-rule card. Preparation: about 7 minutes.

10-15 minutes choosing a complete search; 10 for symmetry; 15 for projective completion and theorem hypotheses; 20-30 for the addition cycle and decoding. These stages can occupy separate meetings.

Launch and facilitation

Choose a column or a square residue and find points obeying $y^2 = x^3 + x + 1$ modulo 5. Can you find them all while doing less than twenty-five unrelated tests?

Student page 1: Find the finite point set

Entry: residue arithmetic and an exhaustive search

All arithmetic is modulo 5. Coordinates are residues 0,1,2,3,4. Place a point exactly when $y^2 = x^3 + x + 1$. The printed grid is a map of labels, not a sampled real curve: do not join the dots. Squares of 0,1,2,3,4 are 0,1,4,4,1.

Can these finite points make one cycle?

1 / prompt and solution

Choose a search organized by columns or square values. Find every point, and give a certificate that none were missed. Predict which points must come in pairs before completing the search.

Solution: For $x=0,1,2,3,4$, the right-hand sides are 1,3,1,1,4. Thus the affine points are $(0,1),(0,4),(2,1),(2,4),(3,1),(3,4),(4,2),(4,3)$, eight in all. The square table contains every y , and the five-column table contains every x , proving completeness. If (x,y) works then $(x,-y)$ works. Here no column has right side 0, so no point has $y=0$ and all pairs are distinct.

1 / hints to offer only after exploration

1. Reuse the same square table in every column.

2 / prompt and solution

Choose one coefficient change: replace the constant 1 by 0, or replace the coefficient of x by 0. Predict a difference, then find the new affine point count and explain why a smooth real-looking sketch would not settle it.

Solution: For $y^2=x^3+x$, right-hand sides for $x=0\dots 4$ are 0,2,0,0,3, so only $(0,0),(2,0),(3,0)$: three affine points. For $y^2=x^3+1$, right-hand sides are 1,2,4,3,0, giving $(0,1),(0,4),(2,2),(2,3),(4,0)$: five affine points. These counts depend on residue arithmetic and square roots in F_5 ; a real drawing cannot supply them. Either choice is acceptable if the full column certificate is correct.

2 / hints to offer only after exploration

1. A zero right-hand side has one square root, whereas 1 and 4 each have two.

Student page 2: Account for the point off the grid

Advanced: projective coordinates, formal partial derivatives and a supplied theorem

Return to $y^2=x^3+x+1$. A projective point $[X:Y:Z]$ is a nonzero triple modulo nonzero scalar multiplication. The homogeneous equation is $Y^2Z=X^3+XZ^2+Z^3$. Supplied theorem: over a field of characteristic other than 2 or 3, a curve $y^2=x^3+ax+b$ with $\Delta=-16(4a^3+27b^2)\neq 0$, completed by $O=[0:1:0]$, has a commutative associative point-addition law with identity O . The next page supplies its formulas.

Can these finite points make one cycle?

3 / prompt and solution

Prove that points with $Z \neq 0$ correspond exactly to the affine points already found, and find every projective point with $Z=0$. Check the discriminant hypothesis for the original curve.

Solution: If $Z \neq 0$, multiply the triple by Z^{-1} to obtain $[x:y:1]$; substitution gives the affine equation, and this representative is unique. If $Z=0$, the equation forces $X^3=0$, hence $X=0$ in the field. Since the triple is nonzero, $Y \neq 0$, and scaling gives the single point $[0:1:0]=O$. The projective total is therefore 9, not a mysterious extra grid point. With $a=b=1$, $\Delta = -16 \cdot 31 \equiv 4 \pmod{5} \neq 0$, and characteristic 5 is neither 2 nor 3, so the supplied group theorem applies.

3 / hints to offer only after exploration

1. Use the field inverse to normalize one nonzero coordinate.

4 / prompt and solution

Why should the theorem's nonzero-discriminant condition not be dropped? Test the curve $y^2=x^3$ at $(0,0)$ using its two partial derivatives. The proposed tangent slope for $y^2=x^3+ax+b$ is $s=(3x^2+a)/(2y)$. Identify what fails in this formula at the tested point.

Solution: For $F(x,y)=y^2-x^3$, partial derivatives $F_x=-3x^2$ and $F_y=2y$ both vanish at $(0,0)$, which lies on the curve. Thus the point is singular; its discriminant is 0. The tangent slope $(3x^2+a)/(2y)$ would be $0/0$, not a defined field quotient. The formula's vertical inverse case alone does not turn this singular cubic into the same nonsingular elliptic-curve theorem. We have not claimed that no other structure can be studied after removing or resolving the singularity.

4 / hints to offer only after exploration

1. At the origin, compare numerator and denominator rather than canceling them.

Student page 3: Try repeated point addition

Advanced: field division and the supplied group law

Use the original curve and arithmetic modulo 5. $O+P=P$. The inverse of (x,y) is $(x,-y)$; inverse pairs add to O . Otherwise, for $P=(x_1,y_1), Q=(x_2,y_2)$, use $s=(y_2-y_1)/(x_2-x_1)$ when $x_1 \neq x_2$; for $P=Q$ with $y_1 \neq 0$ use $s=(3x_1^2+1)/(2y_1)$. Then $P+Q=(x_3,y_3)$, where $x_3=s^2-x_1-x_2$ and $y_3=s(x_1-x_3)-y_1$. Doubling a point with $y=0$ gives O . Division means multiplying by a nonzero residue's inverse.

Can these finite points make one cycle?

5 / prompt and solution

Choose a nonidentity point and repeatedly add it, starting at O . Record the orbit until it returns. Try to find a point that visits every point. If your first choice does not, choose another. Give at least one exact addition or doubling check.

Solution: One successful choice is $P=(0,1)$. The multiples $0P \dots 9P$ are $O, (0,1), (4,2), (2,1), (3,4), (3,1), (2,4), (4,3), (0,4), O$. For doubling, $s=1/2=3$, $x_3=3^2=4$, $y_3=3(0-4)-1=2$, so $2P=(4,2)$. Adding P to $2P$ uses $s=(2-1)/(4-0)=1/4=4$ and gives $(2,1)$. The displayed orbit contains all eight affine points plus O before returning. Other choices: $3P=(2,1)$ and $6P=(2,4)$ have order 3; every other nonidentity point has order 9. A learner's different orbit is checked against the finite group law, not its visual adjacency.

5 / hints to offer only after exploration

1. After finding a point, locate its inverse in your inventory.
2. A small orbit is evidence about the chosen starting point, not a reason to add missing grid dots.

6 / prompt and solution

For a point P whose orbit visits everything, prove that labeling its multiples by integers modulo the return time turns the point law into modular addition. Explain which part is experimental evidence and which part uses the supplied theorem.

Solution: For $P=(0,1)$, the explicit list shows $9P=O$ and that $0P$ through $8P$ are distinct and exhaust the projective set. Associativity from the supplied theorem gives $(rP)+(sP)=(r+s)P$; the identity $9P=O$ makes labels depend only on the residue modulo 9. Distinctness makes the label map bijective. Thus the group is cyclic of order 9. The orbit calculations establish the order and exhaustion in this instance; they do not by themselves prove associativity of the chord/tangent formulas for arbitrary triples. The finite checker additionally verifies all triples here, but the student proof openly uses the supplied theorem.

6 / hints to offer only after exploration

1. Use associativity to regroup repeated sums, then use the return time.

Student page 4: Solve equations in the hidden clock

Advanced continuation: a cyclic-group representation

Use the generator $P=(0,1)$ and the orbit you proved. A point equation may now be translated into an equation on its multiple-of- P label. A solution list must include every point or explain why none can work.

Can these finite points make one cycle?

7 / prompt and solution

Solve $2Q=P$, then solve $3Q=P$. Solve $3Q=O$ and $3Q=(2,1)$. Translate every answer back to actual point coordinates and explain the difference among the four equations.

Solution: Write $Q=kP$ with $k \bmod 9$. For $2Q=P$, $2k \equiv 1$ gives $k=5$, so $Q=(3,1)$. For $3Q=P$, $3k \equiv 1$ has no solution because the left side is divisible by 3 modulo 9. For $3Q=O$, $k=0,3,6$ gives $O, (2,1), (2,4)$. Since $(2,1)=3P$, $3Q=(2,1)$ means $3k \equiv 3$, hence $k=1,4,7$ and $Q=(0,1), (3,4), (4,3)$. The label bijection and congruence classifications prove these lists are complete.

7 / hints to offer only after exploration

1. Which residues can $3k$ have modulo 9?

Optional facilitator extension 1

Gate: Cyclic groups, divisibility and subgroup closure.

Without recomputing the full addition table, classify the order of every point and every subgroup of this nine-point group.

Solution: Under $Q=kP$ the order is $9/\gcd(9,k)$, with O of order 1. Labels 3,6 have order 3, namely $(2,1), (2,4)$; labels 1,2,4,5,7,8 have order 9. Every subgroup containing an order 9 point is the whole group. A nontrivial subgroup without such a point contains $3P$ or $6P$ and therefore contains exactly $\{O, 3P, 6P\}$, unless it is already whole. Thus the subgroups are $\{O\}$, $\{O, (2,1), (2,4)\}$, and all nine points. This exhaustive argument uses only the explicit cyclic representation.

Mathematics and prior use

Finite-field rational points, projective completion, nonsingular Weierstrass equations and a concretely identified cyclic group.

Prior use: Week 4 modular rings are prior preparation. The original AD-16 curve $y^2=x^3-x$ and its seven affine points are not reused: this curve has a different inventory and a cyclic group continuation. AD-15's cubic counterexample uses the same equation over $\mathbb{Q}$, deliberately contrasting ground fields.

Source: J. S. Milne, Elliptic Curves, second edition

I §3, addition law and associativity, printed pp.26-28; I Proposition 4.10, p.32; II Theorem 2.1, p.50 and addition/duplication formulas, pp.51-52. Supply the nonsingular projective group theorem and use equivalent slope formulas. The chosen F5 point set, orbit, orders and equations are independently calculated. Evidence: Read nonsingularity criterion, exceptional inverse/doubling cases and associativity justification on 2026-09-25. The source uses $4a^3+27b^2$ as a nonvanishing criterion; our discriminant is the nonzero multiple $-16(4a^3+27b^2)$.

<https://www.jmilne.org/math/Books/EC2.pdf>

Design the round when closeness begins

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with fraction/algebra gates. Marker motion is accessible earlier, but weighted limits and all-parameter claims need algebra-ready learners. Not classroom-piloted.

An exact formula and a proof of the first round meeting a tolerance, followed by a family of starting pairs designed to cross a chosen threshold.

Prerequisites and preparation

Core: Fractions, simultaneous updates, signed differences and geometric powers. Exact first-entry bounds use inequalities; weighted invariants and parameter classification are the algebraic core.

Extensions: Linear algebra for eigenvectors; real geometric-sequence limits for the full parameter classification. No differential calculus.

Reading and representation: Read two update rules and distinguish old from new coordinates. A number-line trial is an entry; a finite list is not a convergence proof.

Materials: Two labeled markers, printed number line and open old/new ledger; pencil and optional fraction strips. Preparation: about 5 minutes.

10 minutes choose and run a rule; 15 find a preserved quantity; 20 derive the exact time bound; 15 design an initial pair. General parameter classification can be another meeting.

Launch and facilitation

Both markers move at once from their old positions. Let partners choose which marker is more willing to move. Start both trials from the same pair and predict whether they settle at the same place.

Student page 1: Choose who moves farther

Core: fractions and simultaneous movement

At each integer round, $x' = (1-\alpha)x + \alpha y$ and $y' = \beta x + (1-\beta)y$. Both right sides use the OLD pair. Start at $(-3, 9)$. Choose rule A: $\alpha = 1/4, \beta = 1/2$; or rule B: $\alpha = 1/2, \beta = 1/4$.

1 / prompt and solution

Run your chosen rule for two rounds. Compare with a partner using the other rule. Predict the long-term location and explain why averaging the two current positions may mislead.

Solution: A gives $(-3, 9) \rightarrow (0, 3) \rightarrow (3/4, 3/2)$, apparently approaching 1. B gives $(-3, 9) \rightarrow (3, 6) \rightarrow (9/2, 21/4)$, approaching 5. The starting ordinary midpoint is 3 but the first midpoint is $3/2$ under A and $9/2$ under B. Each move uses old values; sequential updates give a different map. The limit predictions are certified later.

Design the round when closeness begins

1 / hints to offer only after exploration

1. Compute both new entries before replacing either old entry.
2. Check the ordinary midpoint after one round.

2 / prompt and solution

Find a nonzero weighted sum $px+qy$ that never changes under rule A. Then find one for rule B. Prove your choices work for every starting pair, and explain why equal shrinkage of the gap does not force equal limits.

Solution: For A, $2x+y$ is invariant: $2(3x/4+y/4)+(x/2+y/2)=2x+y$. For B, $x+2y$ is invariant. Both differences satisfy $x'-y'=(x-y)/4$. If both coordinates share a limit, the invariants force 1 under A and 5 under B for this start. All invariant coefficient pairs satisfy $p\alpha=q\beta$, so these are unique up to scale.

2 / hints to offer only after exploration

1. Compare the coefficients of x and y after one update.

Student page 2: Predict a first round, then design one

Algebra core: a formula and sharp inequalities

Use rule A. Write $d=x-y$ and $L=(2x+y)/3$. “Within ε ” means BOTH $|x-L|\leq\varepsilon$ and $|y-L|\leq\varepsilon$. Round 0 is the initial pair.

3 / prompt and solution

Derive x_n and y_n from L and d_0 without repeated updating. For the start $(-3,9)$, find the FIRST round when both coordinates are within $1/100$ of L . Prove the preceding round fails. Can they be exactly equal at a finite round?

Solution: One step keeps L fixed and replaces d by $d/4$. Solving $2x+y=3L$ and $x-y=d$ gives $x_n=L+d_0/(3\cdot 4^n)$, $y_n=L-2d_0/(3\cdot 4^n)$. Here $L=1, d_0=-12$, so errors are $4/4^n$ and $8/4^n$. At $n=4$ the maximum is $1/32>1/100$; at $n=5$ it is $1/128\leq 1/100$. Error decreases strictly, so 5 is first. A nonzero initial difference never becomes zero in finitely many divisions by 4; equality at every round occurs only for an equal initial pair.

3 / hints to offer only after exploration

1. Track the larger of the two absolute errors.
2. Compare powers of 4 exactly; logarithms are unnecessary.

Design the round when closeness begins

4 / prompt and solution

Design an initial pair whose limit is 2 and whose FIRST round within $1/8$ is round 3. Find one pair, then classify every real initial pair with this property.

Solution: Let $d_0=d$. The invariant forces $x_0=2+d/3$ and $y_0=2-2d/3$. The maximum error is $E_n=2|d|/(3 \cdot 4^n)$. First entry at 3 means $E_2 > 1/8$ and $E_3 \leq 1/8$, equivalently $3 < |d| \leq 12$. Every pair in this family works, and these conditions are necessary. Example $d=6$ gives $(4, -2)$; errors at rounds 2 and 3 are $1/4$ and $1/16$. Strictness at the earlier boundary matters: $|d|=3$ first enters at 2.

4 / hints to offer only after exploration

1. Use two inequalities: failure at 2 and success at 3.

Student page 3: Choose the dynamics

Continuation: real parameters and geometric limits

Now allow $0 \leq \alpha, \beta \leq 1$. Initial coordinates may be any real numbers. A common limit means both coordinates converge to the same finite number.

5 / prompt and solution

For unequal initial coordinates, classify every parameter pair that produces a common limit. Include zero movement, swapping and exact finite meeting. Derive the limit when it exists.

Solution: Put $s=\alpha+\beta$. The difference multiplier is $r=1-s$. If $s>0$, $L=(\beta x_0 + \alpha y_0)/s$ is invariant, and $x_n=L+(\alpha/s)r^n d_0$, $y_n=L-(\beta/s)r^n d_0$. Common convergence occurs exactly when $0 < s < 2$: $|r| < 1$. If $s=1$, $r=0$ and both meet after one round. Otherwise a nonzero difference never becomes zero at finite n . If $s=0$ then $\alpha=\beta=0$ and the unequal pair is fixed. If $s=2$ then $\alpha=\beta=1$ and it swaps forever. Equal starting coordinates stay fixed for every allowed pair and are deliberately excluded from the classification.

5 / hints to offer only after exploration

1. First study the signed difference; then use $\beta x + \alpha y$.
2. Do not divide by $\alpha + \beta$ before treating its zero case.

6 / prompt and solution

Return to the start $(-3, 9)$. Design ALL parameter pairs that give common limit 3 and halve the absolute gap each round. Which preserve the order of the markers and which alternate it?

Solution: The limit equation $(-3\beta + 9\alpha)/(\alpha + \beta) = 3$ forces $\alpha = \beta$, with positive sum. Halving requires $|1 - \alpha - \beta| = 1/2$, hence sums $1/2$ or $3/2$. The two pairs are $(1/4, 1/4)$ and $(3/4, 3/4)$. Their signed multipliers are $1/2$ and $-1/2$, respectively. The first keeps the order; the second swaps order on every round while the gap shrinks. Both have limit 3 and never meet finitely from this unequal start.

Design the round when closeness begins

6 / hints to offer only after exploration

1. The invariant determines the ratio; the gap multiplier determines the sum.

Optional facilitator extension 1

Gate: Bases, eigenvectors and geometric powers.

For $0 < \alpha + \beta < 2$, express the update using the directions $(1, 1)$ and $(\alpha, -\beta)$. Identify the eigenvalues and prove the coordinate formula from this decomposition.

Solution: The directions are independent because their determinant is $-(\alpha + \beta) \neq 0$. $T(1, 1) = (1, 1)$, and $T(\alpha, -\beta) = (1 - \alpha - \beta)(\alpha, -\beta)$. Every starting pair is $L(1, 1) + (d_0/(\alpha + \beta))(\alpha, -\beta)$, giving the stated n th iterate. An eigenvector need not be a unit vector; here the fixed value L is a coordinate, not necessarily the ordinary mean.

Mathematics and prior use

A change of coordinates separates the fixed direction from the decaying or oscillating disagreement. This is an exact two-dimensional spectral calculation, with finite equality distinguished from convergence.

Prior use: The original AD-17 uses symmetric quarter-gap movement from $(0, 8)$. The accepted AP-01 trial and GA-05 worksheets use static harmonic averaging and uniqueness. Here asymmetric simultaneous dynamics preserve a weighted quantity and support exact time-to-tolerance design; it is a related linear idea with a different problem and proof.

Source: Sheldon Axler, Linear Algebra Done Right, fourth edition

§3A definition 3.1, printed p. 52 / PDF p. 66; §5A definitions 5.5 and 5.8, printed pp. 134-135 / PDF pp. 148-149. Use the linear-map and eigenvector framework. The asymmetric update, weighted invariant, threshold problems and parameter proofs are derived independently. Evidence: Read the current official PDF locally on 2026-09-26 after web retrieval timed out; exact arithmetic and general formulas are checked separately.

<https://linear.axler.net/LADR4e.pdf>

Can you preserve area without preserving shape?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Promising with coordinate-area gates; projection/determinant and rigidity arguments are an advanced continuation. A grid count alone is not an adequate core proof.

A complete parameter design for area 1, exact inverses, and a proof separating area preservation from preservation of all lengths.

Prerequisites and preparation

Core: Coordinate vectors, parallelogram area and signed arithmetic. The general proof uses perpendicular projection/dot products; inverse and composition arguments require 2×2 algebra.

Extensions: Integer divisibility and inverse matrices for the lattice continuation.

Reading and representation: Ordered image arrows determine a linear map. Keep signed area, nonnegative physical area, and distance separate.

Materials: Printed coordinate grid, ruler, two colored image-arrow strips, optional tracing paper and scissors; vector/matrix rule card. Preparation: about 8 minutes.

10 minutes choose an image square; 15 compare orientation and collapse; 15 derive the area formula; 20 design and compose area-preserving maps. Later proof pages may be a separate session.

Launch and facilitation

Keep the first arrow fixed and slide the second arrow horizontally. Build the resulting parallelogram before calculating. Which choices could fold the whole plane onto a line?

Student page 1: Slide one image arrow

Entry: vectors, orientation and parallelograms

The unit horizontal arrow maps to $u=(2,1)$; the unit vertical arrow maps to $v=(k,2)$. The full linear rule is $A_k(x,y)=x u+y v$. Traverse the image square in the order $0,u,u+v,v$. Counterclockwise traversal has positive signed area, clockwise negative; a collapsed image has signed area 0.

Can you preserve area without preserving shape?

1 / prompt and solution

Choose $k=1$ or $k=5$. Draw the image square, find its ordinary area, and determine its signed area. Compare your result with the other choice. Justify area by geometry, not by counting partly covered grid squares.

Solution: For $k=1$ the ordered vertices are $(0,0),(2,1),(3,3),(1,2)$, with signed area 3 and ordinary area 3. For $k=5$ they are $(0,0),(2,1),(7,3),(5,2)$, with signed area -1 and ordinary area 1. One geometric certificate uses base u of length $\sqrt{5}$ and the signed perpendicular height $(4-k)/\sqrt{5}$; the general derivation comes later. Equivalently triangulate with exact coordinate triangle areas, preserving orientation. The sign records which side of the directed line through u contains v .

1 / hints to offer only after exploration

1. Find the perpendicular height to the slanted base, or enclose and subtract exact triangles.

2 / prompt and solution

Test $k=4$. Find two distinct inputs with the same output. Describe the whole image and ALL inputs producing an arbitrary reachable output. Explain why there cannot be an inverse on the plane.

Solution: $v=(4,2)=2u$, so $A_4(x,y)=(x+2y)(2,1)$. Its image is the line $X=2Y$. Output $(2s,s)$ has exactly the inputs $(x,y)=(s-2t,t)$, $t \in \mathbb{R}$. For example $(0,0)$ and $(-2,1)$ both map to 0. The distinct-input collision precludes an inverse; no output off that line is reachable.

2 / hints to offer only after exploration

1. Express the second arrow as a multiple of the first.

Student page 2: Turn the picture into a test

Proof gate: dot products and invertibility

For vectors $p=(p_1,p_2), q=(q_1,q_2)$, the dot product is $p \cdot q = p_1q_1 + p_2q_2$. Supplied geometry fact: projection onto a unit normal gives signed perpendicular height. The left normal to a nonzero vector (a,c) is $(-c,a)/\sqrt{a^2+c^2}$.

3 / prompt and solution

For any real matrix $M=[[a,b],[c,d]]$, prove that the signed area of its ordered image square is $ad-bc$ and its ordinary area is $|ad-bc|$. Include the case $(a,c)=(0,0)$. Explain exactly when the image collapses.

Solution: If $u=(a,c) \neq 0$, the signed height of $v=(b,d)$ above the directed base is $v \cdot (-c,a)/||u|| = (ad-bc)/||u||$. Multiplying by base length $||u||$ gives signed area $ad-bc$; taking absolute value gives ordinary area. If $u=0$, the image is a segment or point and $ad-bc=0$. Otherwise the height is 0 exactly when v is parallel to u . Thus collapse is exactly linear dependence of the two columns, equivalently determinant 0.

Can you preserve area without preserving shape?

3 / hints to offer only after exploration

1. Keep the sign of the height until the last step.

4 / prompt and solution

Find ALL real k for which A_k preserves ordinary area of parallelograms. For each, give an explicit inverse and its orientation. Prove that every other k except the collapse value is still invertible.

Solution: $\det A_k = 4 - k$. Ordinary area scale 1 requires $|4 - k| = 1$, so $k = 3$ or 5 . The inverse for $k \neq 4$ is $(1/(4 - k))[[2, -k], [-1, 2]]$, verified by multiplication. For $k = 3$ this is $[[2, -3], [-1, 2]]$, preserving orientation. For $k = 5$ it is $[[2, 5], [1, -2]]$, reversing orientation. Every $k \neq 4$ has this inverse; $k = 4$ is excluded by the earlier collision. The same scale for any parallelogram follows from the composition identity proved next, or the same base-height calculation applied to its columns.

4 / hints to offer only after exploration

1. Area 1 permits either sign; invertibility only requires a nonzero determinant.

Student page 3: Compose, then test rigidity

Continuation: matrix products and a length identity

Write $\det M = ad - bc$ for $M = [[a, b], [c, d]]$. A shear $S_t(x, y) = (x + ty, y)$ has matrix $[[1, t], [0, 1]]$. In $A_3 S_t$, the shear acts first.

5 / prompt and solution

Choose a nonzero t and compare $A_3 S_t$ with $S_t A_3$ on the two coordinate arrows. Prove for arbitrary real 2×2 matrices M, N that $\det(MN) = \det M \cdot \det N$, then explain why both compositions preserve area even when they differ.

Solution: $A_3 = [[2, 3], [1, 2]]$. $A_3 S_t = [[2, 2t + 3], [1, t + 2]]$, whereas $S_t A_3 = [[2 + t, 3 + 2t], [1, 2]]$; for nonzero t they differ in their first columns. Example $t = 1$ gives $[[2, 5], [1, 3]]$ and $[[3, 5], [1, 2]]$, each determinant 1. With $N = [[e, f], [g, h]]$, direct expansion of $(ae + bg)(cf + dh) - (af + bh)(ce + dg)$ cancels $acef$ and $bdgh$ and leaves $(ad - bc)(eh - fg)$. This includes singular cases and proves scale multiplication. Signed area of any parallelogram with column matrix N changes by $\det M$.

5 / hints to offer only after exploration

1. Write all four product entries before expanding.
2. Check the order with the horizontal unit input.

Can you preserve area without preserving shape?

6 / prompt and solution

Area preservation alone does not preserve every length: demonstrate that with A_3. Now decide whether a linear map can preserve ordinary area AND the lengths of both unit coordinate arrows, yet fail to preserve some other length. Prove your answer.

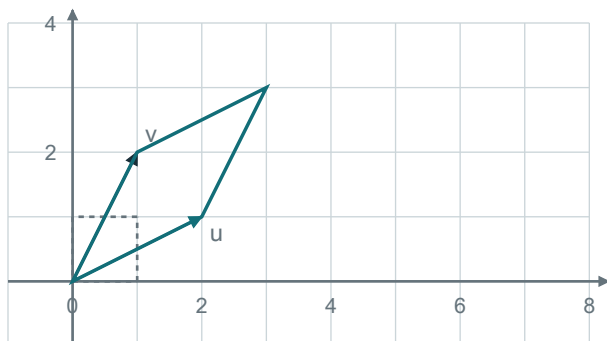
Solution: A_3 sends (1,0) to (2,1), length $\sqrt{5}$ rather than 1. But the proposed stronger failure is impossible. Let image columns u, v each have length 1 and $|\det(u, v)| = 1$. Expanding gives $\det(u, v)^2 = \|u\|^2\|v\|^2 - (u \cdot v)^2$, so $u \cdot v = 0$. Then $\|xu + yv\|^2 = x^2\|u\|^2 + 2xy(u \cdot v) + y^2\|v\|^2 = x^2 + y^2$ for every x, y . Thus all lengths are preserved; either orientation is possible.

6 / hints to offer only after exploration

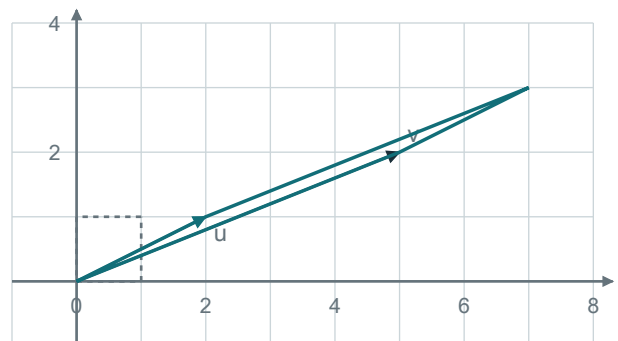
1. Expand the square of the determinant and compare with the product of squared lengths.

The chosen image squares

k = 1; signed area 3



k = 5; signed area -1



The same base $u = (2, 1)$ has signed heights $3/\sqrt{5}$ and $-1/\sqrt{5}$. The second traversal reverses orientation. Both axes use the same scale.

Can you preserve area without preserving shape?

Optional facilitator extension 1

Gate: Integer matrices and determinant multiplication.

Among integer matrices M , prove that M maps $\mathbb{Z}^2$ bijectively onto $\mathbb{Z}^2$ exactly when $\det M = \pm 1$.

Solution: If $\det M = \pm 1$, the displayed 2×2 inverse has integer entries, so M and its inverse carry integer vectors to integer vectors. Conversely, a bijection of $\mathbb{Z}^2$ has preimages of the two unit vectors; place them in an integer matrix N , giving $MN = I$. Hence integer $\det M$ times integer $\det N$ is 1, forcing $\det M = \pm 1$. This concerns bijectivity on the lattice, not merely drawing an integer parallelogram.

Mathematics and prior use

The determinant measures oriented area and detects loss of a direction. The Gram identity explains when area and two preserved edge lengths force a rigid linear map.

Prior use: The original AD-18 compares a unit shear, reflection and coordinate projection. This instance uses an oblique image family and a constrained design problem; Week 1 shape work is preparation, not this linear-map classification. AD-17 studies iteration instead of geometric distortion.

Source: Sheldon Axler, *Linear Algebra Done Right*, fourth edition

§7F7.108-7.111, printed pp. 291-293 / PDF pp. 305-307; §9C and 9.61, printed p. 363 / PDF p. 377. Use the linear volume-scaling framework. The worksheet proves its planar signed-area, inverse and composition formulas directly rather than appealing to singular-value theory. Evidence: Read the current official volume discussion and determinant-volume statement on 2026-09-26; the oblique family and projection calculation are independent.

<https://linear.axler.net/LADR4e.pdf>

Choose bases that reveal what an arrow does

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist with a genuine linear-algebra gate. Arrow cards can organize the proof but do not replace vector-space prerequisites.

An explicit basis decomposition with killed, transmitted and unreachable directions, plus a proof that these three counts completely classify one linear arrow under independent bases.

Prerequisites and preparation

Core: Finite-dimensional vector spaces over $\mathbb{Q}$, linear independence, bases, kernel, image and rank. The general theorem works over any field. This is an advanced linear-algebra investigation.

Extensions: Composition and ranks for the two-arrow counterexample.

Reading and representation: Read coordinate vectors and distinguish a basis vector from a scalar coordinate. Independent choices on the two sides are part of the equivalence rule.

Materials: Printed linear-map rule, source/target vector cards and open basis ledger; pencil. Optional colored arrows represent basis directions, not individual points of a finite graph. Preparation: about 6 minutes.

10 minutes invent collisions and test targets; 20 choose and check bases; 20 prove the block inventory; 10 test the limit of the equivalence rule. General proof can be a separate meeting.

Launch and facilitation

The machine has four inputs and three outputs. Choose different inputs that collide, and a target you believe can never appear. Now invent coordinates that make the machine's behavior immediately visible.

Student page 1: Try to hide and recover directions

Advanced core: exact linear algebra

Over $\mathbb{Q}$, let $T: \mathbb{Q}^4 \rightarrow \mathbb{Q}^3$ be $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_3 + x_4, x_1 + x_2 + x_3 + x_4)$. You may choose a basis independently in the source and target. A basis must be proved independent and spanning.

1 / prompt and solution

Invent two distinct inputs with the same output. Decide whether $(3, -2, 1)$ and $(3, -2, 0)$ are reachable. Then describe ALL inputs giving any reachable target.

Solution: The first two output coordinates may be arbitrary a, b and the third must be $a+b$. Thus $(3, -2, 1)$ is reachable, for example from $(3, 0, -2, 0)$, while $(3, -2, 0)$ is not. All preimages of $(a, b, a+b)$ are $(a-s, s, b-t, t)$, $s, t \in \mathbb{Q}$. For example $(0, 0, 0, 0)$ and $(1, -1, 0, 0)$ collide at 0. This also describes the two-parameter kernel.

Choose bases that reveal what an arrow does

1 / hints to offer only after exploration

1. Compare the third output with the first two.
2. Keep one free choice inside each input-coordinate pair.

2 / prompt and solution

Choose source and target bases so the matrix has as few nonzero entries as possible. Show each basis vector's image, prove both lists are bases, and explain why fewer nonzero entries cannot suffice.

Solution: Take source $u_1=e_1, u_2=e_3, k_1=e_2-e_1, k_2=e_4-e_3$. Their coefficients recover any input uniquely: $x=(x_1+x_2)u_1+(x_3+x_4)u_2+x_2k_1+x_4k_2$. Take target $w_1=(1,0,1), w_2=(0,1,1), w_3=(0,0,1)$; target $(a,b,c)=a w_1+b w_2+(c-a-b)w_3$ uniquely. Images are $w_1, w_2, 0, 0$, so the matrix is $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Rank is 2, while a matrix with fewer than 2 nonzero entries has rank at most 1, proving minimality. Other checked bases are equally valid.

2 / hints to offer only after exploration

1. Use independent image directions as the first target vectors.
2. Complete the source with kernel vectors.

Student page 2: Prove the inventory does not depend on your choices

General proof: basis extension over a field

A block $k \rightarrow 0$ is one killed source direction; $k \rightarrow k$ with identity is one transmitted direction; $0 \rightarrow k$ is one unreachable target direction. A direct sum places blocks side by side in both source and target. A nonzero representation is indecomposable if it is not a direct sum of two nonzero representations.

3 / prompt and solution

For ANY linear map $V \rightarrow W$ between finite-dimensional spaces over a field k , construct bases giving only these blocks. Prove your chosen transmitted source vectors really are independent modulo the kernel, and account for every direction.

Solution: Choose a basis $w_1, \dots, w_r$ of the image and choose v_i with $Tv_i=w_i$. Add a kernel basis $k_1, \dots, k_a$. If $\sum c_i v_i + \sum d_j k_j = 0$, apply T : image independence forces all $c_i=0$; kernel independence forces all $d_j=0$. Any v has $Tv = \sum c_i w_i$; subtracting $\sum c_i v_i$ leaves a kernel vector, proving spanning. Extend the image basis to a target basis by b further vectors. The matrix has r identity entries and zeros elsewhere. Thus $a = \dim \ker T$, $r = \text{rank } T$, $b = \dim W - r$; this includes zero spaces and the zero map.

Choose bases that reveal what an arrow does

3 / hints to offer only after exploration

1. Apply T to a proposed linear dependence.
2. Subtract a vector with the same image to land in the kernel.

4 / prompt and solution

Why are the three displayed block types indecomposable? Prove that any two decompositions have the same number of each type. Do equal inventories also guarantee equivalence of two one-arrow representations?

Solution: For $k \rightarrow 0$ and $0 \rightarrow k$, the sole 1-dimensional space cannot split into two nonzero contributions. For identity $k \rightarrow k$, a nontrivial split would have to put source and target in separate blocks $k \rightarrow 0$ and $0 \rightarrow k$, whose map is zero, contradicting rank 1. Any decomposition has transmitted count r , killed count $\dim V - r$ and unreachable count $\dim W - r$, all invariant under independent invertible bases. Conversely equal counts give the same block matrix after choosing bases by prompt 3; compose these basis isomorphisms to obtain an equivalence. Thus the inventory is both necessary and sufficient.

4 / hints to offer only after exploration

1. Check dimensions on both sides AND rank.

Student page 3: Design an arrow and test the rule's boundary

Continuation: classification versus similarity

Keep source Q^4 and target Q^3 . Later in this page, a SAME-basis change for an endomorphism means replacing a square matrix M by $P^{-1}MP$, with the same invertible P on both sides.

5 / prompt and solution

Design a map $Q^4 \rightarrow Q^3$ with exactly one transmitted direction. Give a basis certificate. Then list every possible killed/transmitted/unreachable inventory for these fixed dimensions and construct a map for each.

Solution: One design is $S(x) = (x_1 + x_2, 2x_1 + 2x_2, 0)$. It has image $\text{span}(1, 2, 0)$, with source e_1 followed by $e_2 - e_1, e_3, e_4$, and target $(1, 2, 0), (0, 1, 0), (0, 0, 1)$. Its inventory is $(3, 1, 2)$. In general $0 \leq r \leq 3$ and the inventory is $(4 - r, r, 3 - r)$, giving $(4, 0, 3), (3, 1, 2), (2, 2, 1), (1, 3, 0)$. Maps $(x_1, \dots, x_4) \mapsto (x_1, \dots, x_r, 0, \dots, 0)$ attain each. No other inventory is possible by prompt 3.

5 / hints to offer only after exploration

1. Rank cannot exceed either dimension; a diagonal rectangular matrix realizes every allowed rank.

Choose bases that reveal what an arrow does

6 / prompt and solution

Someone applies your theorem to say I_2 and $2I_2$ become equal under a SAME-basis change. Audit this claim. Show how independent source/target bases can identify the arrows, and why a same-basis change cannot.

Solution: Over $\mathbb{Q}$ both maps have rank 2 and independent-basis inventory $(0,2,0)$. For $2I$, keep source e_1, e_2 and take target $2e_1, 2e_2$; its matrix becomes I . But $P^{-1} I P = I$ for every invertible P , so I cannot become $2I$ through a same-basis change. Equivalence of one arrow permits independent coordinates and is a different problem from similarity of an endomorphism.

6 / hints to offer only after exploration

1. Compute $P^{-1}IP$ before trying any specific P .

Optional facilitator extension 1

Gate: Composition, rank and compatible basis changes at a shared intermediate space.

Do the separate ranks of two arrows determine a chain $Q \rightarrow Q^2 \rightarrow Q$? Construct two chains with both arrows of rank 1 that cannot be related by independent bases at the three vertices.

Solution: Use $A(t)=(t,0)$. In the first chain $B(x,y)=x$, so BA is identity of rank 1; in the second $C(x,y)=y$, so $CA=0$ of rank 0. Both individual ranks are 1, but the composite rank is preserved under a chain isomorphism: internal changes cancel and only invertible end changes remain. Therefore these chains are inequivalent. This establishes a limitation, not a full two-arrow classification.

Mathematics and prior use

One-arrow quiver representations are classified by domain dimension, target dimension and rank, using independent basis changes. Their three indecomposable types follow from elementary basis extension.

Prior use: The original AD-19 map $Q^3 \rightarrow Q^2$ has no unreachable target block. The new $Q^4 \rightarrow Q^3$ instance has all three types and asks learners to design bases and test the classification scope. Week 2/AD-13 image-kernel work is preparatory; the new destination is representation decomposition.

Source: Pavel Etingof et al., Introduction to Representation Theory, 2011 notes

§1.8 quiver representations; §5.2 Example 5.8, printed p. 81 / PDF p. 81. Use the one-arrow direct-sum framework. The $Q^4 \rightarrow Q^3$ map, explicit basis certificate and design/comparison tasks are new. The proof uses only field-linear basis extension. Evidence: Read the decomposition of kernel, image and complementary target in Example 5.8 on 2026-09-26; independently construct and verify the new bases.

<https://math.mit.edu/~etingof/replect.pdf>

Find what commutes, then explain the cancellations

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. Exact matrix trials are meaningful learner choices for those with the algebra gate; they are not advertised as an elementary movement substitute.

A complete commuting-partner family, a concrete nonassociativity witness and a general Jacobi proof by cancellation of ordered triple products.

Prerequisites and preparation

Core: Rational 2×2 matrix multiplication, subtraction and symbolic algebra. General identities require distributivity and associativity of multiplication, not assumptions about associativity of the bracket.

Extensions: Trace in arbitrary finite dimension for the operator obstruction.

Reading and representation: Products act on column vectors from right to left. Brackets are a newly defined operation and must retain their parentheses.

Materials: Printed matrices N, C , blank product and ordered-word ledgers; pencil. No specialized manipulatives. Preparation: about 5 minutes.

10 minutes choose commuting and noncommuting trials; 15 classify the commuting partners; 20 audit bracket identities; 15 prove Jacobi and classify the center. Advanced pages may be separated.

Launch and facilitation

Choose a matrix B and predict whether applying B before N has the same effect as applying N before B . Use a vector to expose a difference, then search for every B that hides the order.

Student page 1: Choose a partner for one matrix

Advanced core: products on column vectors

Work over $\mathbb{Q}$. Let $N = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. For any matrices A, B define $[A, B] = AB - BA$. Matrix multiplication acts right to left: ABv means apply B first, then A .

Find what commutes, then explain the cancellations

1 / prompt and solution

Choose a nonscalar matrix B that commutes with N , and another that does not. Certify the first by full products and expose the second with an explicit input vector. Check N^2 as a useful clue.

Solution: $N^2=0$. A nonscalar commuting choice is $B=N+I=\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$; both NB and BN equal N . A noncommuting choice is C : $NC=\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$ and $CN=\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. On $e_1=(1,0)$, $NC e_1=(1,-1)$ while $CN e_1=(1,0)$. Any proposed examples are checked by exact products; scalar multiples of I alone do not meet the first request.

1 / hints to offer only after exploration

1. Build a candidate from N and I .
2. For a nonzero difference matrix, one of its columns gives a witness vector.

2 / prompt and solution

Classify EVERY $B=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $[N,B]=0$. Express your answer using I and N , and explain why a few successful trials would not prove completeness.

Solution: $NB-BN=\begin{bmatrix} b+c & 2b+d-a \\ d-a-2c & -b-c \end{bmatrix}$. Vanishing is equivalent to $c=-b$ and $d=a-2b$; after these equations the other entries vanish automatically. Therefore $B=(a-b)I+bN$, with arbitrary rational a,b , or equivalently $sI+tN$. This proves necessity and sufficiency. Trials suggest a family but do not rule out other entries.

2 / hints to offer only after exploration

1. First use a diagonal entry of the difference, then the upper-right one.

Student page 2: Audit the new operation

Proof gate: distinguish two operations

Keep ordinary multiplication and the bracket separate. The trace of a square matrix is the sum of its diagonal entries. A Lie algebra bracket is bilinear, has $[A,A]=0$, and obeys the Jacobi identity studied on the next page.

Find what commutes, then explain the cancellations

3 / prompt and solution

Prove bilinearity and $[A,A]=0$ for arbitrary rational 2×2 matrices. Prove also that every commutator has trace 0. Must a zero bracket mean either matrix is zero?

Solution: For example $[A+B,D]=(A+B)D-D(A+B)=[A,D]+[B,D]$, and $[qA,D]=q[A,D]$; the second argument is analogous. $[A,A]=AA-AA=0$ and $[B,A]=-[A,B]$. For $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B=\begin{bmatrix} e & f \\ g & h \end{bmatrix}$, traces of AB and BA are $ae+bg+cf+dh$ and $ea+fc+gb+hd$, equal over $\mathbb{Q}$, so $\text{trace } [A,B]=0$. A zero bracket does not force either matrix to vanish: I commutes with every B , including nonzero N .

3 / hints to offer only after exploration

1. Expand one identity without changing factor order.
2. Pair scalar terms in the two diagonal sums.

4 / prompt and solution

Compare $[[N,C],C]$ with $[N,[C,C]]$. Use the result to decide whether the bracket is associative. Explain why this does not contradict associativity of ordinary matrix multiplication.

Solution: $[N,C]=\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$. Bracketing this with C gives $[[0,1],[-1,0]]$, while $[C,C]=0$ and $[N,0]=0$. Hence the bracket is not associative. Ordinary multiplication remains associative; nested brackets contain sums and subtractions of differently ordered products, and are a distinct operation.

4 / hints to offer only after exploration

1. Compute $[C,C]$ before multiplying any longer expression.

Student page 3: Why every Jacobi sum vanishes

General proof: ordered triple products

For arbitrary matrices A,B,D of the same size, the Jacobi expression is $[[A,B],D]+[[B,D],A]+[[D,A],B]$. Ordinary products remain in their original order; associativity lets us omit parentheses inside each three-letter product.

Find what commutes, then explain the cancellations

5 / prompt and solution

Prove the Jacobi identity for ALL rational square matrices by expanding the expression and pairing cancellations. Then prove the related identity $[A, BD] = [A, B]D + B[A, D]$. State where associativity of multiplication is used.

Solution: The three expansions are $ABD - BAD - DAB + DBA$; $BDA - DBA - ABD + ADB$; $DAB - ADB - BDA + BAD$. Every ordered word cancels with its opposite, giving 0 in every dimension. Associativity identifies, for example, $(AB)D$ with $A(BD)$; factors are never commuted. For the second identity, the right side is $(AB - BA)D + B(AD - DA) = ABD - BAD + BAD - BDA = ABD - BDA = [A, BD]$. Checking a single triple would not establish either general identity.

5 / hints to offer only after exploration

1. Treat ABD and BAD as different words.
2. Each expansion has four terms; cancel only identical words.

6 / prompt and solution

Call a matrix central if it commutes with EVERY rational 2×2 matrix. Prove that the central matrices are exactly scalar multiples of I . Use only the tests with N and C for necessity, then verify sufficiency. What is the center inside the trace-zero matrices?

Solution: Commuting with C forces the off-diagonal entries b, c to be 0. The earlier N -centralizer condition then forces $d = a$. Thus any central matrix is aI . Conversely aI commutes with every matrix by scalar multiplication. Inside the trace-zero subspace, $\text{trace}(aI) = 2a$, so over $\mathbb{Q}$ only $a = 0$ remains. Commuting with N alone is weaker: N itself passes that test but fails the C test. For the center of the trace-zero Lie algebra itself, use the trace-zero tests N and $C - (1/2)I$. The latter has the same commutator with any matrix as C , so these tests again force a scalar matrix; trace zero then forces that scalar to be zero.

6 / hints to offer only after exploration

1. Combine two necessary tests; do not mistake one centralizer for the whole center.

Find what commutes, then explain the cancellations

Optional facilitator extension 1

Gate: Finite sums and trace; polynomial differentiation only for the optional contrast.

Can finite rational square matrices A, B satisfy $[A, B] = I$? Prove your answer in any positive dimension n .

Solution: No. In dimension n , $\text{trace}(AB) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \text{trace}(BA)$, so every commutator has trace 0. But $\text{trace } I = n \neq 0$ over $\mathbb{Q}$. This finite-dimensional obstruction is compatible with infinite-dimensional polynomial operators D and multiplication by X satisfying $[D, M_X] = I$; the finite trace argument does not apply there.

Mathematics and prior use

Associative matrix multiplication induces a bilinear alternating bracket satisfying Jacobi. Centralizers and the scalar center distinguish commutation with one operator from commutation with all.

Prior use: The original AD-20 uses the E, F, H bracket table. The new entry uses a different nilpotent N and a centralizer design problem; the structural identity is shared and acknowledged. Week 3 composition prepares order conventions but does not supply matrix subtraction or a Lie-algebra proof.

Source: Pavel Etingof et al., Introduction to Representation Theory, 2011 notes

§1.9 Definition 1.39 and Example 1.40, printed p. 15 / PDF p. 15; displayed $\mathfrak{sl}(2)$ example on that page. Use the associative-commutator and Lie-algebra framework. The N -centralizer, repeated- C associativity witness and center proof are derived here. Evidence: Read the definition, associative-algebra example and trace-zero matrix discussion on 2026-09-26; ordered-product expansion and chosen matrices are independently verified.

<https://math.mit.edu/~etingof/replect.pdf>

Make a catalog that remembers every choice

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the finite catalog; promising with function-language gates for the universal proof. The counted catalog alone is not the full categorical destination.

A complete compatible-pair catalog and a proof that every compatible assignment has exactly one record-preserving map into it; duplicate records then explain failure of uniqueness.

Prerequisites and preparation

Core: Named objects, ordered pairs and compatible assignments. The universal-property proof needs functions and separate existence/uniqueness arguments; it can be explained with tokens before formal notation.

Extensions: Finite counting and fiber sizes for the projection/count extension.

Reading and representation: Color names accompany all marks so color vision is unnecessary. Reusing a catalog entry for different tokens is permitted; no one-to-one matching rule is imposed.

Materials: Printed person and badge cards with names and color words, reusable token markers, blank catalog area and projection arrows. Preparation: about 8 minutes.

10 minutes explore pairings; 15 construct and check a catalog; 10 optimize one recoloring; 15 send token assignments through it; 15-20 prove the universal and uniqueness claims.

Launch and facilitation

Choose one person and one badge of the same color. Design a catalog that remembers exactly which person and which badge were chosen. Challenge a partner with repeated names and repeated colors.

Student page 1: Build and improve the catalog

Core: named choices and color agreement

People: Aster and Birch are red; Cedar and Dahlia are blue; Elm is green. Badges: r,s are red; t is blue; u,v are green; w is gold. A legal entry is an ordered (person,badge) pair with matching colors. Names remain distinct even when colors agree.

Make a catalog that remembers every choice

1 / prompt and solution

Choose a catalog layout and record EVERY legal entry, once each. Prove completeness. Identify any person or badge that appears in no entry. Can one slot per color preserve both names?

Solution: The entries are (Aster,r),(Aster,s),(Birch,r),(Birch,s),(Cedar,t),(Dahlia,t),(Elm,u),(Elm,v). These are all pairs within the red, blue, green color fibers, hence complete, with 8 entries. Every person appears; only badge w appears nowhere because no person is gold. A slot storing only a color merges distinct person/badge pairs, so it cannot recover both names.

1 / hints to offer only after exploration

1. Choose one person at a time, but retain every badge of that color.

2 / prompt and solution

You may recolor ONLY badge w to red, blue or green. Choose a color to make the catalog as large as possible. Find every best choice and certify that no permitted recoloring can do better.

Solution: Recoloring w adds one pair for each person of its new color; it removes none because w was unused. Red and blue each add 2 pairs, giving 10, while green adds 1, giving 9. Thus red and blue are exactly the two maximizing choices. This compares all three permitted colors, so it proves optimality.

2 / hints to offer only after exploration

1. Count the people in the chosen color, not the badges already there.

Student page 2: Send tokens without losing either record

Continuation: functions and forced placements

Return w to gold. A token can carry one person name and one badge name. Tokens may repeat the same assignments, and several tokens may land in the same catalog entry. A placement must preserve both names.

3 / prompt and solution

Create compatible assignments for at least three tokens, including two tokens with identical records. Place them in your catalog and explain why the placements are forced. Then create an incompatible token assignment and explain exactly why no placement exists.

Solution: For example x_1, x_2 both carry (Aster,r), and x_3 carries (Elm,v); the first two map to the same entry and the third to (Elm,v). Equal assignments need not have different destinations. Every compatible token's names specify one existing entry, so existence and uniqueness hold. An incompatible example (Cedar,u) has blue/green colors; no entry preserves both names. Any learner-chosen examples follow the same test.

Make a catalog that remembers every choice

3 / hints to offer only after exploration

1. No rule says the token-to-catalog function must be injective.

4 / prompt and solution

For arbitrary sets A, B, C and functions $f: A \rightarrow C, g: B \rightarrow C$, define $P = \{(a, b) : f(a) = g(b)\}$. Its projections return the first and second records. Given any set X and functions $u: X \rightarrow A, v: X \rightarrow B$ with $f \circ u = g \circ v$, prove there is EXACTLY ONE $h: X \rightarrow P$ with projections u, v . Include X empty.

Solution: For every $x \in X$ set $h(x) = (u(x), v(x))$. The compatibility equation makes this pair a member of P , proving existence, and its projections are u, v . Any competing h' with those projections must have first component $u(x)$ and second component $v(x)$, so $h'(x) = h(x)$ for all x . Thus $h' = h$. If X is empty, the unique empty function gives the same conclusion, even when P is empty.

4 / hints to offer only after exploration

1. First show the proposed pair belongs to P ; only then prove it is forced.

Student page 3: Test a rival catalog

Proof continuation: uniqueness of the object

A rival catalog Q has two record maps $q_A: Q \rightarrow A, q_B: Q \rightarrow B$ whose colors agree. It has the universal property if EVERY compatible token assignment has exactly one record-preserving map into Q . Merely containing a record for every compatible pair is not enough.

5 / prompt and solution

Start with your catalog P . Make Q by adding a second, distinct copy of $(Aster, r)$, with identical person and badge records. Which part of the universal property fails? Give the smallest token test. Also explain what fails if the original $(Aster, r)$ entry is removed instead.

Solution: For the duplicate catalog, a one-token set $X = \{x\}$ assigned $Aster$ and r has two possible placements, so uniqueness fails although every compatible assignment can still be placed. If the original entry is removed without a replacement, the same one-token assignment has no placement, so existence fails. An empty token set cannot detect either fault because it has one empty map to any catalog.

5 / hints to offer only after exploration

1. Test one token whose records pick the altered pair.

Make a catalog that remembers every choice

6 / prompt and solution

Suppose a rival Q really does have the universal property. Prove that there is exactly one bijection $P \rightarrow Q$ preserving both records. Prove it is a bijection using the two universal properties, without comparing cardinalities.

Solution: Apply Q 's property to $X=P$ with P 's projections to get a unique $h:P \rightarrow Q$. Apply P 's property to $X=Q$ with q_A, q_B to get $k:Q \rightarrow P$. The composites kh and id_P both preserve P 's two records; uniqueness for P forces $kh = \text{id}_P$. Likewise $hk = \text{id}_Q$. Thus h is bijective with inverse k . Any other record-preserving map $P \rightarrow Q$ equals h by Q 's uniqueness, so in particular no other such bijection exists. No finiteness assumption is used.

6 / hints to offer only after exploration

1. Build a map each way, then compare each composite with the identity.

Optional facilitator extension 1

Gate: Finite products, disjoint sums and the definition of a surjective function.

For finite inventories, prove the catalog count is $\sum_c |A_c| |B_c|$. Characterize when each projection is onto its whole inventory.

Solution: Each compatible pair lies in exactly one disjoint fiber product $A_c \times B_c$, giving the sum. Projection $P \rightarrow A$ is onto exactly when $B_{\{f(a)\}}$ is nonempty for every $a \in A$; equivalently every color used by A is used by B . The other projection is onto exactly when every color used by B is used by A . Empty inventories obey these statements vacuously in the appropriate direction. In the original inventory the projection to people is onto and the projection to badges misses w .

Mathematics and prior use

The pullback of two maps to a common set is a set of compatible ordered pairs with a unique mediating function. Existence, uniqueness and uniqueness of the representing object are separate arguments.

Prior use: The original AD-21 uses a four-pair color catalog. This larger inventory has repeated fibers on both sides, an optimization choice and a duplicate-record audit. Week 5 combines constraints; the universal mediator and uniqueness up to record-preserving isomorphism remain the new categorical content.

Make a catalog that remembers every choice

Source: Brendan Fong and David Spivak, Seven Sketches in Compositionality

§3.5.3 Theorem 3.95 and Example 3.99, printed pp. 112-113 / PDF pp. 124-125. Use the set-pullback and colored-pair framework. Invent a new inventory, recoloring optimization and duplicate-record/unique-isomorphism tests. Evidence: Read the finite-limit tuple formula and pullback example on 2026-09-26; independently enumerate the new catalog and prove its mediator property.

https://ocw.mit.edu/courses/18-s097-applied-category-theory-january-iap-2019/a4175d61479a35340d6307ae5e48ef5a_18-s097iap19textbook.pdf

Which face cards erase the same edge pattern?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for parity-ready learners at the finite game; the matrix/quotient continuation has an honest linear-algebra gate. No general topological-invariance claim is inferred.

A complete classification by two preserved outer-edge marks, with constructive erasing sequences and a minimum face-card purchase proof.

Prerequisites and preparation

Core: Parity, edge toggles and a completeness argument. Formal homology uses F_2 vector spaces, boundary maps and quotients; the finite reachability proof can precede that notation.

Extensions: Kernel in degree 2 for a supplied cap-face extension.

Reading and representation: Names label graph vertices and triangular faces. O is a genuine center vertex where all spokes meet. A drawn triangle is not an available move unless its face card is listed.

Materials: Printed square-with-center graph, eight edge markers, face cards OAB,OBC,OCD,ODA in a separate style, blank copies for patterns. Preparation: about 8 minutes.

10 minutes freely toggle; 15 classify closed patterns; 20 sort reachability classes; 15 choose extra face cards and certify minimality. Boundary-matrix interpretation may be another meeting.

Launch and facilitation

Choose marked edges with even incidence at every vertex. Initially only the opposite face cards OAB and OCD can be played. Try to erase a partner's pattern, then invent a quick certificate for a failed attempt.

Student page 1: Make a closed pattern

Core: parity on an exact graph

Vertices A,B,C,D form a square in that order; O is its center. Edges are AB,BC,CD,DA,OA,OB,OC,OD. O is a vertex, not an overpass. A pattern marks any subset of edges and is CLOSED when each vertex meets an even number of marked edges. Toggling means marked $\leftrightarrow$ unmarked. Initially the only available face moves toggle the three edges of OAB or OCD.

Which face cards erase the same edge pattern?

1 / prompt and solution

Choose a nonempty closed pattern other than a single triangle boundary. Check every vertex. Try each available face move and explain why any legal sequence keeps the pattern closed.

Solution: The outer square $AB+BC+CD+DA$ is one example: each outer vertex has degree 2 and O has degree 0. Either available face has two incident boundary edges at each of its three vertices, so toggling it changes each incidence parity by $0 \bmod 2$. Other vertices are untouched. Therefore every face move preserves closedness, and induction gives the same for every sequence. Learner choices are accepted exactly when every incidence is even.

1 / hints to offer only after exploration

1. Count selected incident edges, not all drawn edges.

2 / prompt and solution

Find EVERY closed pattern without checking all subsets separately. Let a,b,c,d be the marks on AB,BC,CD,DA . What must the four spoke marks be? Prove every choice of a,b,c,d gives exactly one closed pattern.

Solution: At A , $OA=a+d$; at B , $OB=a+b$; at C , $OC=b+c$; at D , $OD=c+d$, all modulo 2. Their sum at O is $2(a+b+c+d)=0$, so O imposes no new constraint. Thus every choice of four bits gives a unique closed pattern, 16 total. If T_1, T_2, T_3, T_4 denote boundaries of OAB, OBC, OCD, ODA , then the pattern is $aT_1+bT_2+cT_3+dT_4$; the outer edge of each triangle proves independence.

2 / hints to offer only after exploration

1. Solve the parity equations at the outer vertices first.

Student page 2: Classify what the available cards can change

Core proof: an invariant and a construction

Keep only face cards OAB and OCD available. Two closed patterns count as equivalent when some legal sequence turns one into the other. A card may be used repeatedly; twice has no net effect.

Which face cards erase the same edge pattern?

3 / prompt and solution

Classify ALL equivalence classes of closed patterns. Give both a certificate preventing moves between different classes and a procedure connecting any two patterns in the same class. Do not stop after finding one loop that cannot be erased.

Solution: OAB toggles a and OCD toggles c in the four-bit representation. Neither changes b or d , which are the marks on BC and DA. Thus (b,d) is invariant and has four possible values. Conversely, if two patterns share b,d , toggle OAB exactly when their a values differ and OCD exactly when c differs. This connects them. Therefore the four classes are $\{bT_2+dT_4+aT_1+cT_3 : a,c \in F_2\}$ indexed by $(b,d) \in F_2^2$; each has 4 patterns. The erasable class is $b=d=0$.

3 / hints to offer only after exploration

1. Look for outer edges that no available move touches.
2. An invariant needs a converse construction to prove completeness.

4 / prompt and solution

Choose a target to erase: the outer square, or the closed loop $AB+BC+OC+OA$. You may buy missing face cards OBC and ODA. Find ALL minimum purchases for each target, with a legal sequence and a reason fewer cards cannot work.

Solution: The outer square is $T_1+T_2+T_3+T_4$, so both OBC and ODA must be bought: its BC mark can change only via OBC and its DA mark only via ODA. Playing all four cards erases it, so the unique minimum purchase is both missing cards. The second loop is T_1+T_2 , so OBC alone is the unique minimum purchase; play OAB and OBC. Buying ODA alone cannot change its BC mark. These arguments prove necessity and sufficiency, not merely a successful attempt.

4 / hints to offer only after exploration

1. Which card is the only one touching each marked outer edge?

Student page 3: Name the quotient you have computed

Advanced: F_2 boundaries and homology

Over F_2 , C_0 has the five vertices as basis, C_1 the eight edges, and C_2 only the available filled triangles OAB, OCD. The boundary ∂_1 of an edge is the sum of its endpoints; ∂_2 of a triangle is the sum of its boundary edges. Define $H_1 = \ker \partial_1 / \text{im } \partial_2$ once you have proved $\text{im } \partial_2 \subseteq \ker \partial_1$. Use vertex order A,B,C,D,O; edge order AB,BC,CD,DA,OA,OB,OC,OD; and face order OAB,OCD.

Which face cards erase the same edge pattern?

5 / prompt and solution

Write the two boundary matrices in the displayed basis orders and verify $\partial_1\partial_2=0$. Use your earlier classification to give bases and dimensions for $\ker \partial_1$ and $\text{im } \partial_2$. Identify H_1 explicitly and say what its zero element means in the game.

Solution: With vertex order A,B,C,D,O and edge order AB,BC,CD,DA,OA,OB,OC,OD, the rows of ∂_1 are: $A=(1,0,0,1,1,0,0,0)$, $B=(1,1,0,0,0,1,0,0)$, $C=(0,1,1,0,0,0,1,0)$, $D=(0,0,1,1,0,0,0,1)$, $O=(0,0,0,0,1,1,1,1)$. The columns of ∂_2 are $T_1=(1,0,0,0,1,1,0,0)$ and $T_3=(0,0,1,0,0,0,1,1)$. Each product column is 0 because triangle vertices appear twice. The kernel has basis T_1, T_2, T_3, T_4 by prompt 2; image has independent basis T_1, T_3 . Hence $H_1 \cong \mathbb{F}_2^2$ via the BC,DA marks. Its zero element is the whole class of erasable patterns, not just the empty drawing.

5 / hints to offer only after exploration

1. Each matrix column is the boundary of its named basis object.
2. Use the unique outer-edge coordinate of each triangle.

6 / prompt and solution

Now choose ANY subset S of the four possible triangular face cards. Classify the resulting quotient without recomputing a large matrix each time. Which cycle coordinates survive, and when can every closed pattern be erased?

Solution: The cycle basis $T_1, \dots, T_4$ is unchanged because the graph is unchanged. Available boundaries are exactly the distinct basis vectors indexed by S , so their span has dimension $|S|$. Two cycles are equivalent exactly when their coefficients agree outside S . Thus $H_1 \cong \mathbb{F}_2^{4-|S|}$, represented by the unfilled triangles' outer-edge coordinates. Every cycle is erasable exactly when all four cards are available. With no cards all 16 cycles remain distinct; intermediate cases have $2^{4-|S|}$ classes. A visible closed loop need not represent a nonzero class.

6 / hints to offer only after exploration

1. Changing face cards changes the boundary image, not the closed-pattern space.

Boundary matrices and quotient coordinates

Facilitator / worked certificate

 ∂_1

	AB	BC	CD	DA	OA	OB	OC	OD
A	1	0	0	1	1	0	0	0
B	1	1	0	0	0	1	0	0
C	0	1	1	0	0	0	1	0
D	0	0	1	1	0	0	0	1
O	0	0	0	0	1	1	1	1

 ∂_2

	OAB	OCD
AB	1	0
BC	0	0
CD	0	1
DA	0	0
OA	1	0
OB	1	0
OC	0	1
OD	0	1

The four triangular cycles T_1, T_2, T_3, T_4 form a kernel basis because their outer-edge coordinates are the four unit vectors. Available moves span T_1, T_3 . The unfilled-face coordinates b and d identify the four classes in F_2^2 .

The zero class consists of every pattern erasable using the available cards, not just the empty edge pattern. Buying OBC and ODA erases the outer square; buying OBC alone erases $T_1 + T_2$.

Optional facilitator extension 1

Gate: F2 kernels in degree 2 and the definition of homology.

Allow all four triangle cards and add a formal cap face attached along the outer square ABCD. There are no 3-dimensional cells. Compute $H_2 = \ker \partial_2$ and H_1 over F_2 for this enlarged chain complex.

Solution: The four triangle boundaries are independent in C_1 and the cap boundary equals their sum. For coefficients $a_1, \dots, a_4, e$, zero boundary means $a_i + e = 0$ for every i , so $\ker \partial_2$ is one-dimensional, generated by the sum of all five faces. With $C_3 = 0$ this is H_2 . The image in C_1 remains the full four-dimensional cycle space, so $H_1 = 0$. This is a calculation for the explicitly supplied attachment; the visible planar regions alone would not specify the extra cap.

Boundary matrices and quotient coordinates

Mathematics and prior use

Closed edge patterns are $\ker \partial_1$; legal triangle boundaries span $\text{im } \partial_2$. Their quotient is the exact move-equivalence set and has a concrete coordinate invariant.

Prior use: The original AD-22 has a square with one diagonal and four cycles. This new square-center complex has 16 cycles and opposite filled triangles, leading to a two-coordinate class invariant and a face-purchase problem. Week 2 toggles and AD-05 cycle language prepare the mechanism but do not include cycles modulo filled-face boundaries.

Source: The Stacks Project, Complexes

§12.13, chain-complex definition and the homology formula immediately after Lemma 12.13.3. Use the chain/boundary/quotient framework. The wheel graph, exact cycle parameterization and legal-face classification are independently derived over F_2 . Evidence: Read $d^2=0$ and $H_i=\ker d_i/\text{im } d_{i+1}$ on 2026-09-26; every edge pattern and face subset is independently checked.

<https://stacks.math.columbia.edu/tag/010V>

How much free space completes this module?

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

Advanced/specialist. Paired counters faithfully represent isomorphism classes only after the idempotent/module argument. The hard gate is preserved rather than hidden.

An explicit smallest free complement and a justified computation $K_0(Q \times Q) = \mathbb{Z}^2$, including a nonfree projective and a virtual class not represented by an actual module.

Prerequisites and preparation

Core: Finite-dimensional rational vector spaces, ring/module action, direct sums and isomorphism. Projective means a direct summand of a finite free module here. Counters alone are only a preparatory model.

Extensions: Abstract commutative monoids and cancellation for the warning example.

Reading and representation: Symbolic definitions are essential. Distinguish an actual module, its isomorphism class and a formal difference; negative coordinates are not negative-dimensional spaces.

Materials: Printed two-control definition card, paired-space diagrams, optional two-color counters and open complement/formal-difference ledgers. Preparation: about 8 minutes.

15 minutes choose modules and test controls; 20 prove splitting and construct free complements; 15 compare stable additions; 20 establish the universal group completion. This advanced family may use two meetings.

Launch and facilitation

Two independent rational knobs scale two components. Choose spaces with the same total dimension but different responses to the knobs. Then ask what smallest additional space makes each into a free module.

Student page 1: Let two controls distinguish equal totals

Advanced core: actual module action

Let $R = Q \times Q$ with componentwise ring operations and identity $(1, 1)$. On $M(a, b) = Q^a \oplus Q^b$, let $(r, s) \cdot (u, v) = (ru, sv)$. Here a, b are nonnegative integers. An R -linear isomorphism is an invertible additive map preserving this action. Set $e_1 = (1, 0), e_2 = (0, 1)$.

How much free space completes this module?

1 / prompt and solution

Choose two different dimension pairs with the same total dimension and test whether their modules can be R -isomorphic. Then compare $P=M(3,1)$ and $Q=M(1,3)$. Find a control-based invariant that proves your answer, rather than relying on the names of the two components.

Solution: For example $M(4,2)$ and $M(3,3)$ have equal total 6 but differ in the dimension of e_1M , respectively 4 and 3. Any R -linear isomorphism φ satisfies $\varphi(e_i m) = e_i \varphi(m)$, so it carries $e_i M$ bijectively onto $e_i N$ and preserves its rational dimension. P and Q therefore differ because their e_1 dimensions are 3 and 1 (also e_2 dimensions 1 and 3). Equal total dimension alone forgets the independent controls. If a learner chooses identical ordered pairs, the modules are isomorphic and the intended distinguishing choice should be revised.

1 / hints to offer only after exploration

1. Apply e_1 and e_2 separately; each kills one component.

2 / prompt and solution

For ANY finitely generated unital R -module M , prove $M = e_1 M \oplus e_2 M$ and show both summands are finite-dimensional rational vector spaces. Prove that the ordered pair of dimensions completely classifies M , including the converse construction of an R -isomorphism.

Solution: The idempotents satisfy $e_1 + e_2 = 1, e_1 e_2 = 0, e_i^2 = e_i$. Thus $m = e_1 m + e_2 m$. An element in both summands is fixed by e_1 and killed by e_1 , so the intersection is 0. Rational scalars act through (q,q) , making each summand a Q -space. If $m_1, \dots, m_n$ generate M over R , then $e_i m_j$ span $e_i M$ over Q , so its dimension is finite. On $e_1 M$ the action (r,s) is multiplication by r , and on $e_2 M$ it is multiplication by s . Choosing rational bases gives an R -isomorphism with $M(a,b)$. Conversely equal ordered dimensions permit separate rational linear isomorphisms on the components; their direct sum is R -linear. Necessity follows from the control invariant in prompt 1.

2 / hints to offer only after exploration

1. Use the identities of the two idempotents before choosing a basis.

Student page 2: Buy the smallest free complement

Advanced: free versus projective modules

A finite free R -module is $R^n, n \geq 0$. In this investigation “finitely generated projective” means a module that has a complement C with $M \oplus C \cong R^n$ for some finite n . Direct sum adds the two component dimensions.

How much free space completes this module?

3 / prompt and solution

Find the smallest n for which $P=M(3,1)$ has a complement in R^n , and exhibit that complement. Then solve this problem for EVERY $M(a,b)$. Use the result to prove every finitely generated R -module is projective and classify which are free.

Solution: R^n has pair (n,n) . For P , the inequalities $n \geq 3$ and $n \geq 1$ force $n \geq 3$; complement $M(0,2)$ attains R^3 . In general $n \geq \max(a,b)$ is necessary, and at $n=\max(a,b)$, $C=M(n-a,n-b)$ gives an explicit direct-sum isomorphism to R^n by grouping coordinates. Thus the smallest free rank is $\max(a,b)$, every $M(a,b)$ is projective, and it is free exactly when $a=b$. The zero module has minimum $n=0$ and zero complement.

3 / hints to offer only after exploration

1. Each control sees dimension n in a free module.

4 / prompt and solution

Can adding the SAME module to $P=M(3,1)$ and $Q=M(1,3)$ make them isomorphic? Can adding some free R^n make $M(1,0)$ free? Prove both answers, and then find a nonfree addition that does make $M(1,0)$ free.

Solution: Adding $M(c,d)$ gives pairs $(3+c,1+d)$ and $(1+c,3+d)$, still unequal, so no common addition repairs P versus Q . $M(1,0) \oplus R^n$ has pair $(n+1,n)$, never equal coordinates, so it is not free for any n . But adding $M(0,1)$ yields $M(1,1)=R$. The obstruction concerns adding a free module, not adding an arbitrary complement; projective need not mean stably free in this example.

4 / hints to offer only after exploration

1. Compare coordinates before and after adding the same quantities.

Student page 3: Allow differences without inventing negative spaces

Advanced: the universal group completion

The monoid of projective isomorphism classes has direct sum as addition. Its K_0 is the abelian group generated by symbols $[M]$, subject to $[M \oplus N] = [M] + [N]$ and $[0] = 0$. An actual class $[M]$ and a formal difference $[P] - [Q]$ are different kinds of description.

How much free space completes this module?

5 / prompt and solution

Compute $z = [M(3,1)] - [M(1,3)]$. Invent a different pair of actual modules giving the same z and justify equality using the relations. Decide whether z is the class of any actual module. More generally, identify exactly which signed pairs can be represented by an actual module.

Solution: z has coordinate difference $(2, -2)$. For example add $M(2,0)$ to both modules: $[M(5,1)] - [M(3,3)] = z$, since the added class cancels in the group. A negative second coordinate cannot be the dimensions of an actual module, so z is genuinely virtual. The coordinate invariant is justified on the defining relations because dimensions add. Every pair with both coordinates nonnegative is represented by $M(a,b)$, and no pair with a negative coordinate is. Prompt 6 proves that no extra invariant is hidden.

5 / hints to offer only after exploration

1. Add the same actual module to the positive and negative sides.

6 / prompt and solution

Prove $K_0(R) \cong \mathbb{Z}^2$, rather than merely giving a map to $\mathbb{Z}^2$. Show every signed pair occurs and that coordinate differences determine formal classes uniquely. Then prove the universal statement: every additive map $\mathbb{N}^2 \rightarrow G$ to an abelian group extends uniquely to a group homomorphism $\mathbb{Z}^2 \rightarrow G$.

Solution: Every actual class is a $[e_1 R] + b [e_2 R]$ by the module classification. Hence every formal class is uniquely forced to have form $u [e_1 R] + v [e_2 R]$, $u, v \in \mathbb{Z}$: the dimension map sends this expression to (u, v) , so no two distinct pairs can coincide. All pairs occur by taking positive and negative parts as actual modules. This proves inverse homomorphisms between $K_0(R)$ and $\mathbb{Z}^2$. For an additive $\phi: \mathbb{N}^2 \rightarrow G$, set $g_1 = \phi(1, 0)$, $g_2 = \phi(0, 1)$; then $\phi(a, b) = ag_1 + bg_2$. The only possible extension is $\Psi(u, v) = ug_1 + vg_2$, using additive inverses for negative integers. It is a well-defined homomorphism extending ϕ , proving existence and uniqueness. In this ring cancellation also means $[P] = [Q]$ iff $P \cong Q$; that conclusion is not asserted for arbitrary rings.

6 / hints to offer only after exploration

1. The two one-component modules generate every actual class.
2. Use the dimension homomorphism to rule out an extra relation.

How much free space completes this module?

Optional facilitator extension 1

Gate: Commutative monoids, group cancellation and a universal property.

Does every commutative monoid embed in its group completion? Test the monoid $\{0, e\}$ with 0 as identity and $e+e=e$.

Solution: No. In any abelian group image, $\phi(e)+\phi(e)=\phi(e)$ implies $\phi(e)=0$ by cancellation, so both monoid elements have the same image. The universal group completion is the zero group: every map to any group is the unique zero map. This contrasts with $\mathbb{N}^2$, whose coordinate cancellation permits an injective map into $\mathbb{Z}^2$. The module computation relied on that special classification.

Mathematics and prior use

Central orthogonal idempotents split modules into two rational vector spaces. Their dimension monoid is $\mathbb{N}^2$, projectivity is certified by free complements, and its universal additive completion is $\mathbb{Z}^2$.

Prior use: The original AD-23 already states $K_0(Q \times Q) = \mathbb{Z}^2$; the ring and mechanism are deliberately retained. The new investigation asks for minimal free complements, stable-addition obstructions and alternate descriptions of one virtual class. No existing elementary counter lesson is claimed to establish the module theorem.

Source: Charles Weibel, The K-book, Chapter II

§1 definition and Proposition 1.1, chapter pp. 1-2; §2 definition of K_0 , p. 5; product-ring statement before Example 2.1.4, p. 6. Use the projective-class monoid and universal group-completion framework. The worksheet independently proves the $Q \times Q$ splitting, smallest complement and virtual-class tests. Evidence: Read the universal construction, stable equality discussion and product-ring statement on 2026-09-26; the chosen module dimensions and all minimality claims are derived directly.

<https://sites.math.rutgers.edu/~weibel/Kbook/Kbook.II.pdf>

Count necklaces when every bead count is fixed

Facilitator guide / exact answers and staged hints / not classroom-piloted

The investigation and a satisfying stop

First-pilot candidate for the six-bead catalog and symmetry explanation; the general n, r formula has an explicit counting/gcd gate. Not classroom-piloted.

A complete fixed-weight necklace catalog, a proof of the rotation average and a real example where adding reflections merges two previously different types.

Prerequisites and preparation

Core: Arranging six beads, exact counting and rotation equivalence. The fixed-pair proof needs orbit/stabilizer reasoning, introduced concretely. The n -bead formula uses gcd and binomial coefficients.

Extensions: Gcd/divisibility for equal-sized orbits at coprime bead counts.

Reading and representation: Two named colors are fixed. Rotate in the plane only until a later explicit reflection rule. Labeled strings and unlabeled necklaces are different collections.

Materials: Six-position circular mats with a removable top marker, three red and three blue counters per learner, open catalog area and rotation ledger. Preparation: about 8 minutes.

10 minutes build freely; 20 classify rotation types; 15 audit naive division; 20 derive the fixed-pair count; 15 test reflection and a new bead number. Do not reveal later counts early.

Launch and facilitation

Every necklace has three red and three blue beads. A turn is allowed, but no flipping and no renaming colors. Build a catalog that another pair cannot enlarge with a new necklace.

Student page 1: Build without counting the same turn twice

Core: six beads and a precise equivalence rule

Use exactly three red beads and three blue beads on six positions. Two necklaces are the same if a rotation makes all named colors match. Do not flip the ring or exchange color names. For recording, 1 means red and 0 means blue; strings read clockwise from a temporary top mark.

Count necklaces when every bead count is fixed

1 / prompt and solution

Choose a catalog method and find every different necklace. Prove that none is missing and that no two entries are rotations of one another. A bare list without a completeness argument is not enough.

Solution: One representative list is 000111,001011,001101,010101. A completeness certificate uses the clockwise blue gaps after the three red beads: triples of nonnegative integers summing 3, considered up to cyclic permutation. The types are (3,0,0),(2,1,0),(2,0,1),(1,1,1). These exhaust the partitions of 3 into three ordered gaps modulo cyclic rotation; the two orders of 2,1,0 are not cyclic permutations. They give precisely the four displayed rotation types. An exhaustive labeled-string catalog with checked rotations is also valid.

1 / hints to offer only after exploration

1. Record how many blue beads occur between successive red beads.
2. Turning changes the starting gap, not their cyclic order.

2 / prompt and solution

Restore the top mark. Count all labeled strings with three red beads. For each necklace in your catalog, count its DISTINCT rotations. Explain why dividing the labeled count by 6 does not give the necklace count.

Solution: There are $C(6,3)=20$ labeled strings, choosing the red positions. The first three representative orbits have 6 distinct rotations each; 010101 has 2. The orbit sizes sum $6+6+6+2=20$. Division by 6 incorrectly assumes every orbit has 6 elements, and $20/6$ is not an integer. The alternating pattern repeats after a two-position turn, so it is the exact obstruction.

2 / hints to offer only after exploration

1. Count actual different strings in an orbit, not the six attempted turns.

Student page 2: Count the patterns fixed by each turn

Proof core: count the same pairs twice

Let r_j move each bead forward j positions modulo 6, for $j=0,\dots,5$. A labeled pattern is FIXED by r_j when the resulting labeled string is unchanged. Count only strings with exactly three red beads.

Count necklaces when every bead count is fixed

3 / prompt and solution

Find the number of patterns fixed by every r_j , with a reason for each turn type. Average the six counts and compare with your catalog. Why must a fixed pattern have constant color along each position-cycle of the turn?

Solution: The fixed counts for $j=0,1,2,3,4,5$ are 20,0,2,0,2,0. Identity fixes all 20. Turns 1 and 5 make one 6-cycle, so a fixed pattern has 0 or 6 reds. Turns 2 and 4 make two 3-cycles, so exactly one cycle must be red, giving 2 choices. Turn 3 makes three 2-cycles, so its red count must be even, excluding 3. Their sum 24 divided by 6 is 4. Equality under the turn propagates a bead's color along its entire position-cycle; choosing one color per cycle is also sufficient.

3 / hints to offer only after exploration

1. Draw the permutation of positions for each turn.
2. The red beads must fill complete cycles.

4 / prompt and solution

Explain why this average **MUST** count necklaces. Count pairs (turn, labeled pattern fixed by that turn) in two ways. Prove that the patterns in any one rotation orbit contribute exactly 6 pairs, even when the orbit has fewer than 6 members.

Solution: Counting by turns gives $\sum_j \text{Fix}(r_j)=24$. For a fixed pattern x , its stabilizer consists of the turns fixing it. The map from six turns to the orbit, $r \mapsto r \cdot x$, has equal fibers: turns r,s give the same result iff $s^{-1}r$ fixes x , so each fiber has size $|\text{Stab}(x)|$. Thus $|\text{Orbit}(x)| \cdot |\text{Stab}(x)|=6$. Stabilizer sizes are equal for all patterns in that orbit (here rotations commute; conjugation gives the general group argument). Summing stabilizer sizes over the orbit therefore contributes 6. Each orbit contributes equally, so total fixed pairs divided by 6 is exactly the number of orbits. This proves the count rather than guessing a successful average.

4 / hints to offer only after exploration

1. For a chosen target rotation of x , compose with one turn that reaches it.

Student page 3: Change the rules, then generalize

Continuation: reflections and gcd

For the first task on this page **ONLY**, allow turning the ring over as well as rotating it, while keeping color names fixed. For the next task return to rotations only.

Count necklaces when every bead count is fixed

5 / prompt and solution

Which of your six-bead necklaces merge when reflections become legal? Give an actual reflected pair and prove the new complete count. Explain why a coincidence for four beads would not settle this case.

Solution: 000111 and 010101 each reflect into their own rotation orbits. The types 001011 and 001101 reflect into each other's orbit, giving 3 bracelet types in all. For example reverse 001011 to 110100, a rotation of 001101. Gap triples make completeness clear: reflection reverses their cyclic order, interchanging (2,1,0) and (2,0,1), while (3,0,0) and (1,1,1) stay equivalent to themselves. Thus exactly one pair of rotation types merges. The four-bead coincidence from the original atlas is not a general rule.

5 / hints to offer only after exploration

1. Reverse the cyclic order of the blue-gap triple.

6 / prompt and solution

For $n \geq 1$ positions and exactly r red beads, $0 \leq r \leq n$, derive the number fixed by a j -position turn using $g = \gcd(n, j)$. State a rotations-only counting formula, justify it, and use it for eight positions with four red beads.

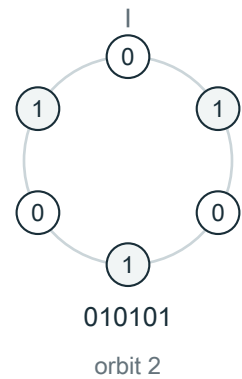
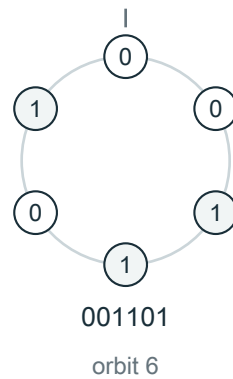
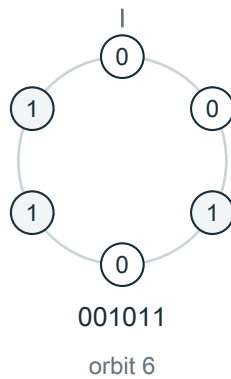
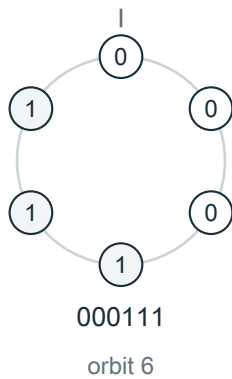
Solution: The j -turn has g cycles, each of length $L = n/g$. A fixed pattern is a union of full red cycles, so the count is 0 unless L divides r ; if L divides r it is $C(g, r/L)$. Here $\gcd(n, 0) = n$, so the identity count is $C(n, r)$. The number of necklaces is $(1/n) \sum_{j=0}^{n-1} F_j$, with F_j as just derived, by the fixed-pair argument. For $n=8, r=4$, the counts in order $j=0, \dots, 7$ are 70, 0, 2, 0, 6, 0, 2, 0, giving $80/8=10$. Endpoints $r=0$ or $r=n$ give one necklace because every turn fixes the unique constant string.

6 / hints to offer only after exploration

1. Equal-length position cycles force a divisibility condition before any binomial choice.

A complete rotation catalog

Facilitator / worked certificate



Colors are encoded by the printed 0/1 labels. The two middle necklaces merge under reflection. Fixed counts for turns $j=0, \dots, 5$ are 20, 0, 2, 0, 2, 0; their mean is 4. The orbit sizes total 20 labeled strings.

Optional facilitator extension 1

Gate: Gcd, divisibility, stabilizers and binomial coefficients.

Prove that if $\gcd(n, r) = 1$ with $0 < r < n$, every fixed-weight rotation orbit has size n . Deduce a shortcut count and apply it to seven beads with three red.

Solution: If a nonidentity turn fixes a pattern, its cycle length $L = n / \gcd(n, r) > 1$ divides both n and r , contradicting coprimality. Thus only the identity stabilizes any pattern, and orbit-stabilizer gives size n for every orbit. The count is $C(n, r) / n$. For 7, 3 it is $35 / 7 = 5$. The coprime hypothesis is sufficient here; without it, orbit sizes may vary and naive division is unsafe.

Mathematics and prior use

An orbit contributes exactly the group size to the fixed-pair count. Rotation cycles also impose divisibility on the allowed red count, explaining the fixed-weight Burnside formula.

Prior use: The original AD-24 counts unrestricted four-bead binary necklaces. The new six-bead fixed-weight instance has a chiral pair and unequal orbit sizes; it supports both a fixed-weight general formula and an actual rotation/reflection distinction. Week 3 return times are preparation, not this orbit-counting argument.

A complete rotation catalog

Source: Thomas Judson, Abstract Algebra: Theory and Applications

§14.3 Theorem 14.3.3 and its fixed-pair proof; supporting orbit-stabilizer Theorem 14.1.11. Use the finite-action double count. Six-bead weight 3, the reflected pair and eight-bead weight 4 are new exact instances solved independently. Evidence: Read the theorem and fixed-point/stabilizer double count on 2026-09-26; enumerate all fixed-weight strings and actual rotation/reflection orbits separately.

<https://judsonbooks.org/aata-files/aata-html/actions-section-burnsides-counting-theorem.html>